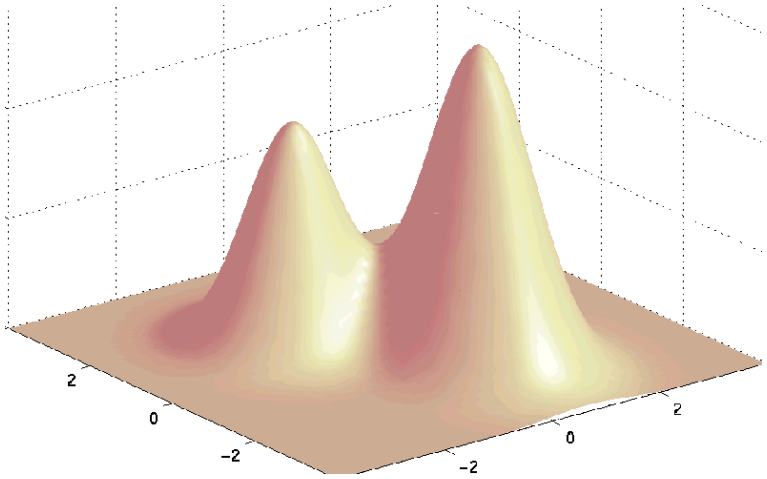
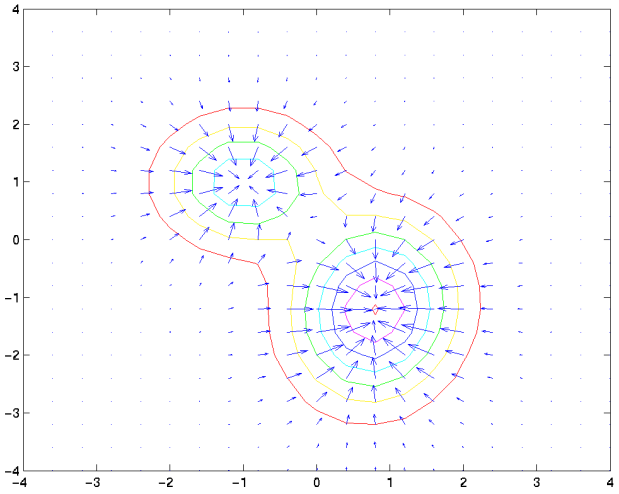
Mathematics of Machine Learning

Optimisation



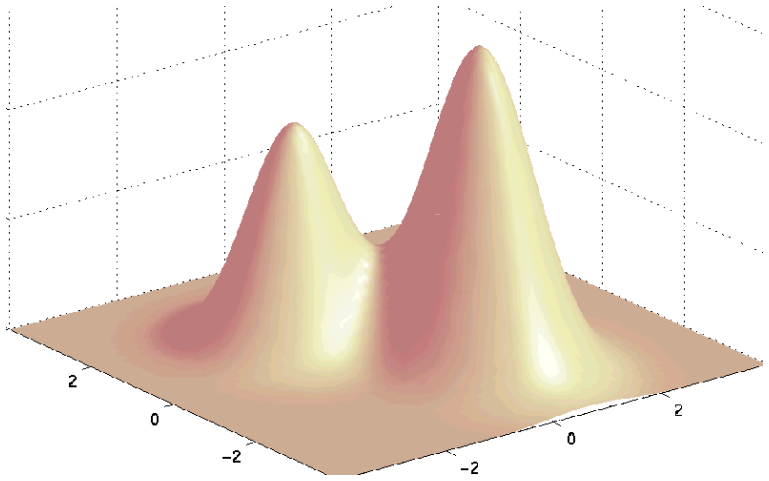


$$z = \mathrm{e}^{-(x+1)^2-(y-1)^2} + 0.6\mathrm{e}^{-(x-1)^2-0.5(y+1)^2+0.1(x-3)(y-3)}$$

Gradient descent, quadratic minima, differing length scales

Outline

1. **Motivation**
2. Gradient Descent
3. Why Gradient Descent is Difficult



ML = Optimisation

- Many learning machines can be thought of as functions of the form

$$\hat{y} = f(\boldsymbol{x}|\boldsymbol{w})$$

(or more generally $\hat{\boldsymbol{y}} = \boldsymbol{f}(\boldsymbol{x}|\boldsymbol{w})$)

- Given an input pattern (set of features) $\boldsymbol{x}$ the learning machine makes a prediction $\hat{y}$
- We try to choose the parameters $\boldsymbol{w}$ so that the predictions are good
- In practice training a learning machine comes down to optimising some loss function

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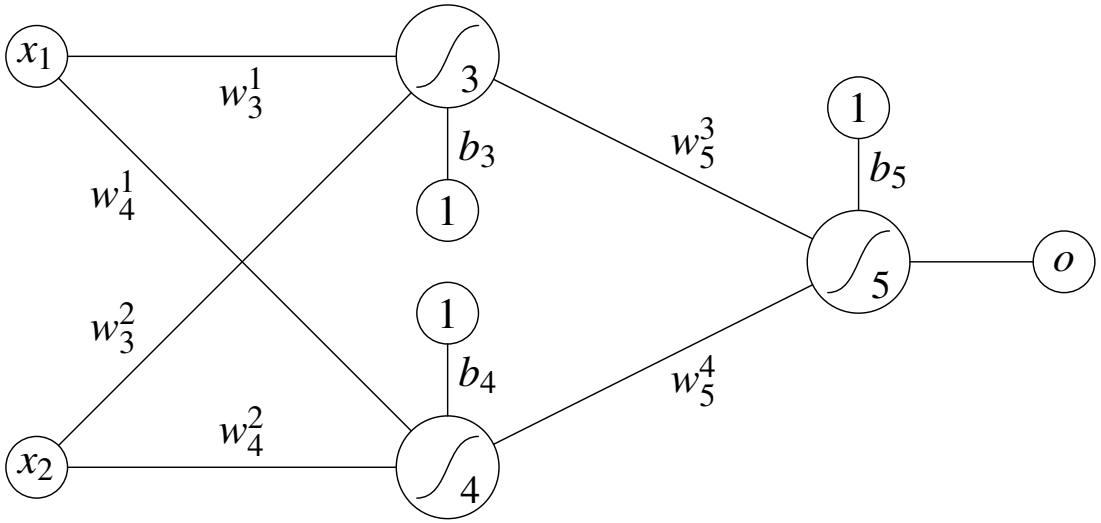
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MLP

- We can depict a neural network such as an MLP by a diagram



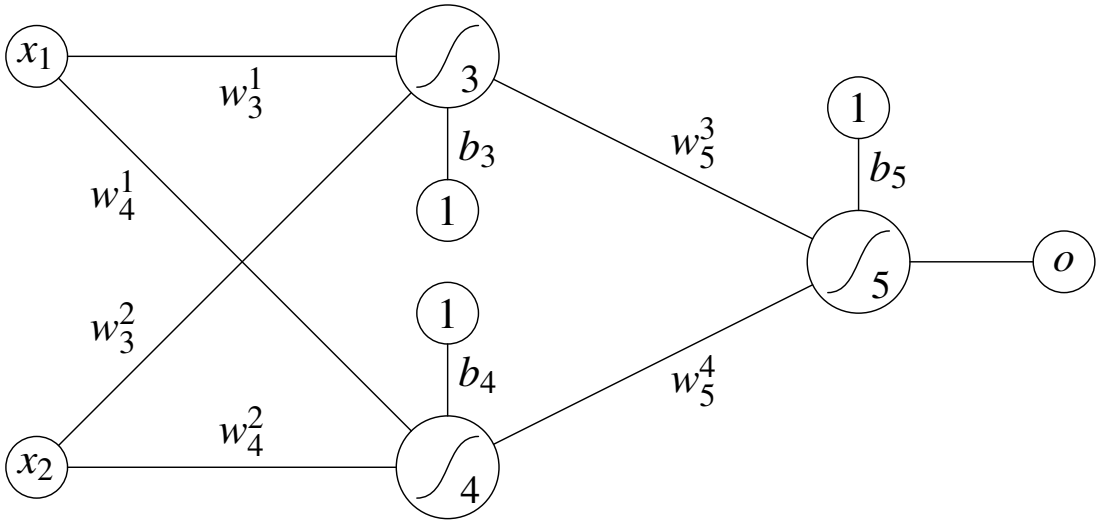
- Stands for the function ($\hat{y} = f(\boldsymbol{x}|\boldsymbol{w})$)

$$\hat{y} = g\big(w_5^3 g(w_3^1 x_1 + w_3^2 x_2 + b_3) + w_5^4 g(w_4^1 x_1 + w_4^2 x_2 + b_4) + b_5\big)$$

where, for example, $g(V) = \frac{1}{1+\mathrm{e}^{-V}}$

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Training

- Given a (labelled) training dataset

$$\mathcal{D} = \left\{ (\boldsymbol{x}_k, y_k) \middle| k = 1, \ldots, m \right\}$$

- We define an error or loss function that we want to minimise

$$L(\boldsymbol{w}|\mathcal{D}) = \frac{1}{m} \sum_{k=1}^{m} \big(f(\boldsymbol{x}_k|\boldsymbol{w}) - y_k \big)^2$$

- We then use the machine with the weights $\boldsymbol{w}^*$ which minimise $L(\boldsymbol{w}|\mathcal{D})$

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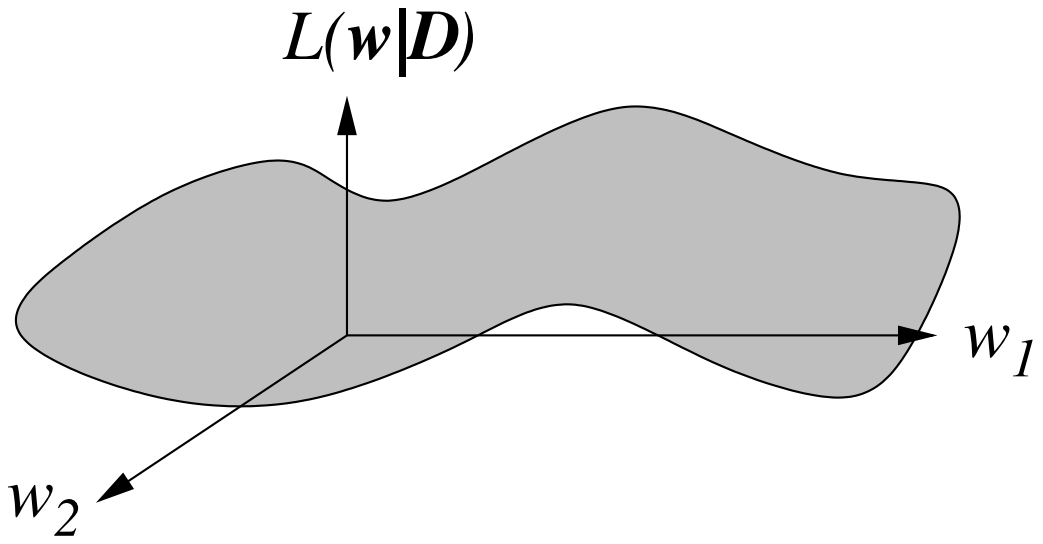
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Computing Gradients

- $L(\boldsymbol{w}|\mathcal{D})$ is a complex function of the weights $\boldsymbol{w}$

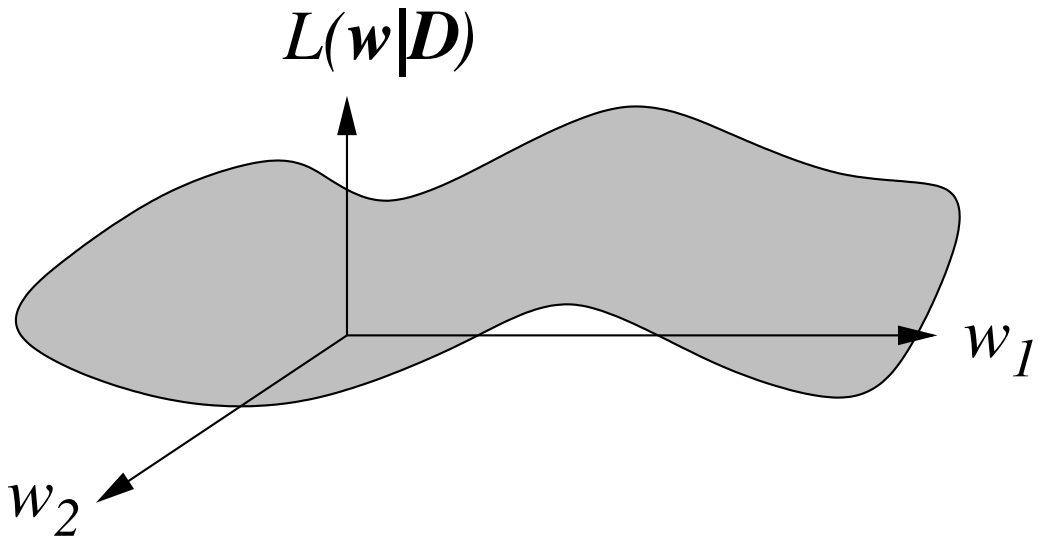


- To minimise $L(\boldsymbol{w}|\mathcal{D})$ we compute the gradient $\boldsymbol{\nabla} L(\boldsymbol{w}|\mathcal{D})$

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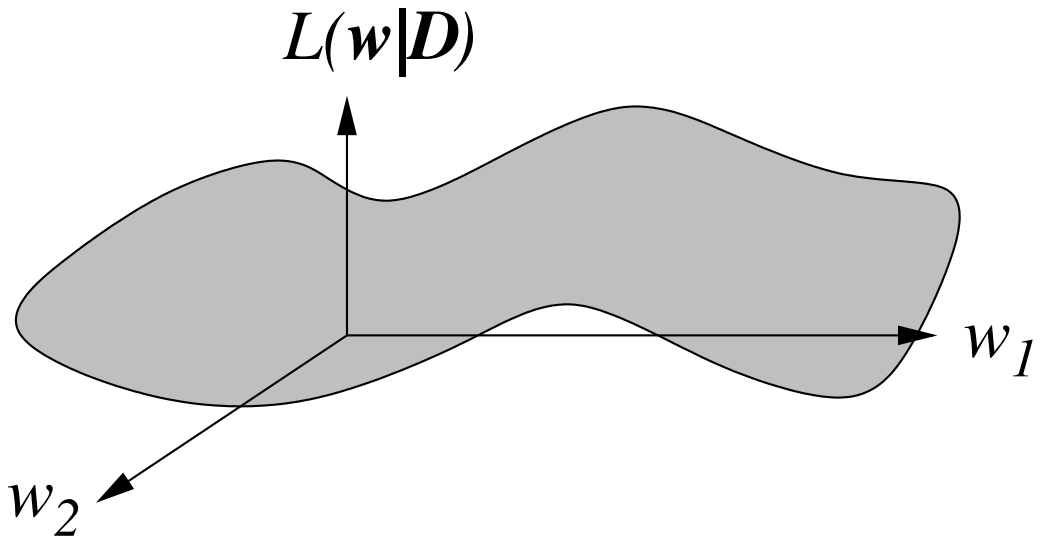


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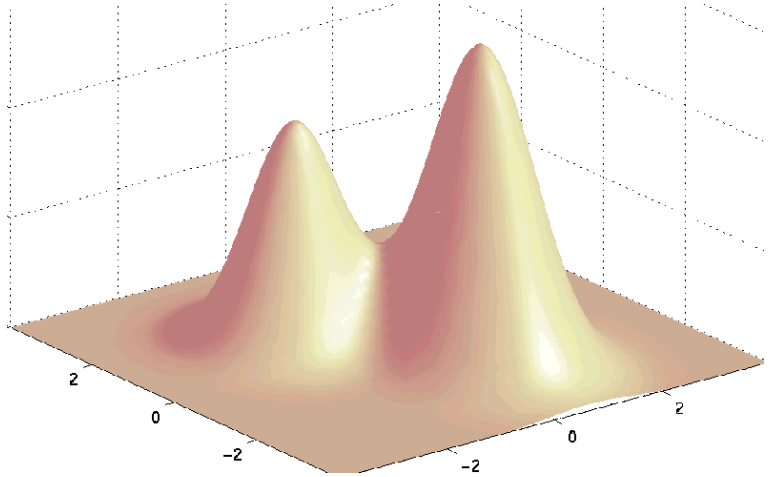


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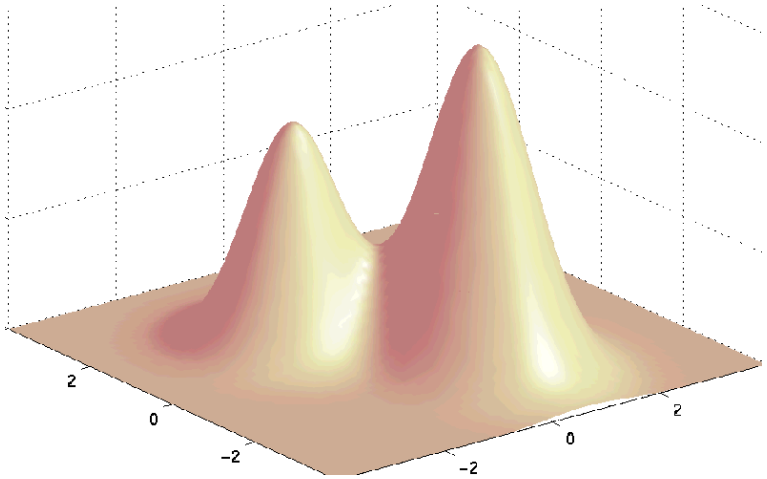
1. Motivation
2. **Gradient Descent**
3. Why Gradient Descent is Difficult

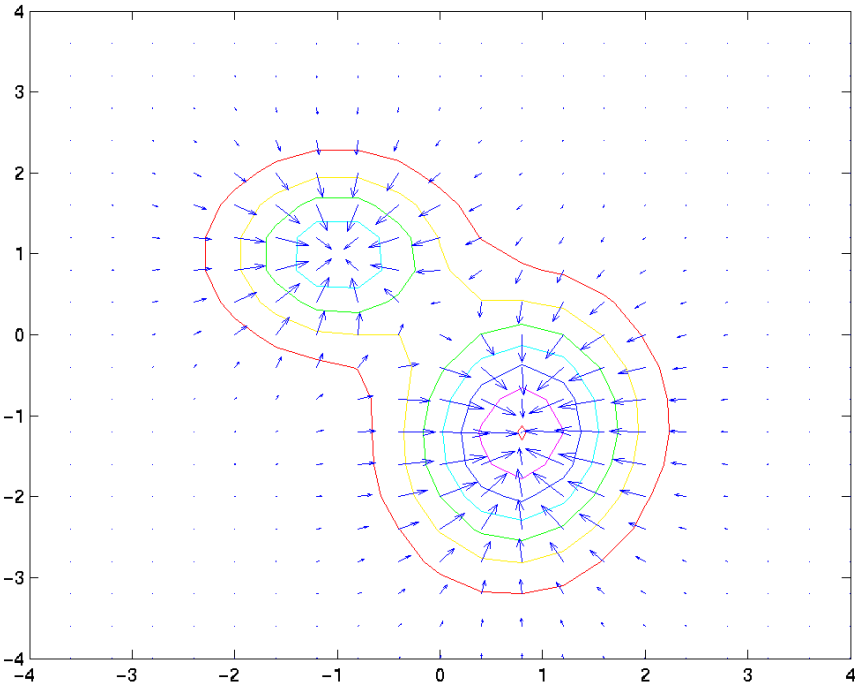


Gradient Optimisation

- At a maximum or minimum (or a saddle point) $\boldsymbol{\nabla} L(\boldsymbol{w}|\mathcal{D}) = \mathbf{0}$
- E.g.

$$L = \mathrm{e}^{-(x+1)^2-(y-1)^2} + 0.6\mathrm{e}^{-(x-1)^2-0.5(y+1)^2+0.1(x-3)(y-3)}$$

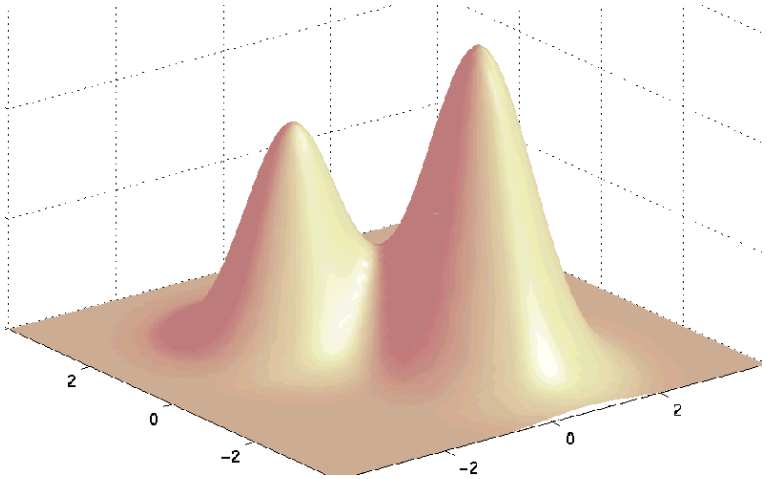


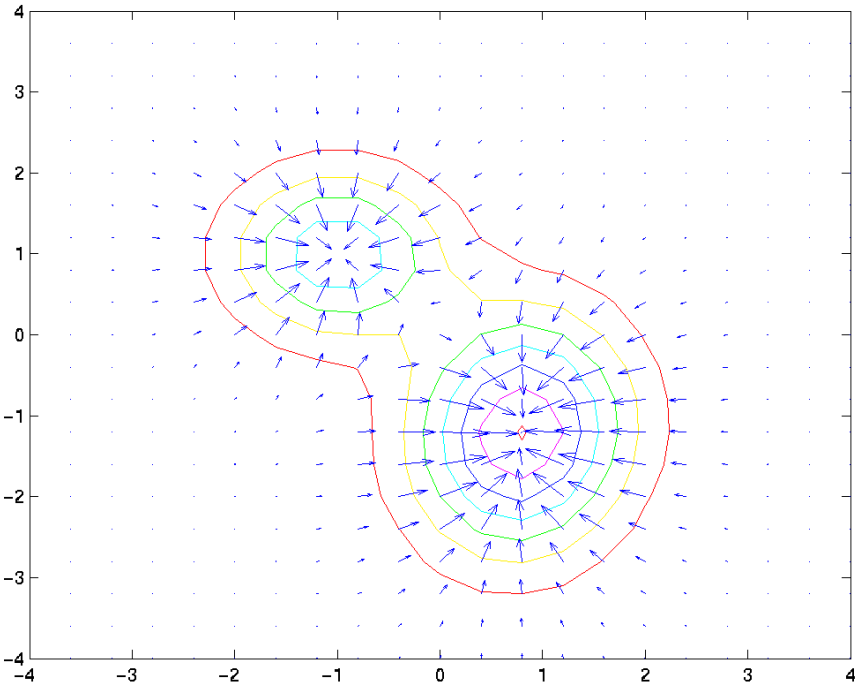


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Gradient Descent

- For a simple function $L(\boldsymbol{w}|\mathcal{D})$ we can solve $\boldsymbol{\nabla} L(\boldsymbol{w}|\mathcal{D}) = \mathbf{0}$ explicitly. E.g. the linear perceptron
- For non-linear functions we usually can't solve this set of simultaneous equations
- We can find a maximum or minimum **iteratively**
- If we know the gradient then we can follow the gradient
  - ⋆ Maximisation: $\boldsymbol{w} \rightarrow \boldsymbol{w}' = \boldsymbol{w} + r\boldsymbol{\nabla} L(\boldsymbol{w}|\mathcal{D})$
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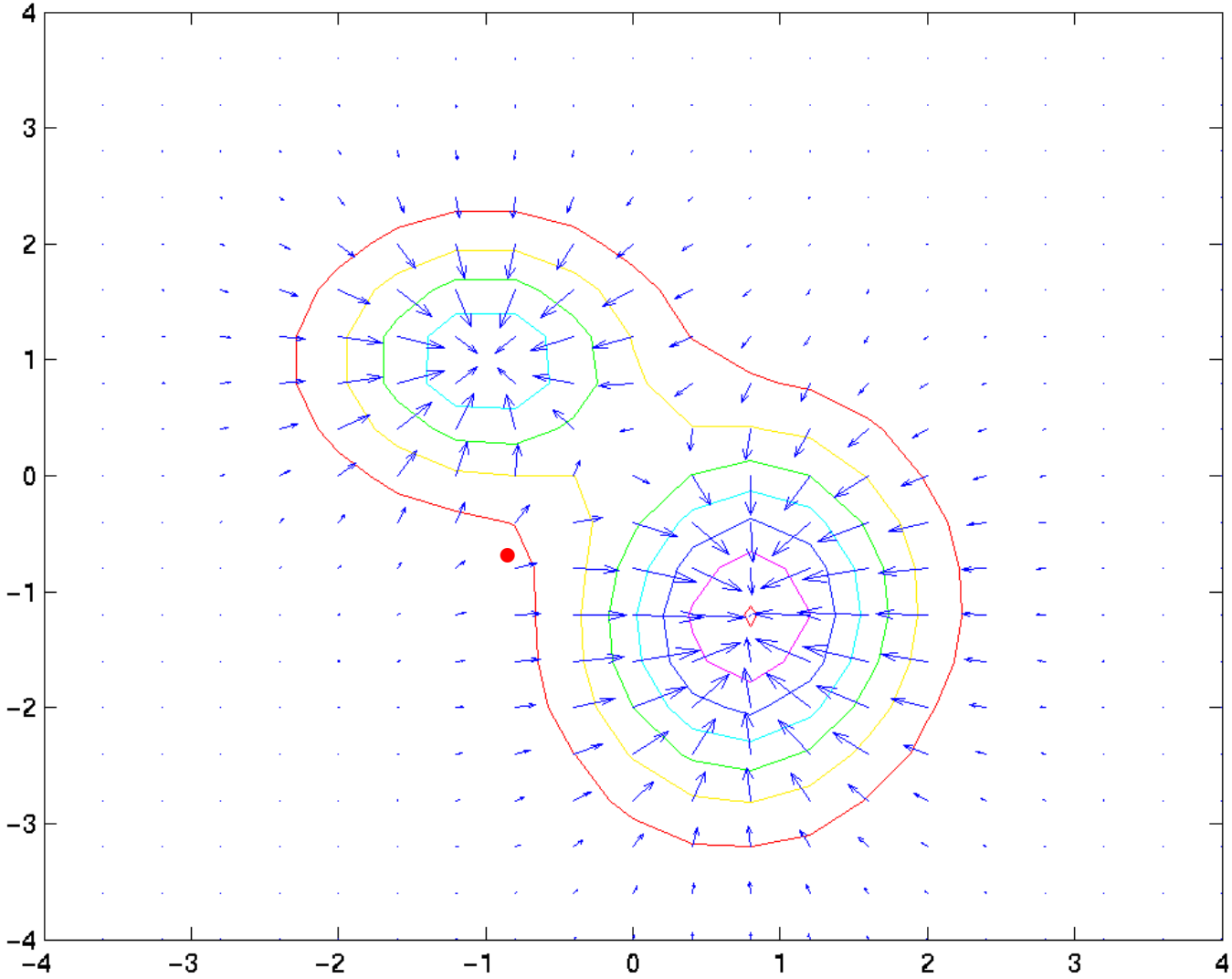
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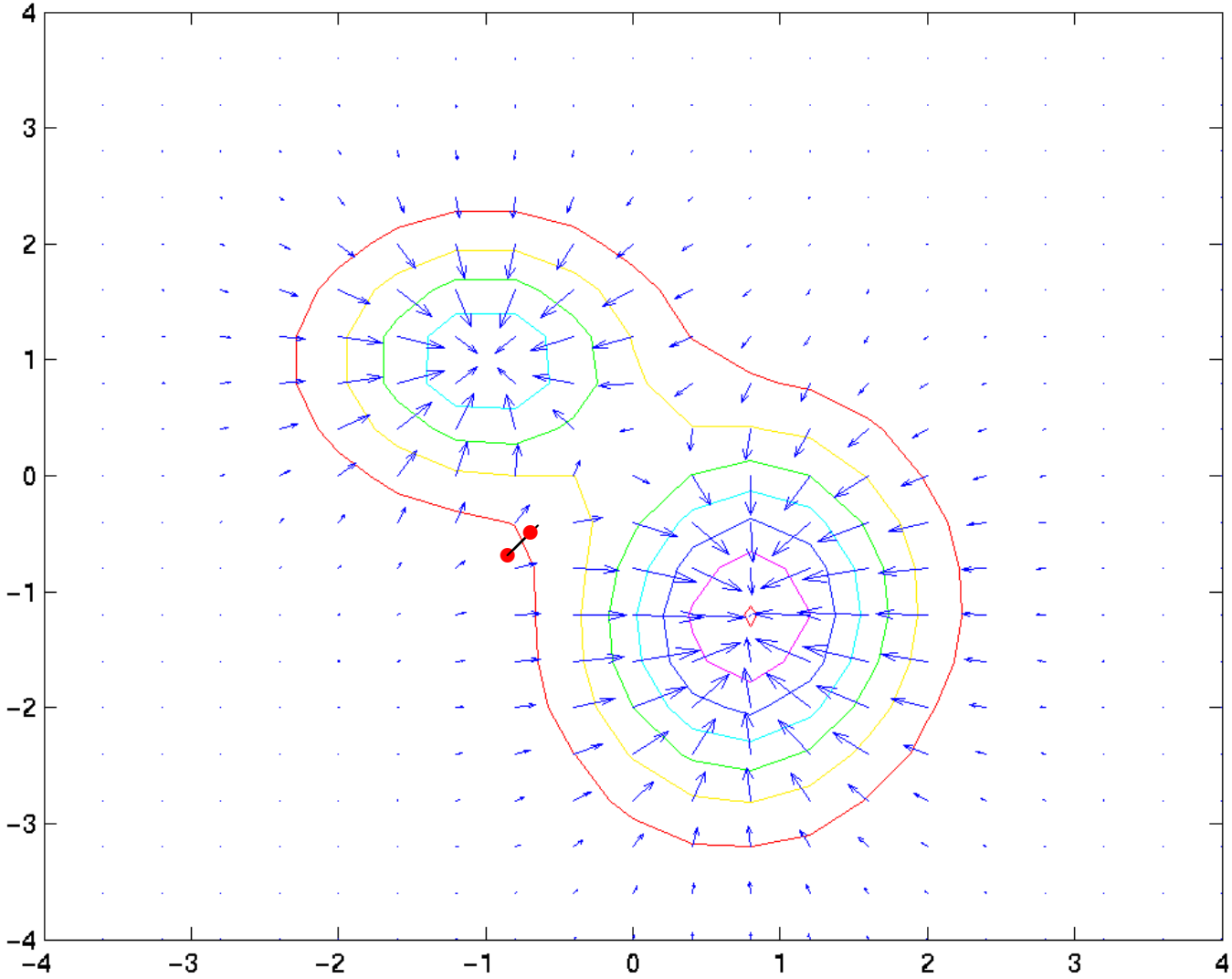
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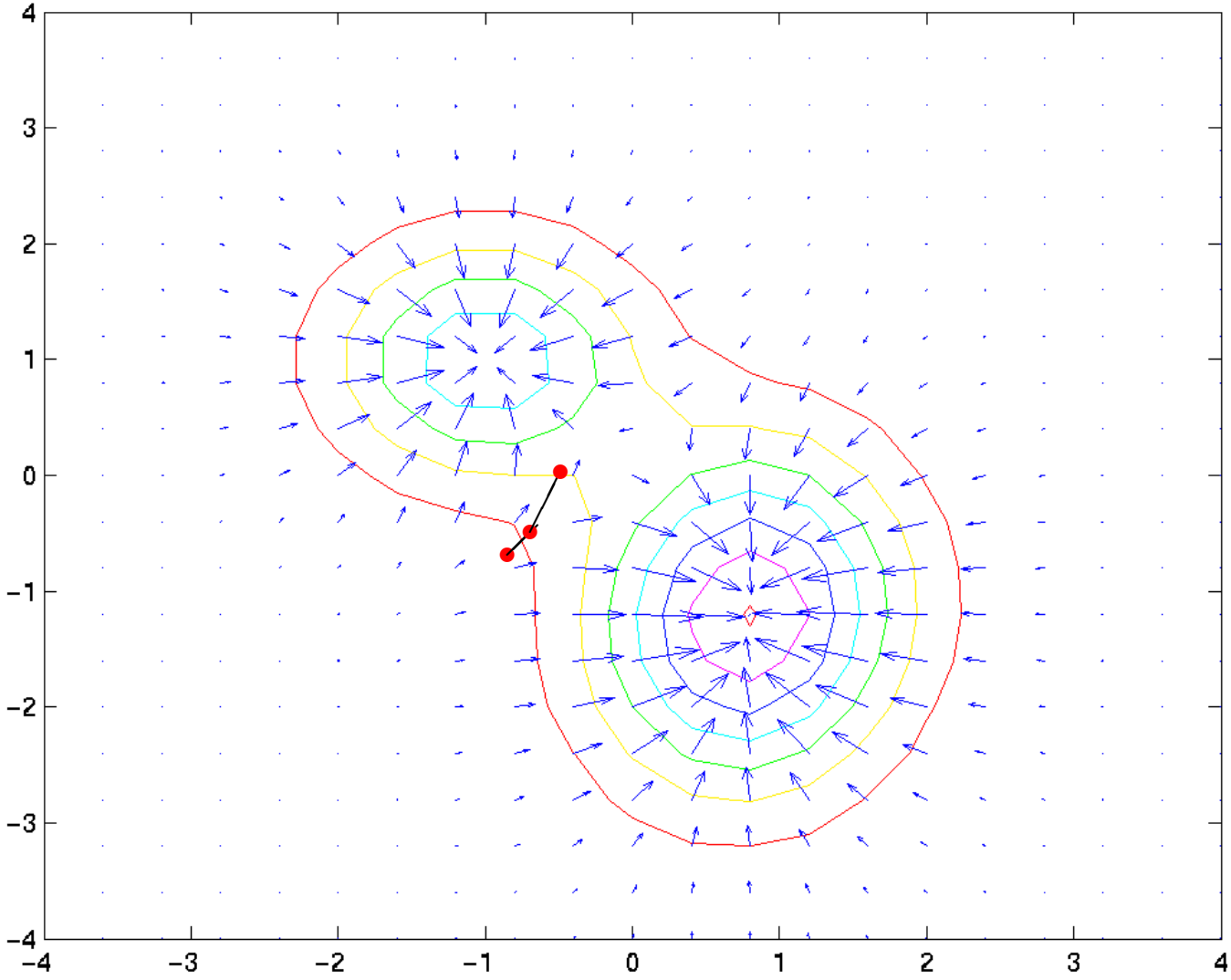
Hill-Climbing



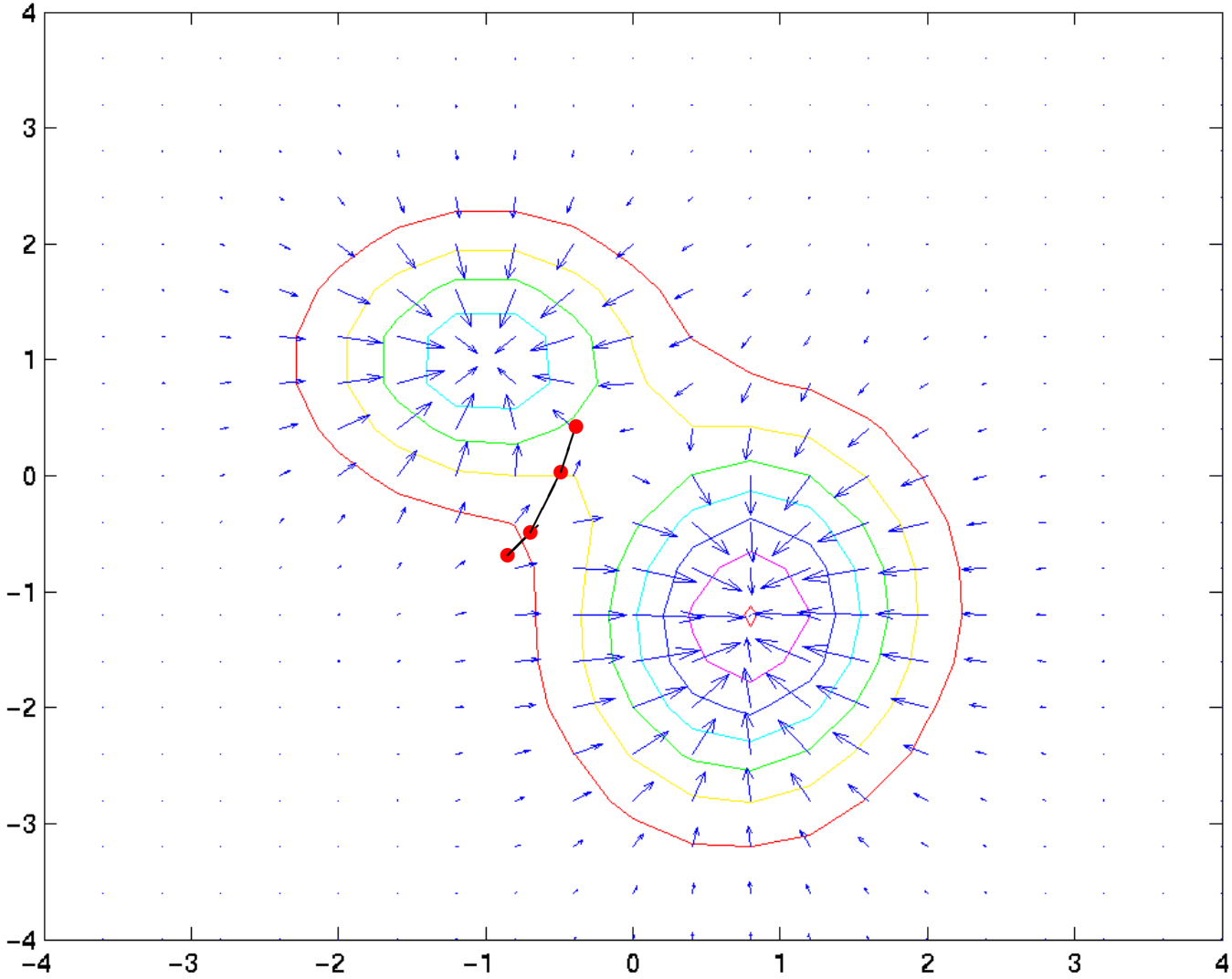
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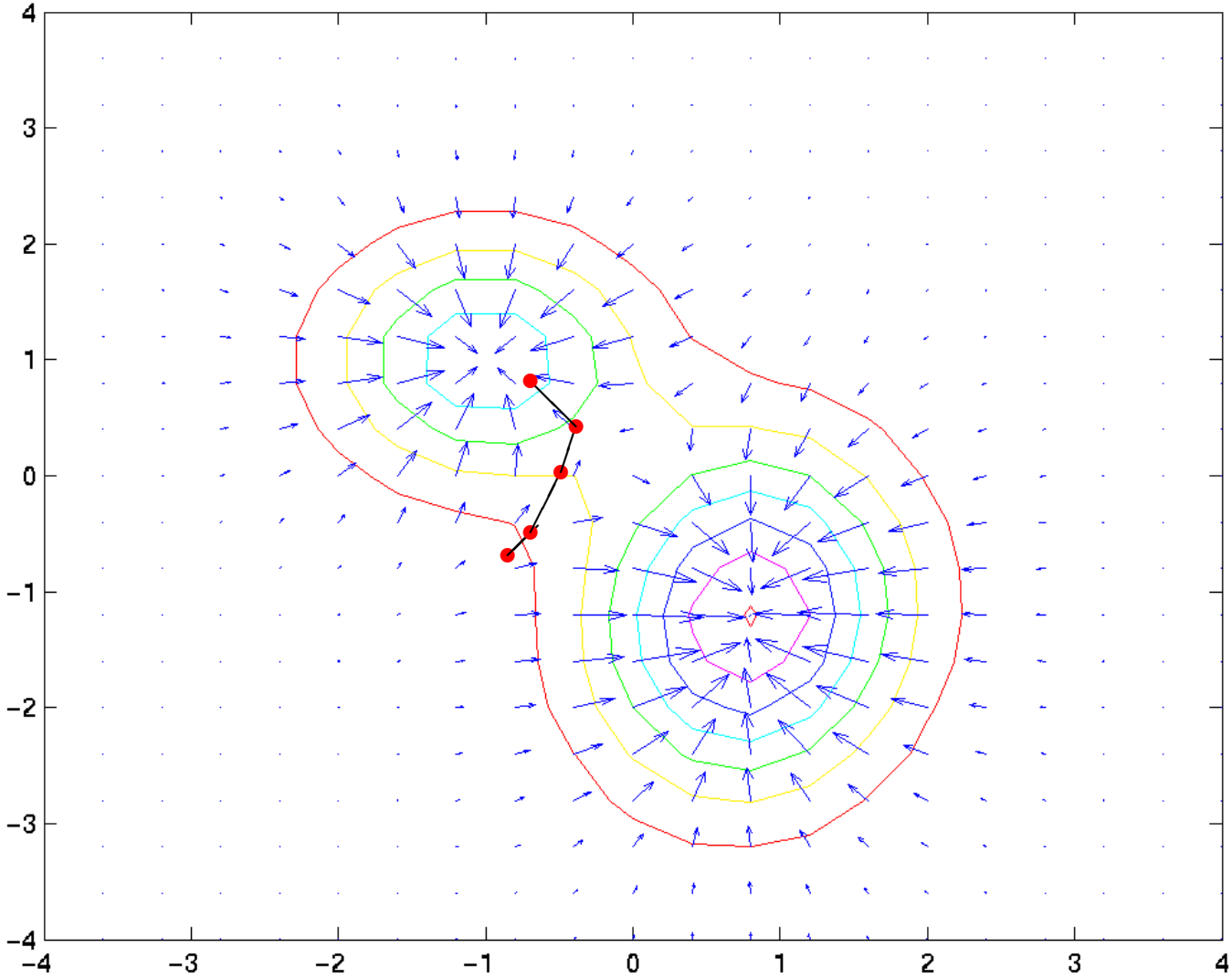
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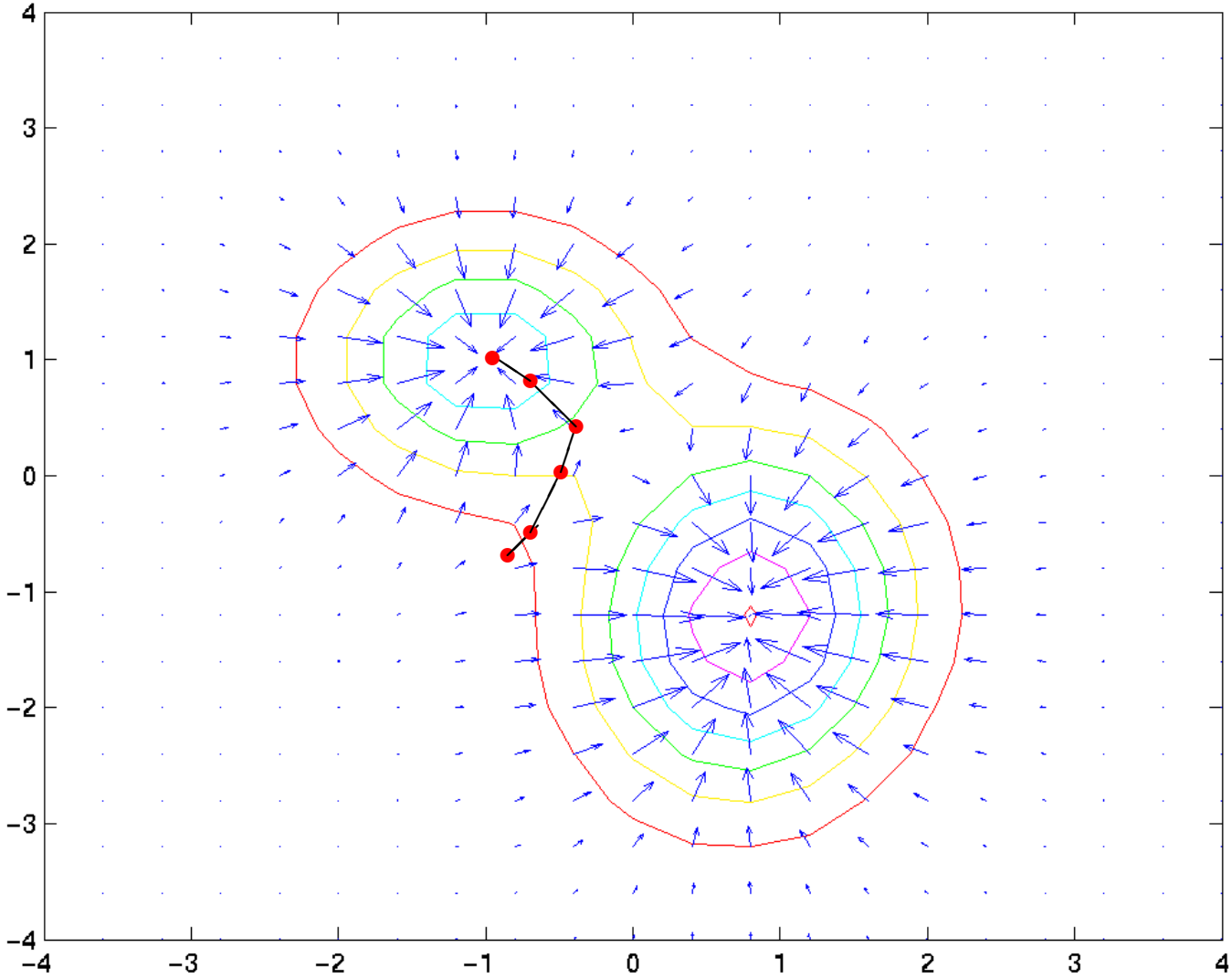
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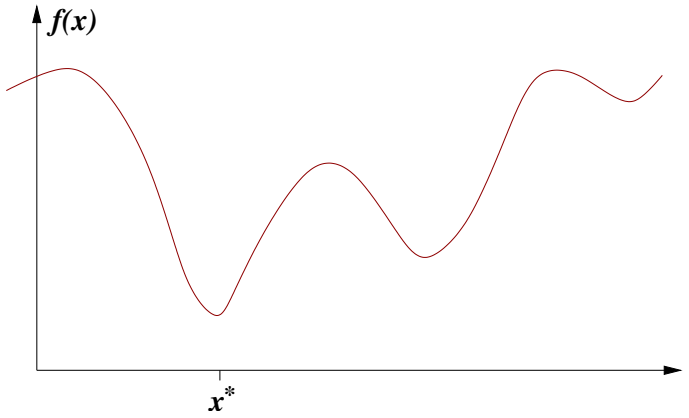


Hill-Climbing



What Goes Right

- Almost all minima are quadratic (Morse's theorem)



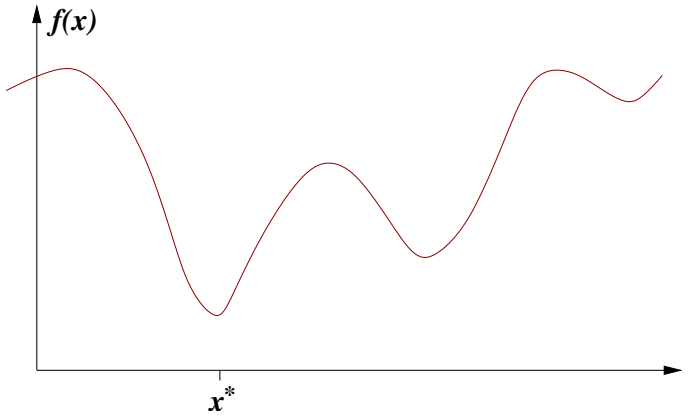
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- If $x - x^*$ is sufficiently small the higher order terms are negligible

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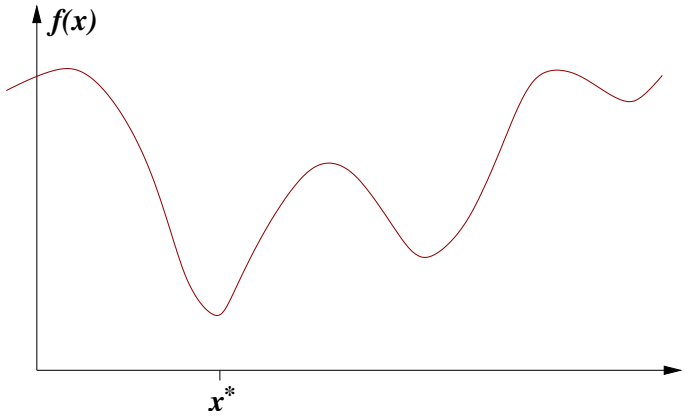
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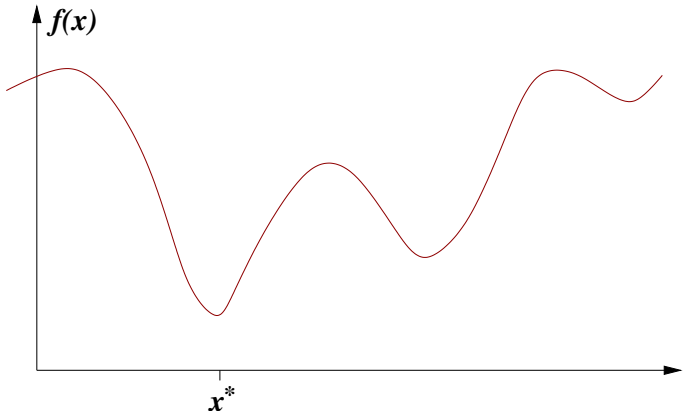
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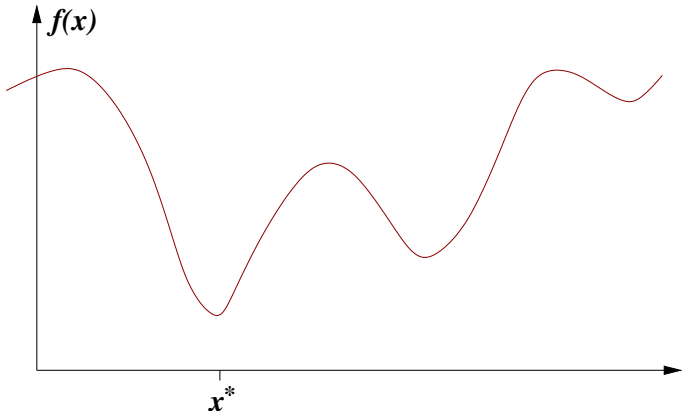
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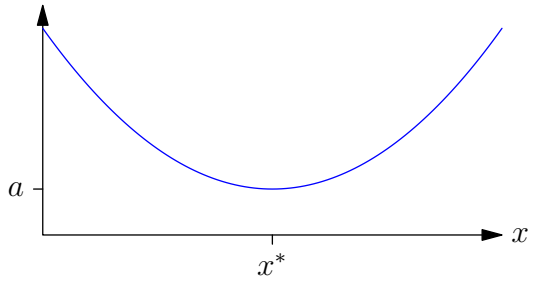
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Newton's Method

- If we were in a quadratic minimum

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- so

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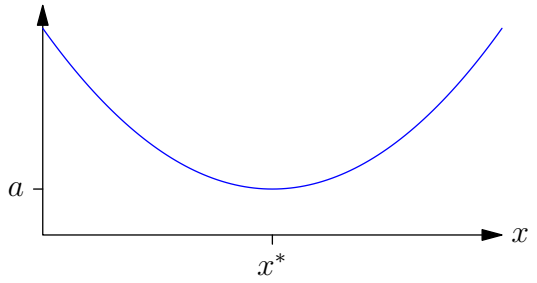
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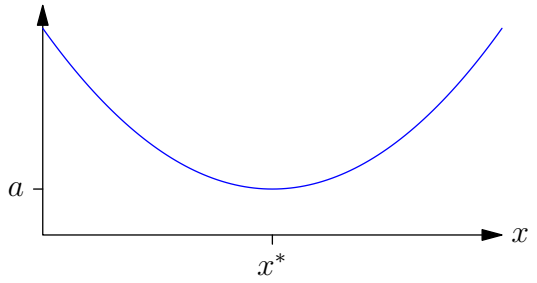
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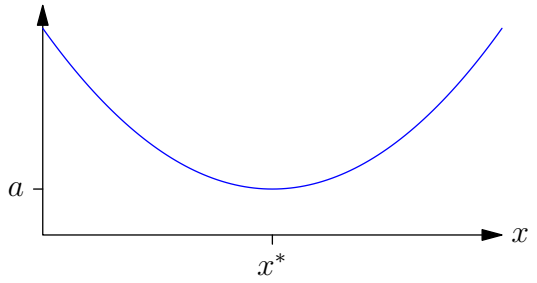
- or

$$x^* = x - \frac{f'(x)}{f''(x)}$$

Newton's Method

- If we were in a quadratic minimum

$$f(x) = a + \frac{b}{2}(x - x^*)^2$$



- then

$$f'(x) = b(x - x^*), \qquad f''(x) = b$$

- so

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- or

$$x^* = x - \frac{f'(x)}{f''(x)}$$

Newton's Method

- This is Newton's method
- For non-quadratic functions Newton's method converges **quadratically** provided we are sufficiently close to a minimum
- If we are at a distance $x - x^* = \epsilon$ from the minimum then after one iteration we will be a distance of order ϵ^2
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Taylor's Expansion in High Dimensions

- We can generalise these results to many dimensions
- The Taylor expansion of a function $f(\boldsymbol{x})$ about $\boldsymbol{x}_0$

$$f(\boldsymbol{x}) = f(\boldsymbol{x}_0) + (\boldsymbol{x} - \boldsymbol{x}_0)^\mathsf{T} \boldsymbol{\nabla} f(\boldsymbol{x}_0) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}_0)^\mathsf{T} \mathbf{H} (\boldsymbol{x} - \boldsymbol{x}_0) + \cdots$$

where $\mathbf{H}$ is the **Hessian** matrix with elements

$$H_{ij} = \frac{\partial^2 f(\boldsymbol{x}_0)}{\partial x_i \partial x_j}$$

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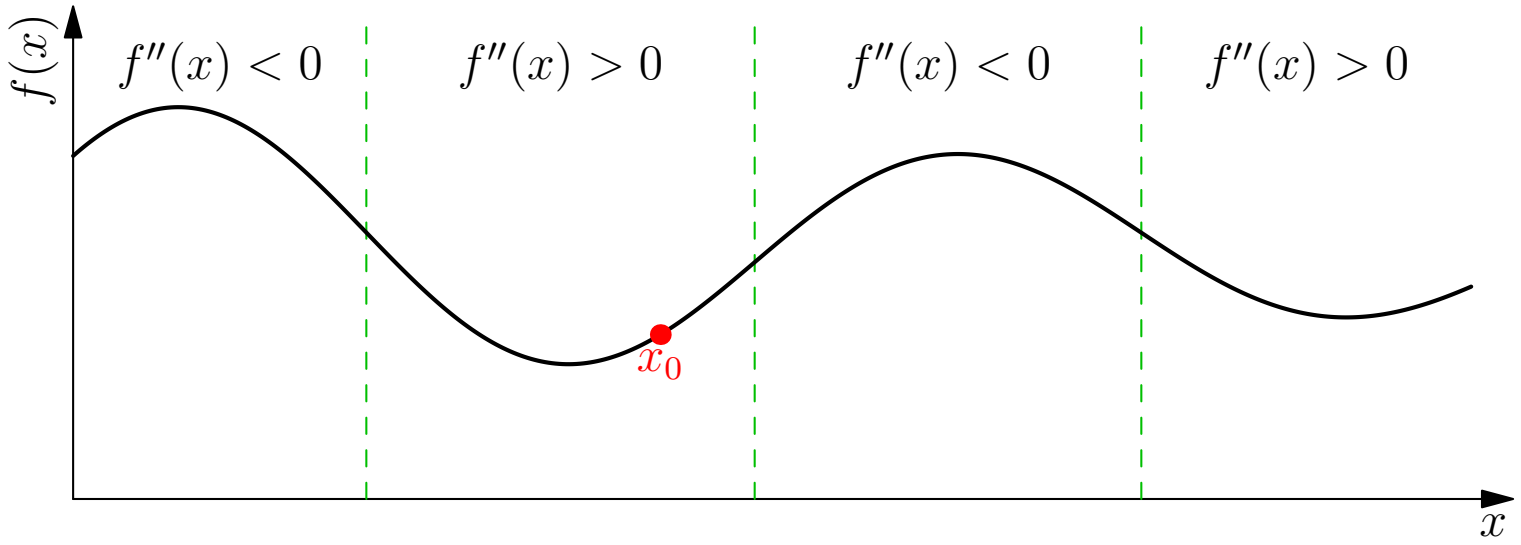
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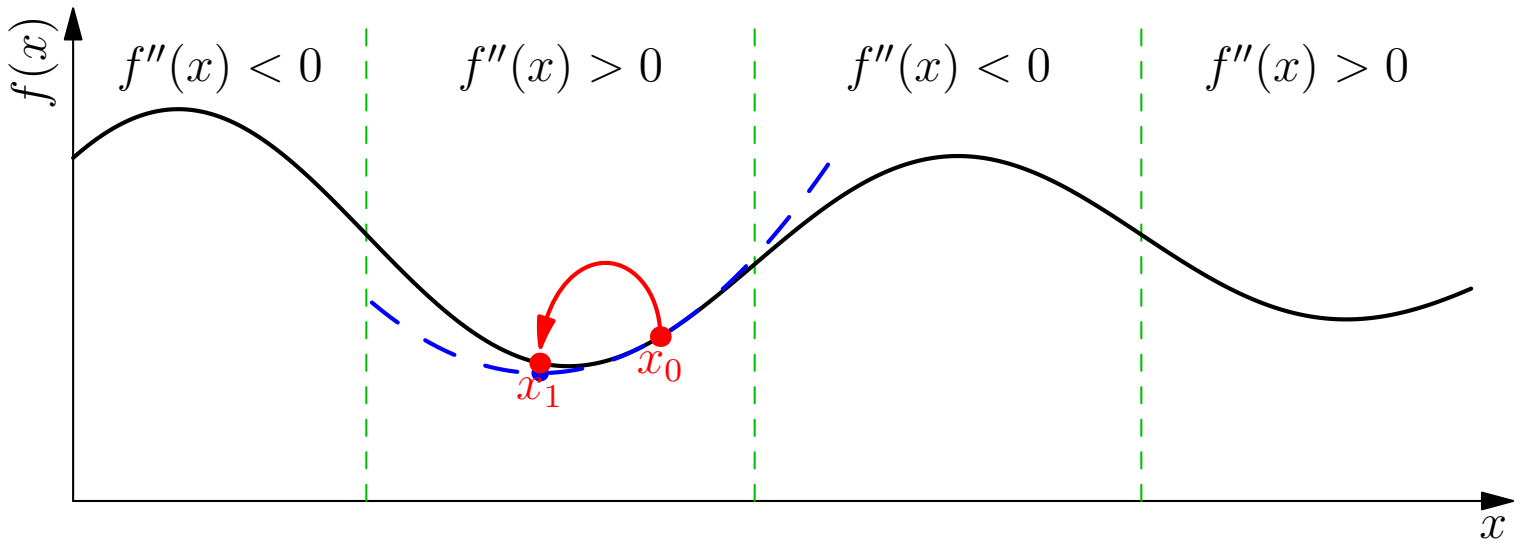
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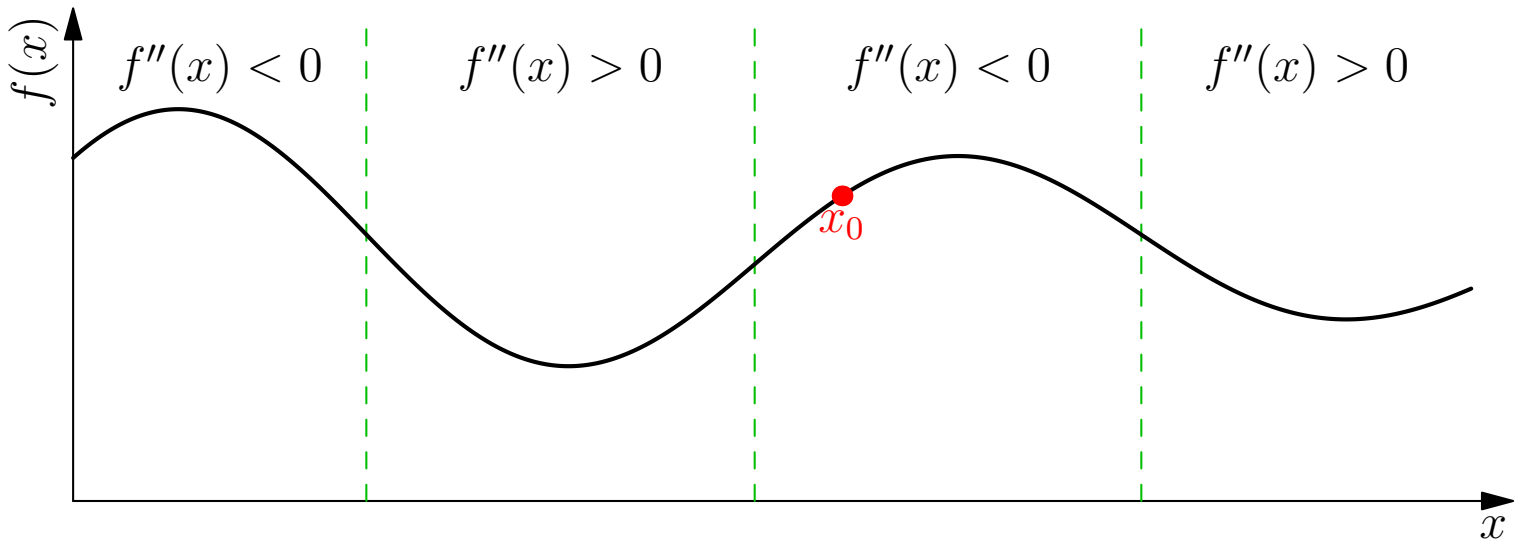
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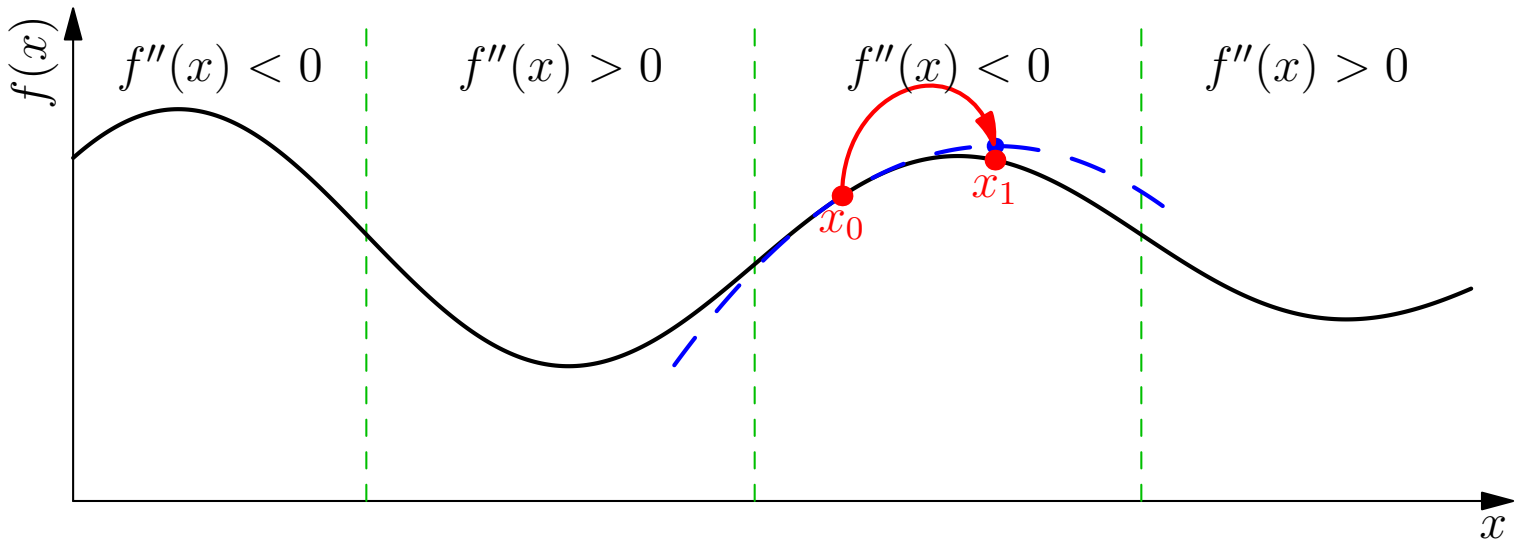
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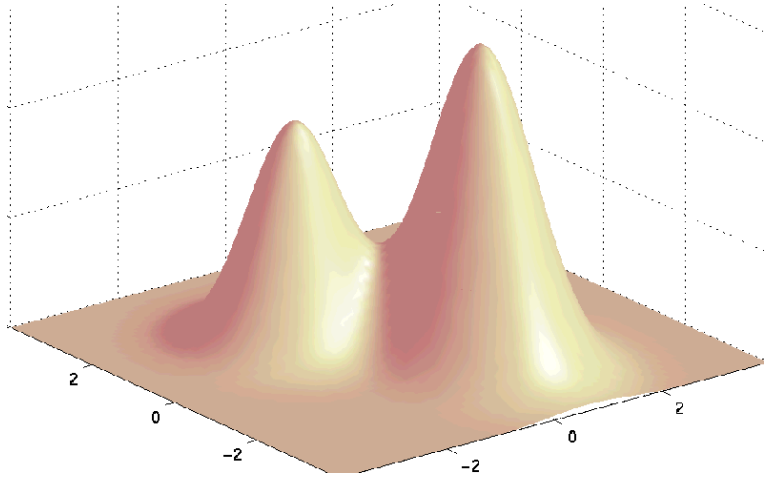
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Outline

1. Motivation
2. Gradient Descent
3. **Why Gradient Descent is Difficult**



Step Size

- Gradient descent

$$\boldsymbol{x}' = \boldsymbol{x} - r\boldsymbol{\nabla} f(\boldsymbol{x})$$

- Need to choose the learning rate or step size, r
- Too small a step takes lots of time
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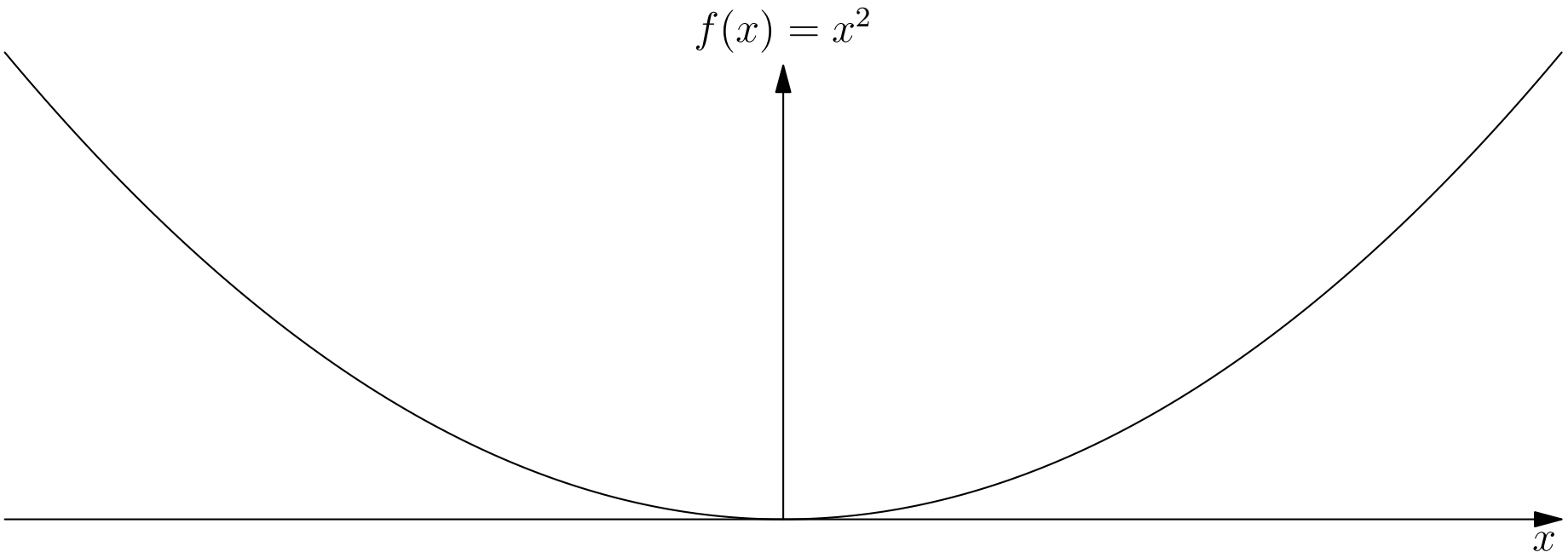
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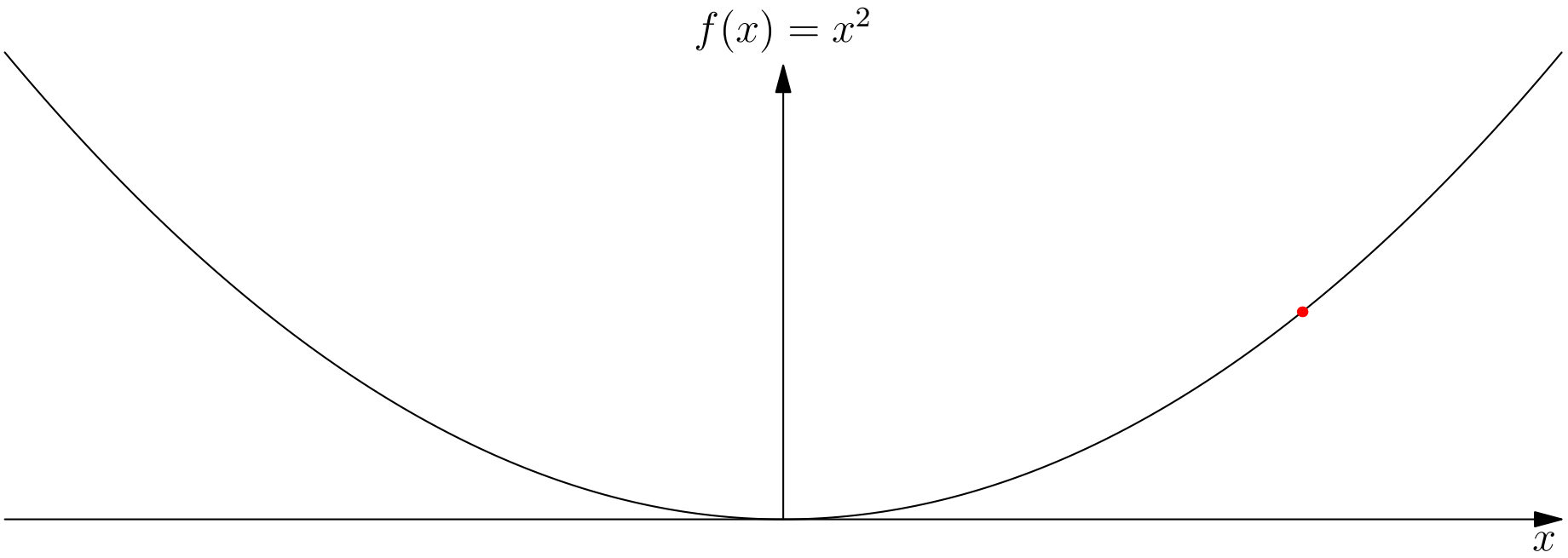
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$$x \leftarrow x - r\, f'(x)$$



Step Size

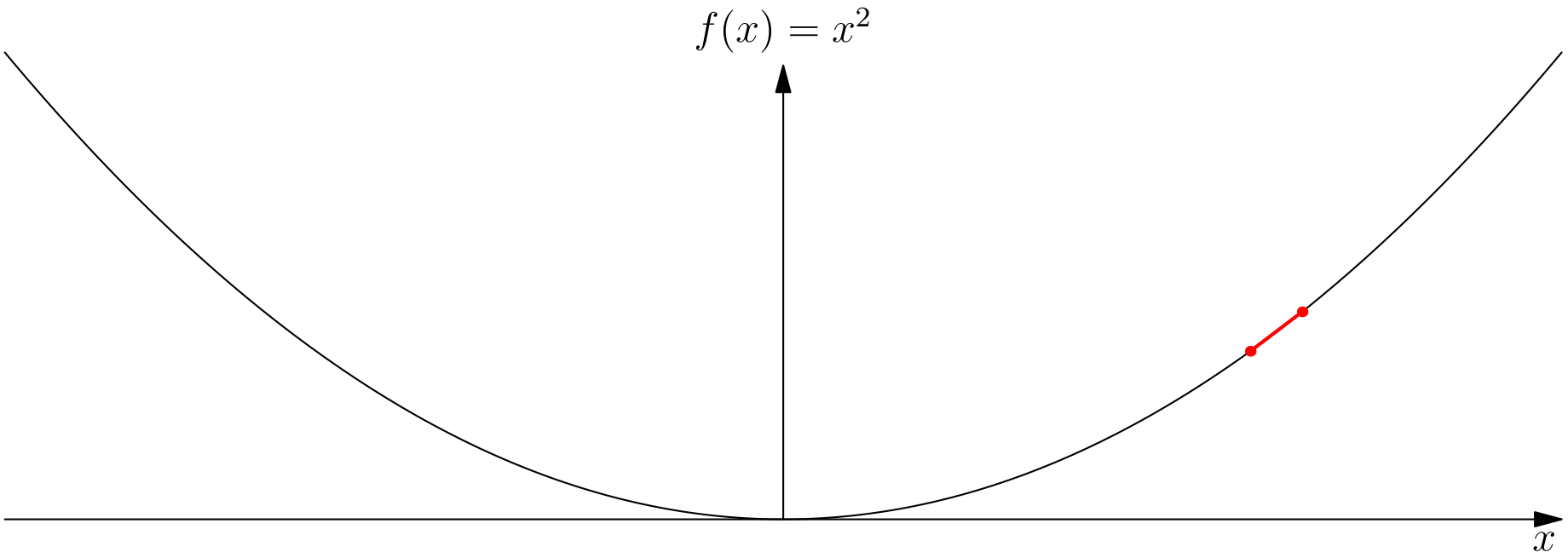
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$\mathrm{r} = 0.05$

Step Size

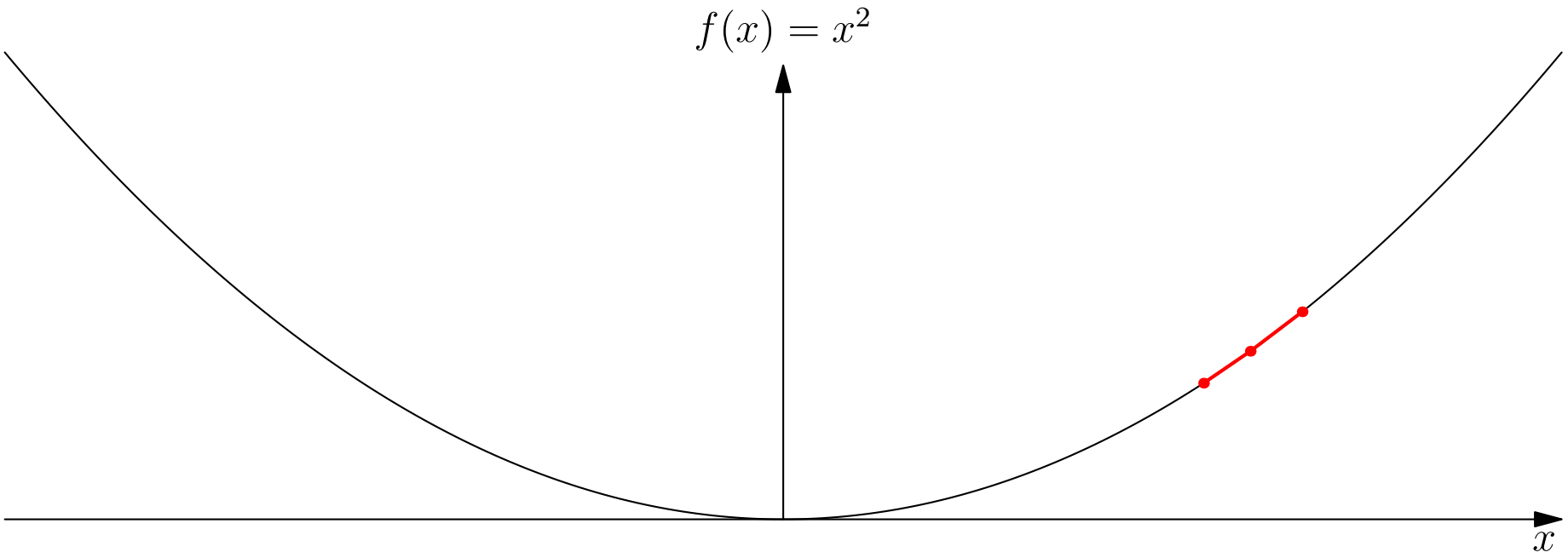
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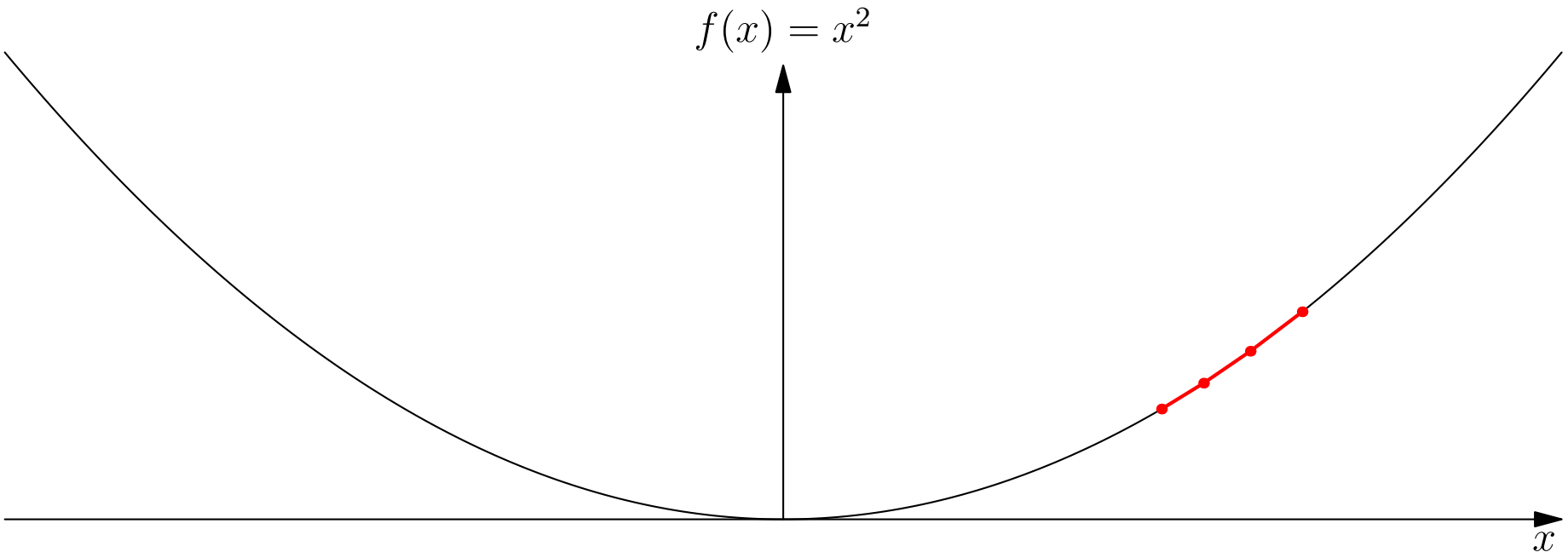
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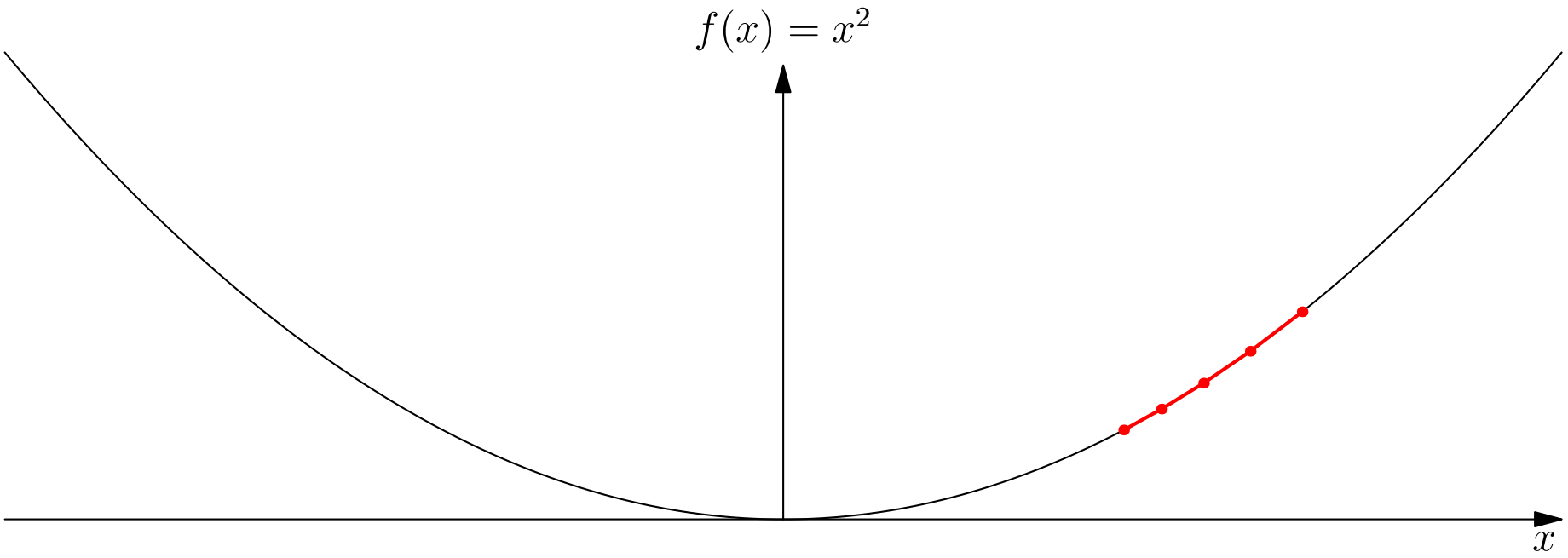
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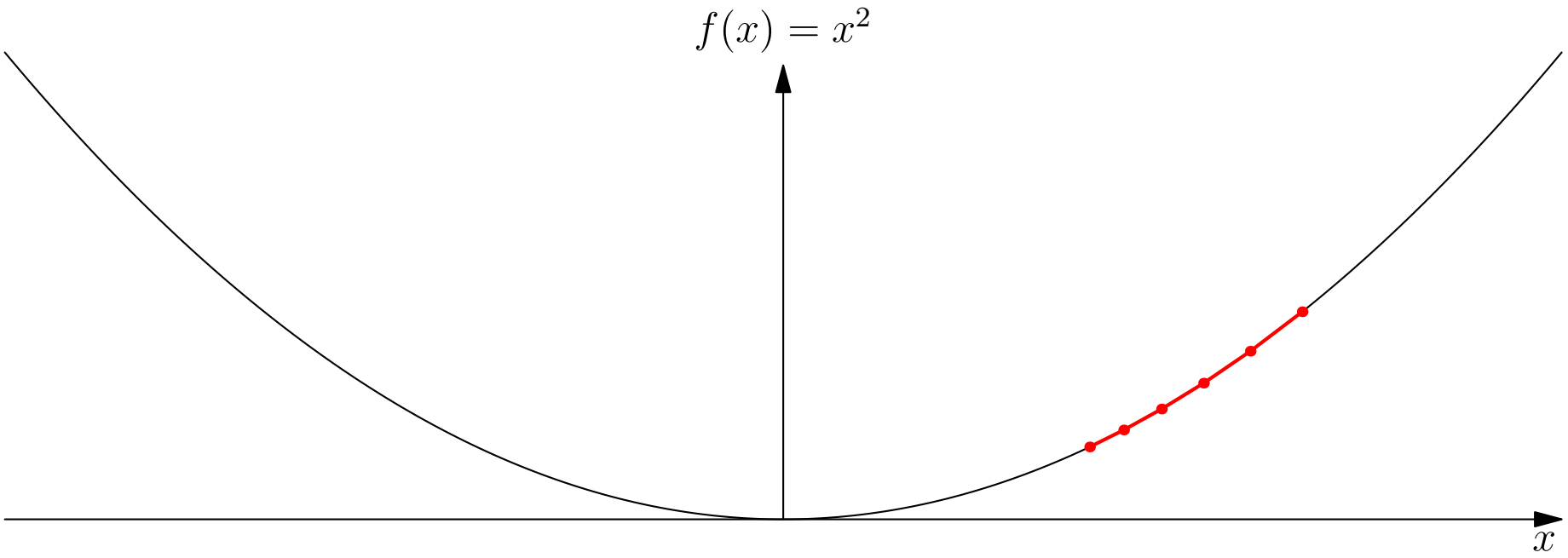
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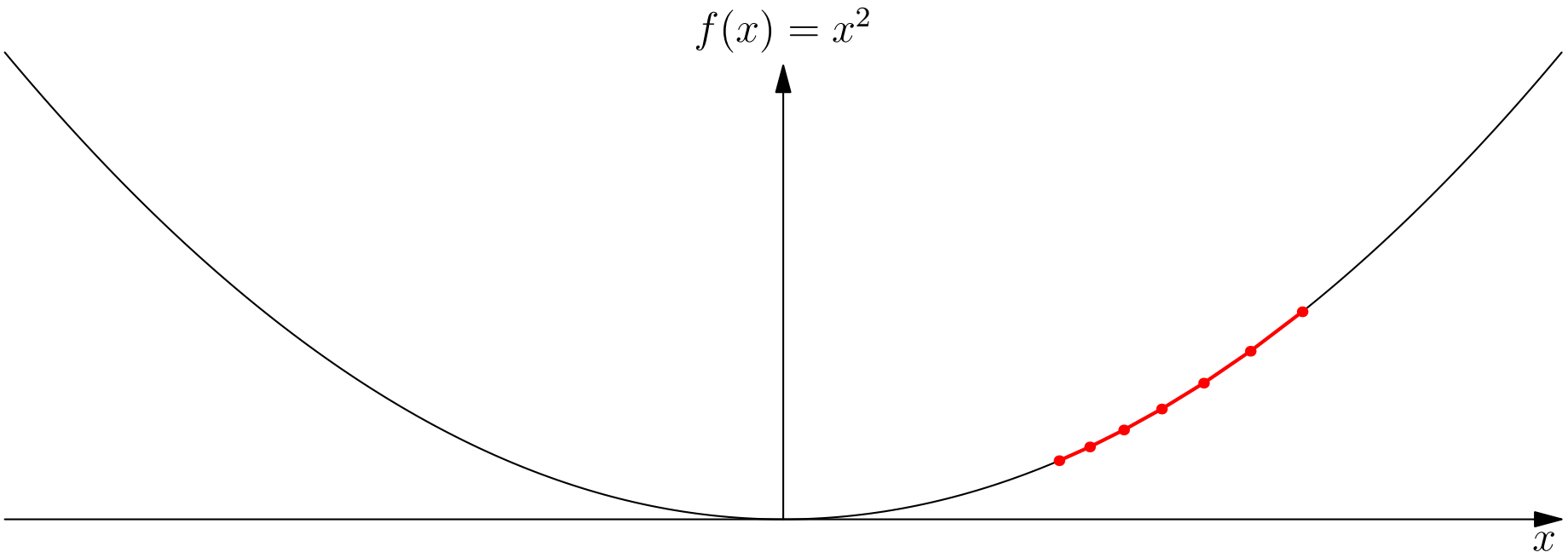
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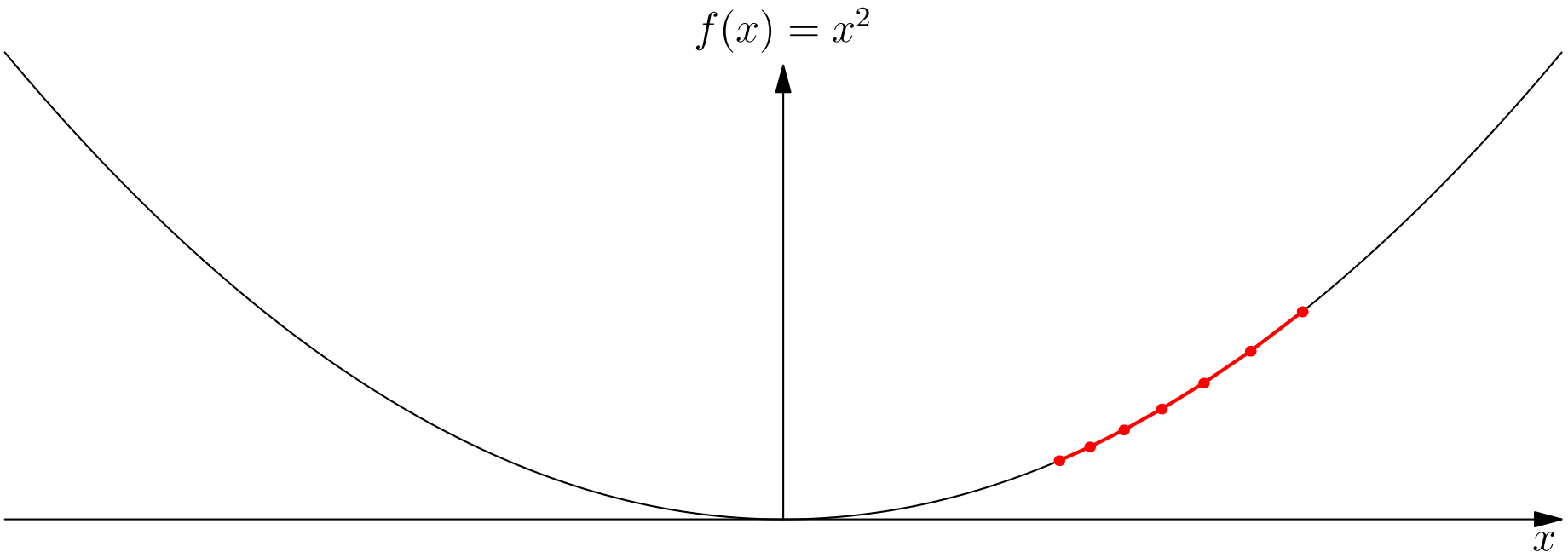
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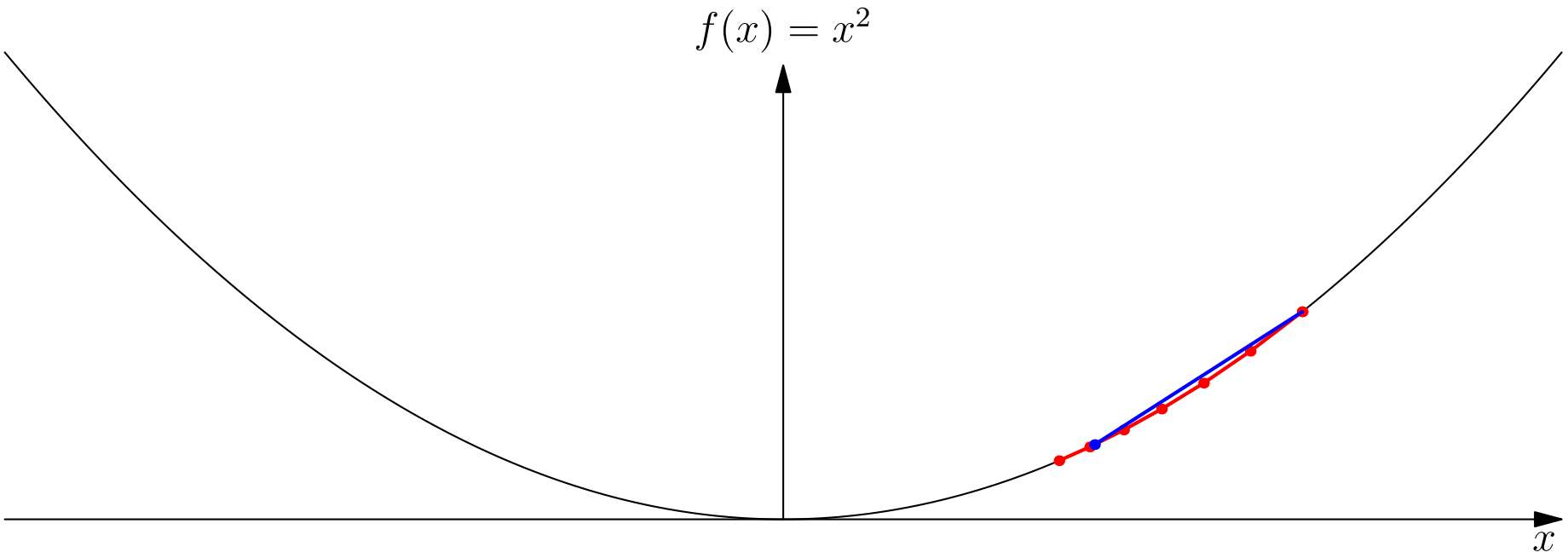
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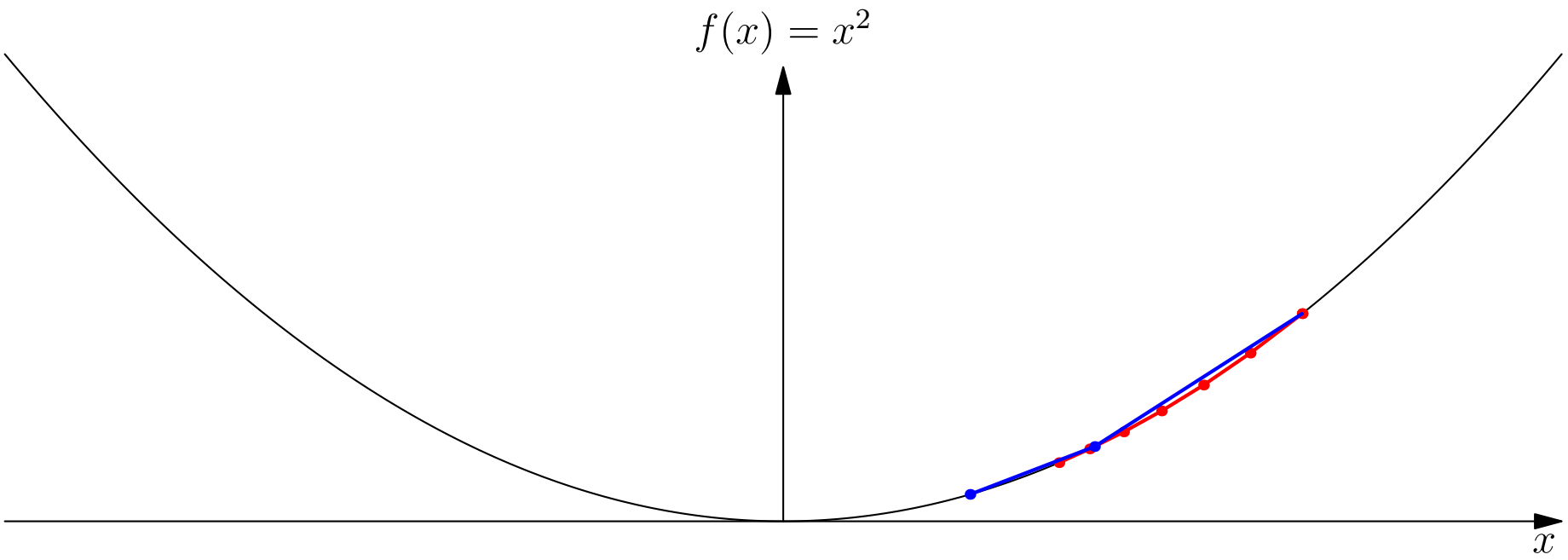
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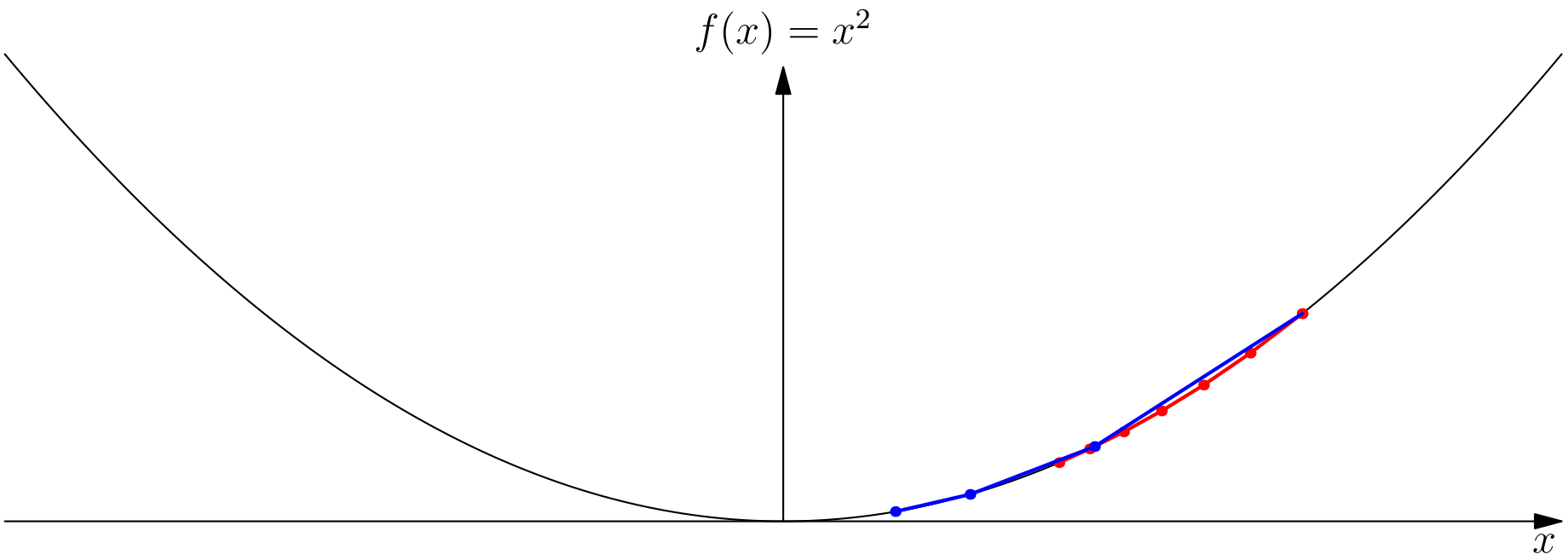
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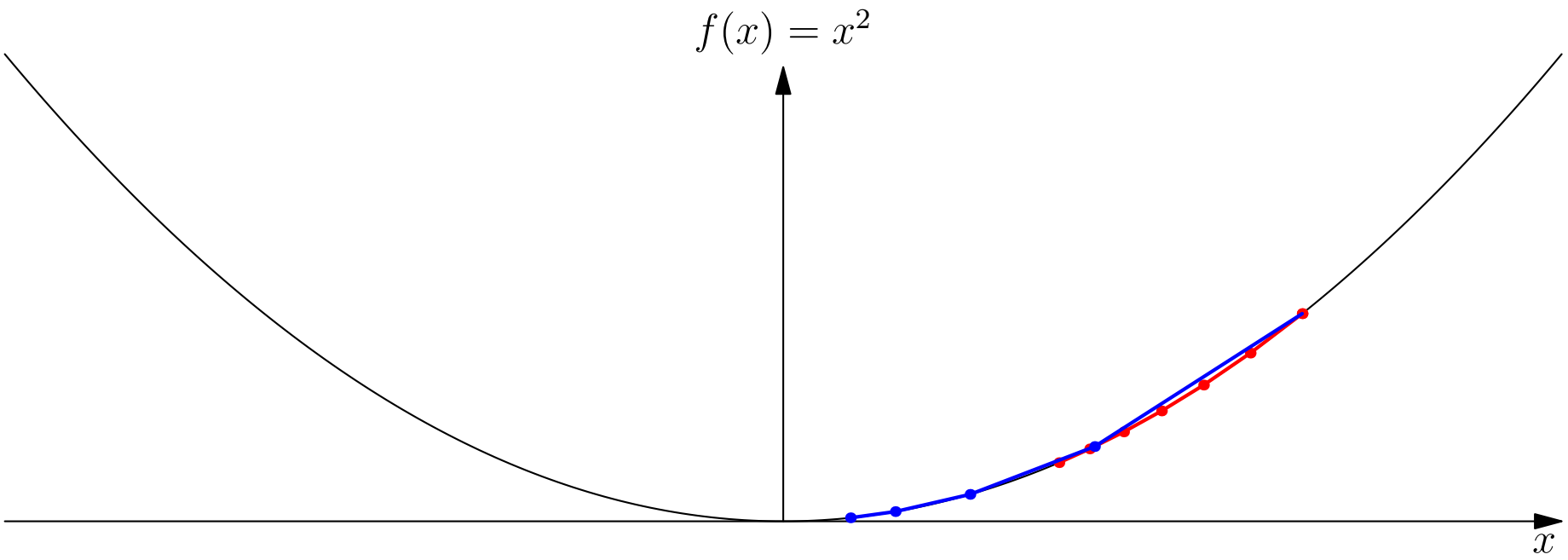
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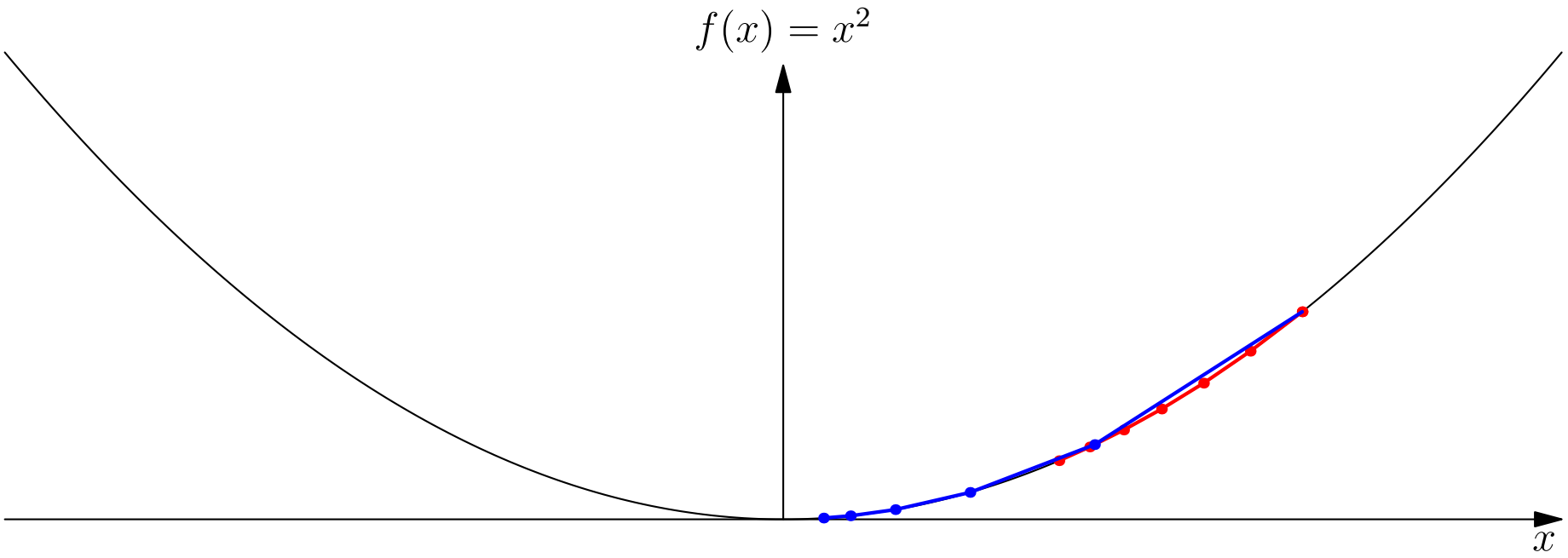
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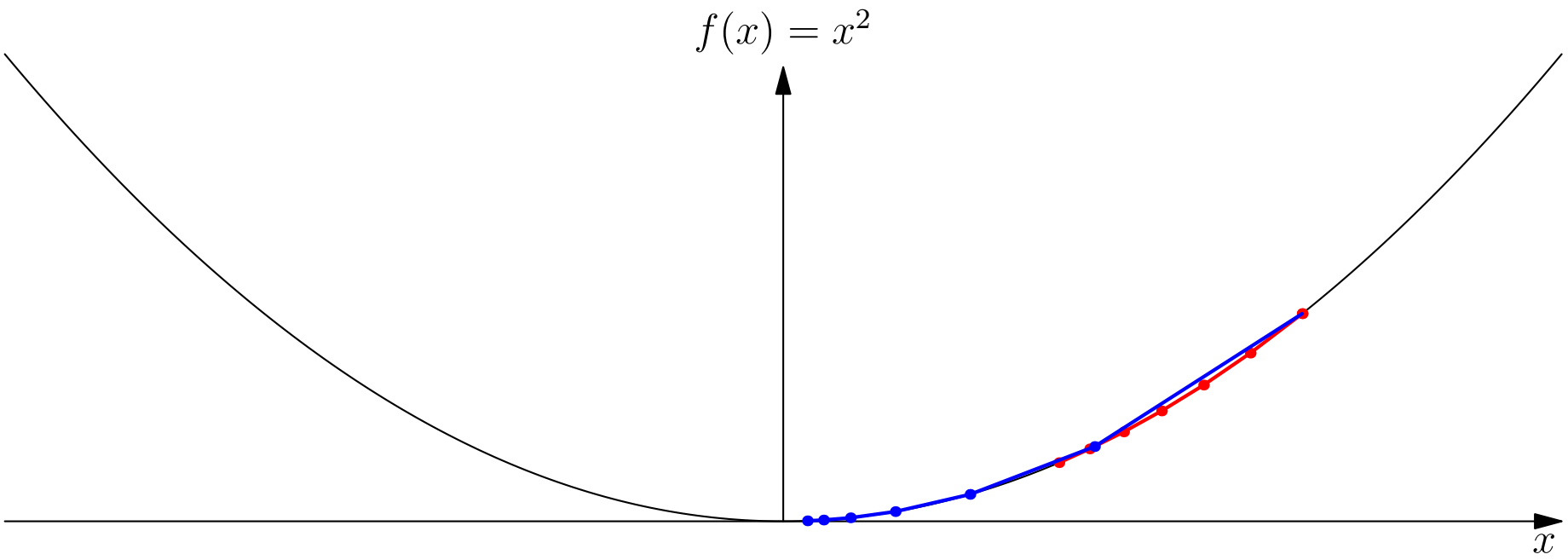
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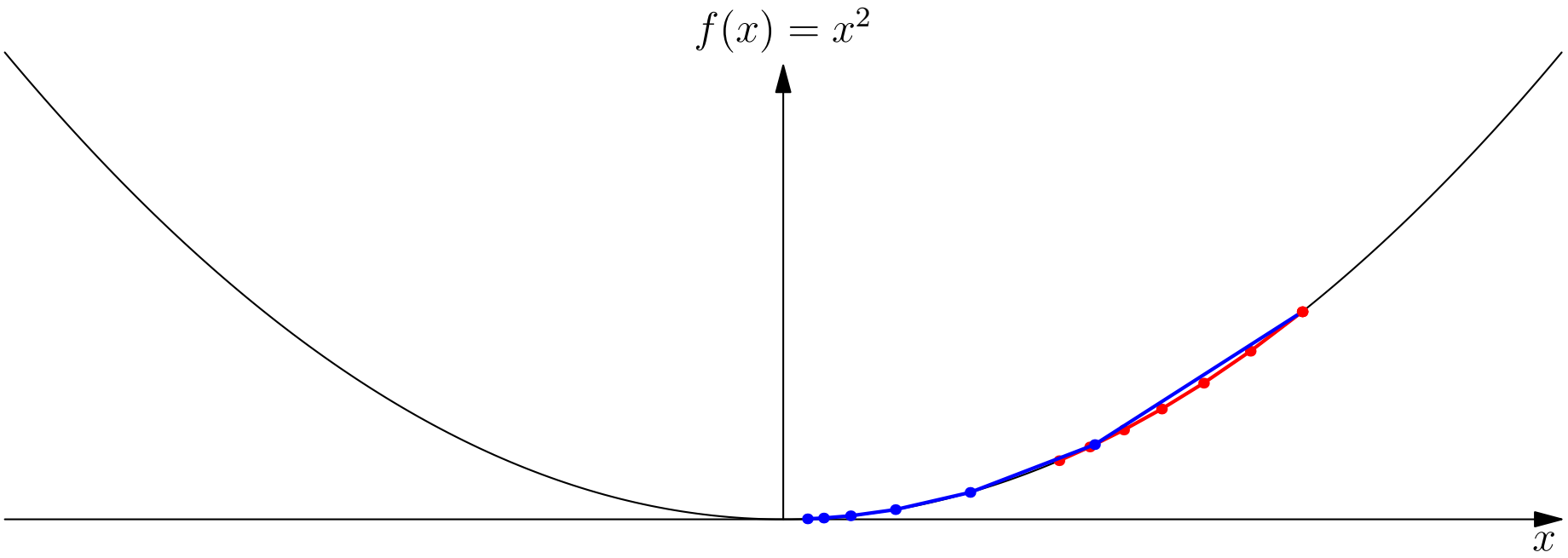
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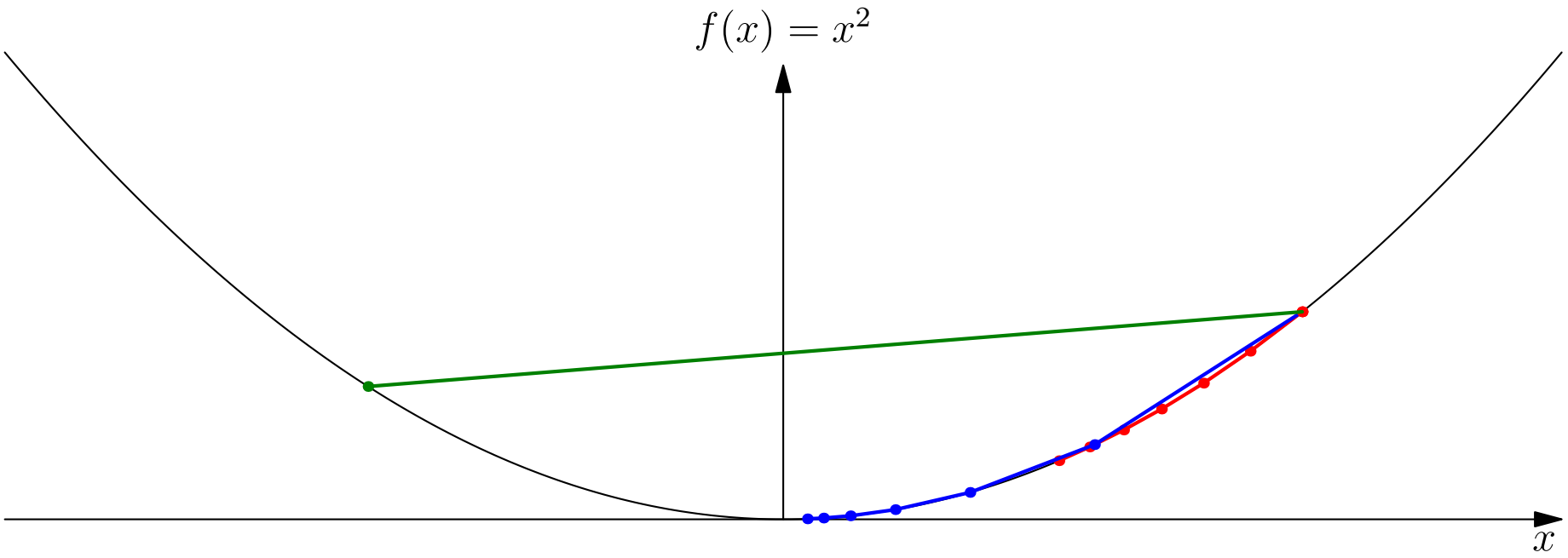
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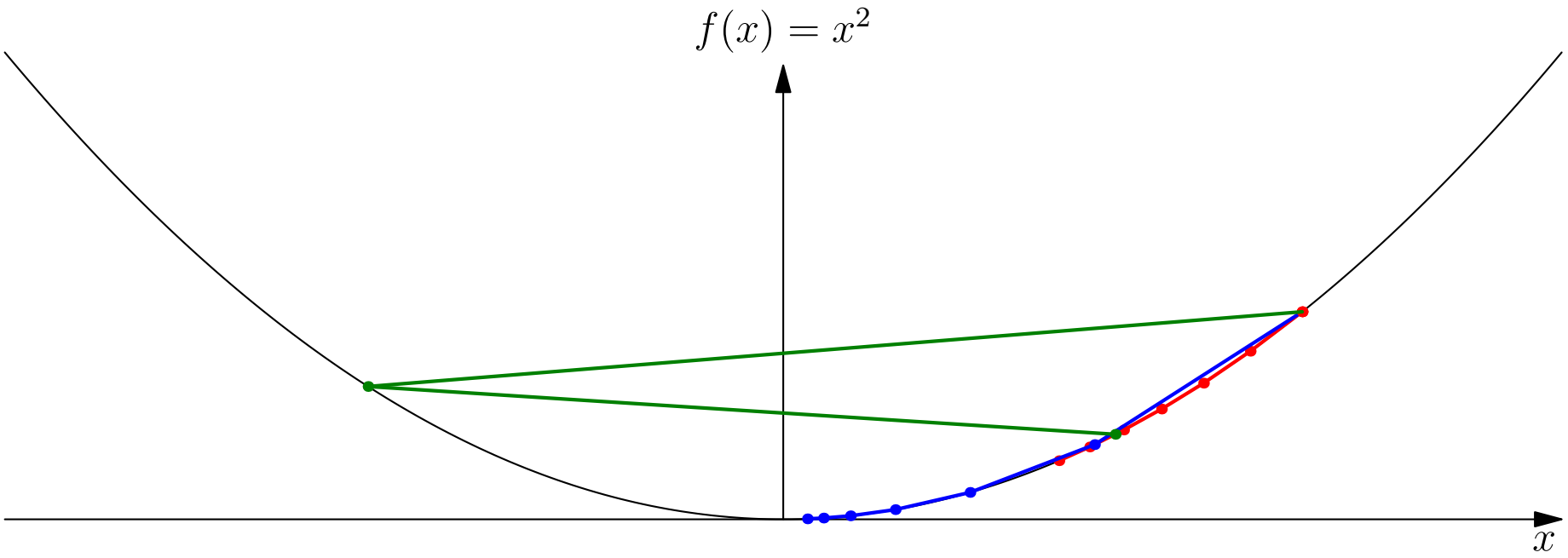
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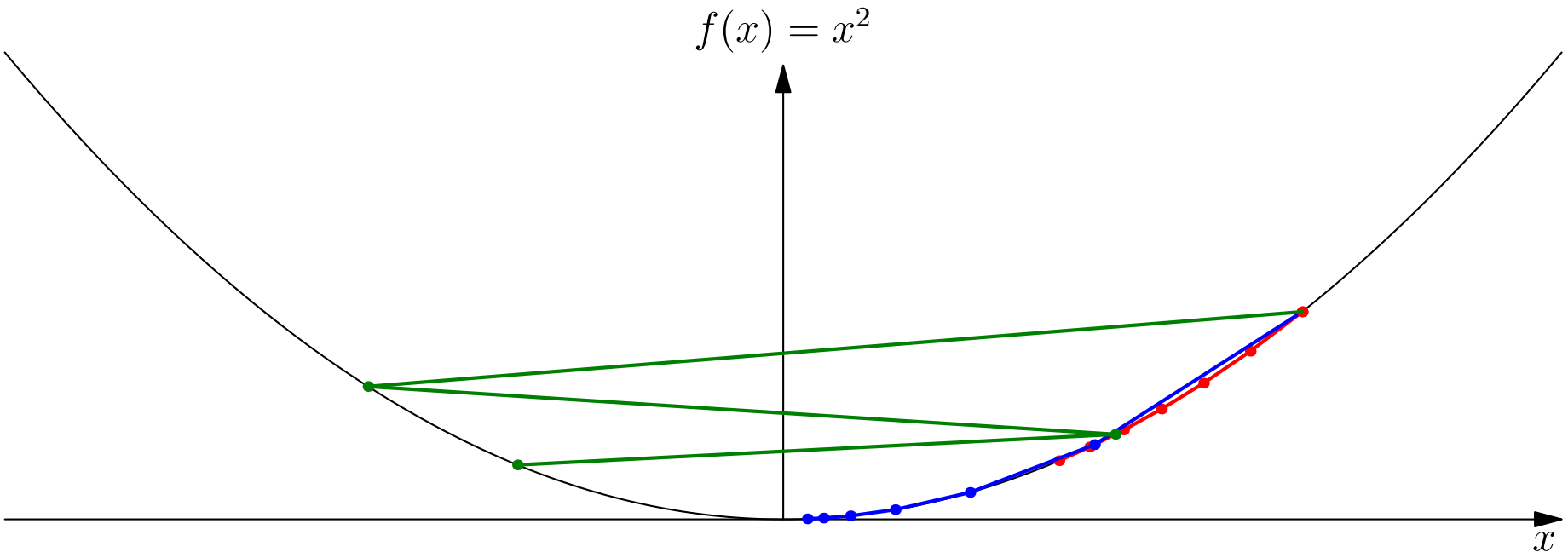
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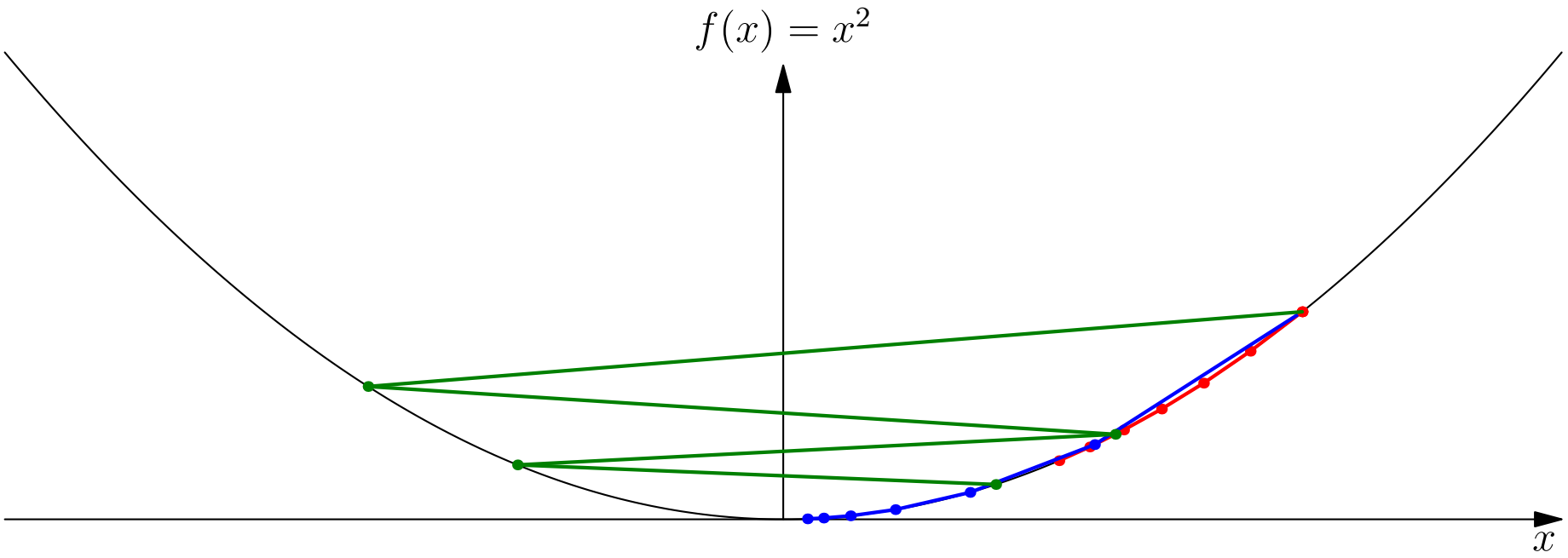
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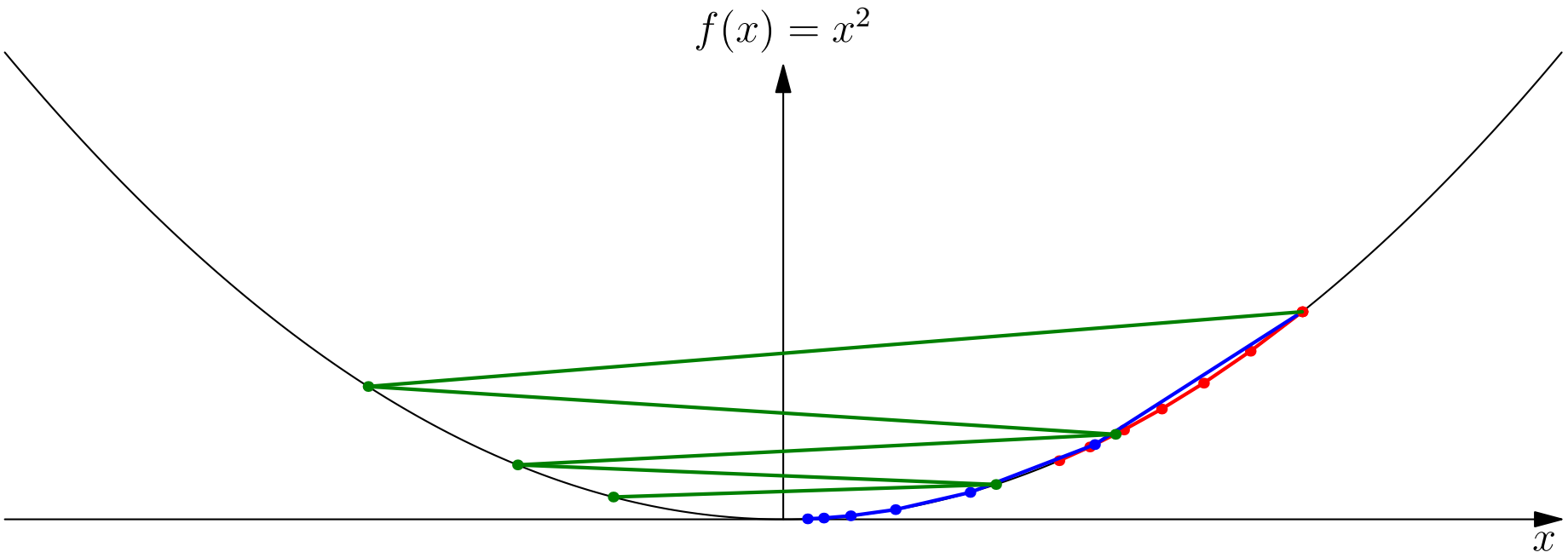
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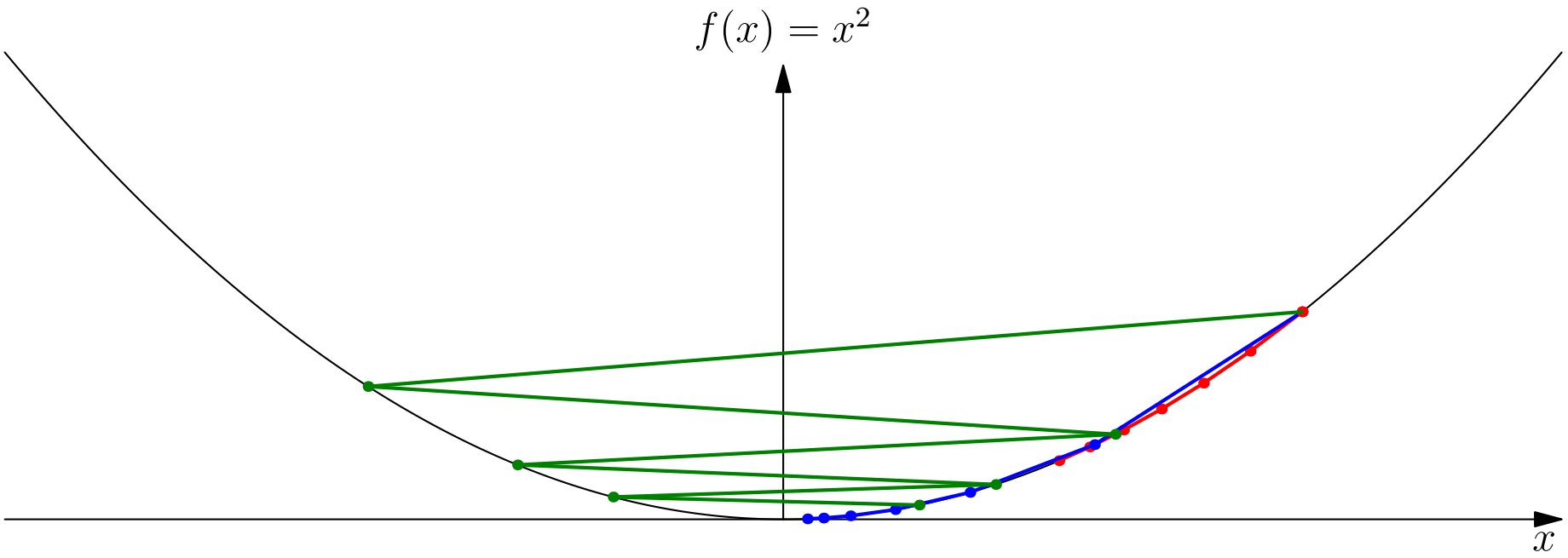
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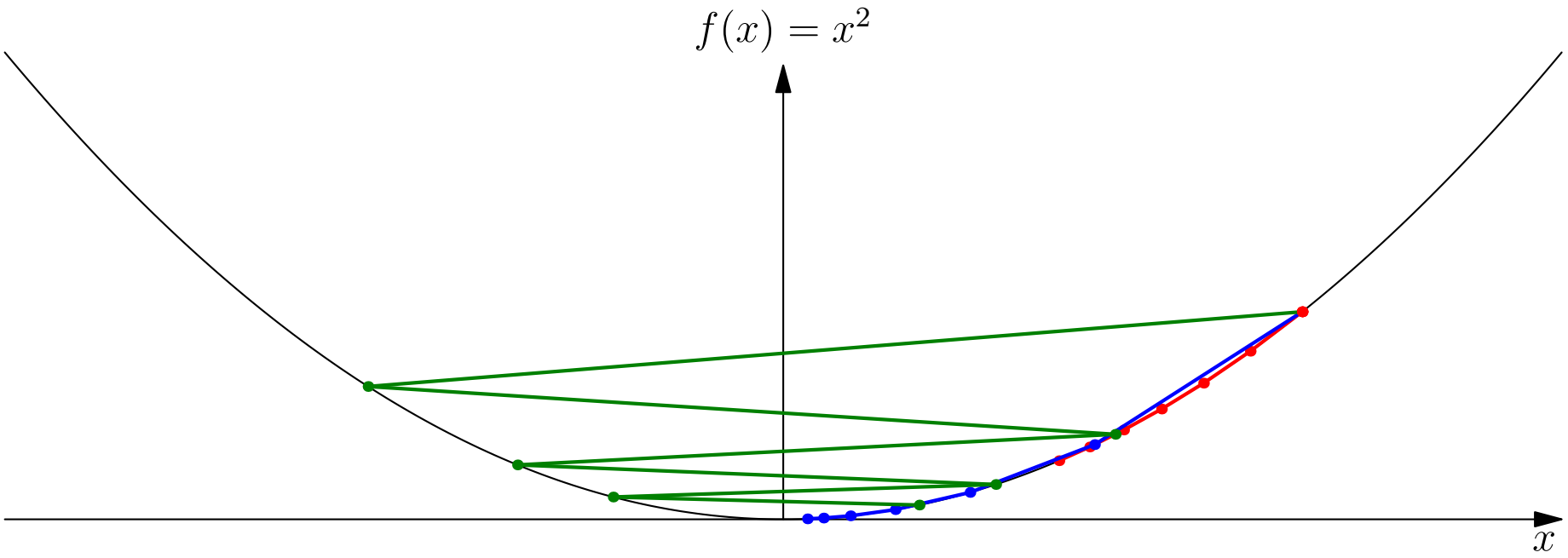
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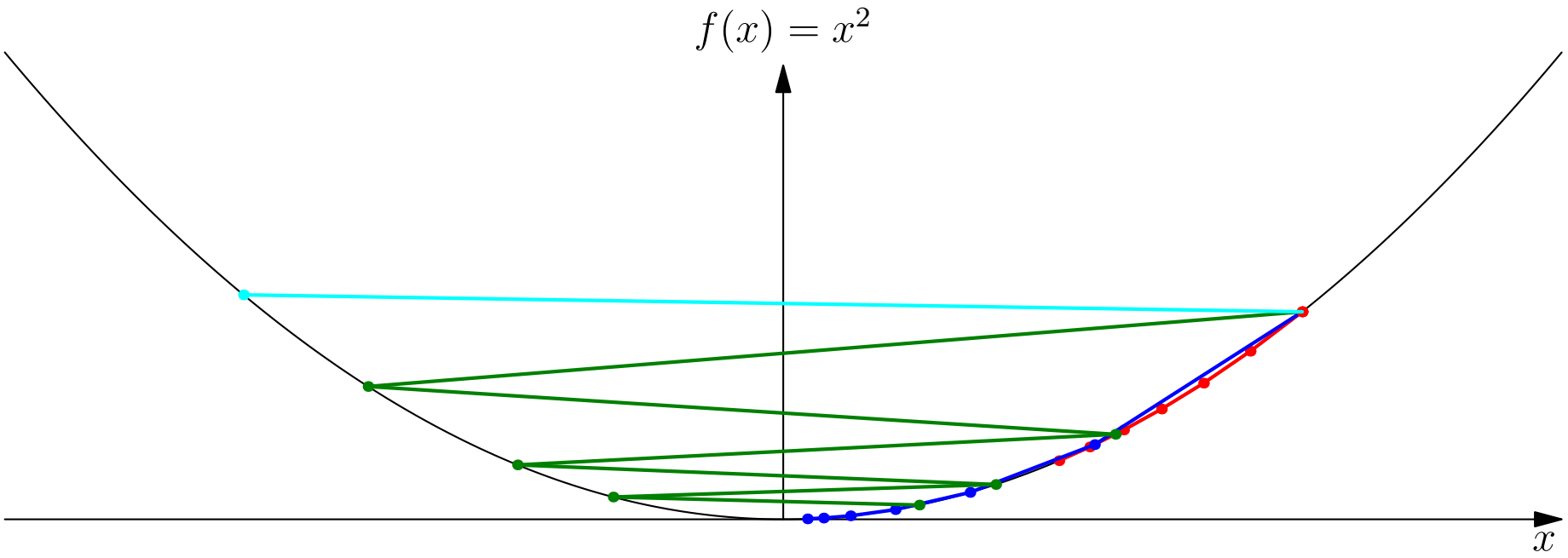
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r $= 0.05$ r $= 0.2$ r $= 0.9$ r $= 1.02$

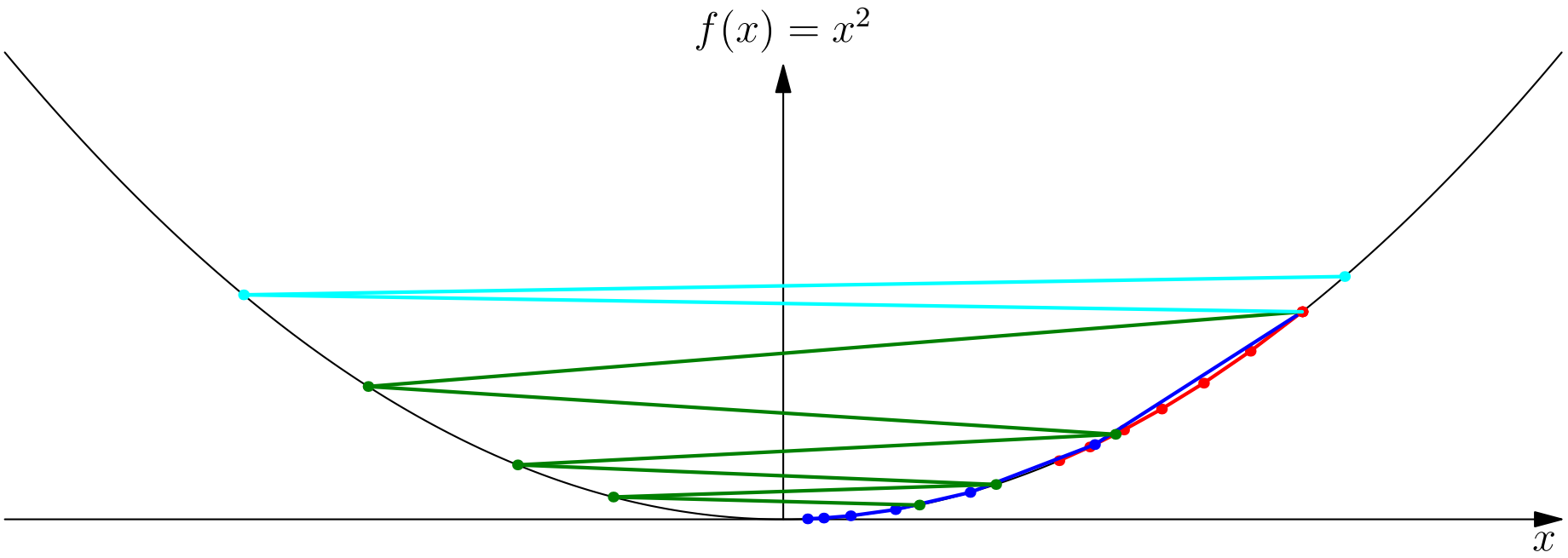
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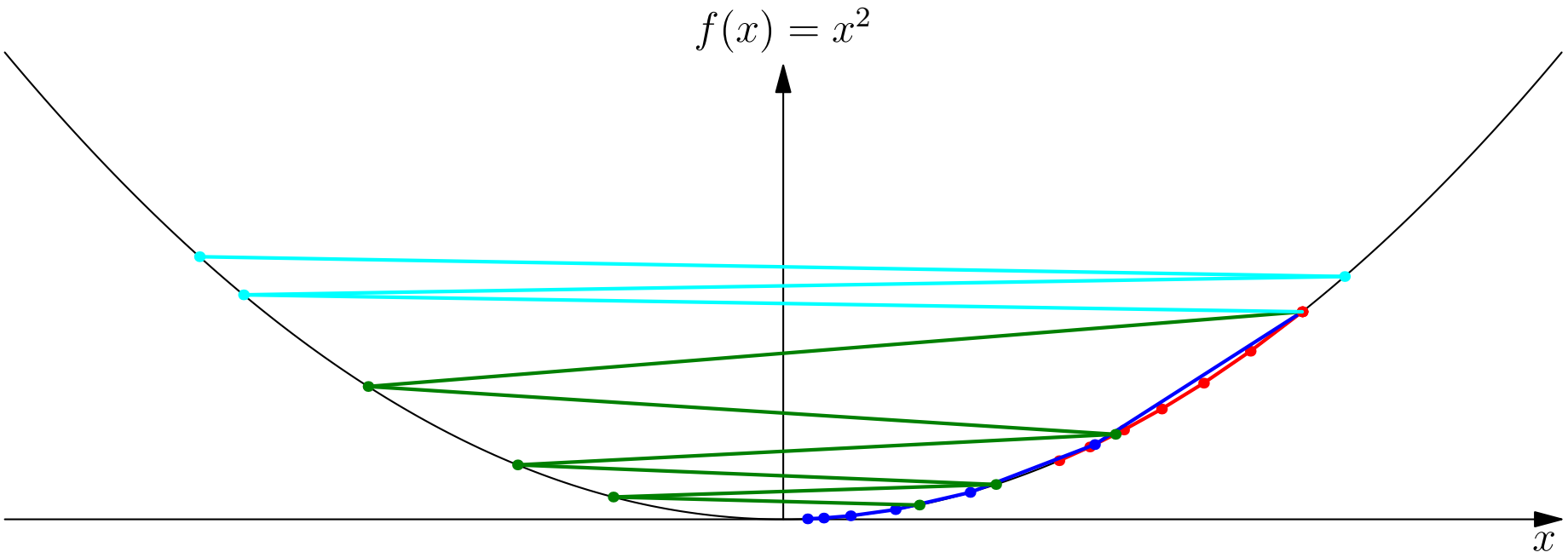
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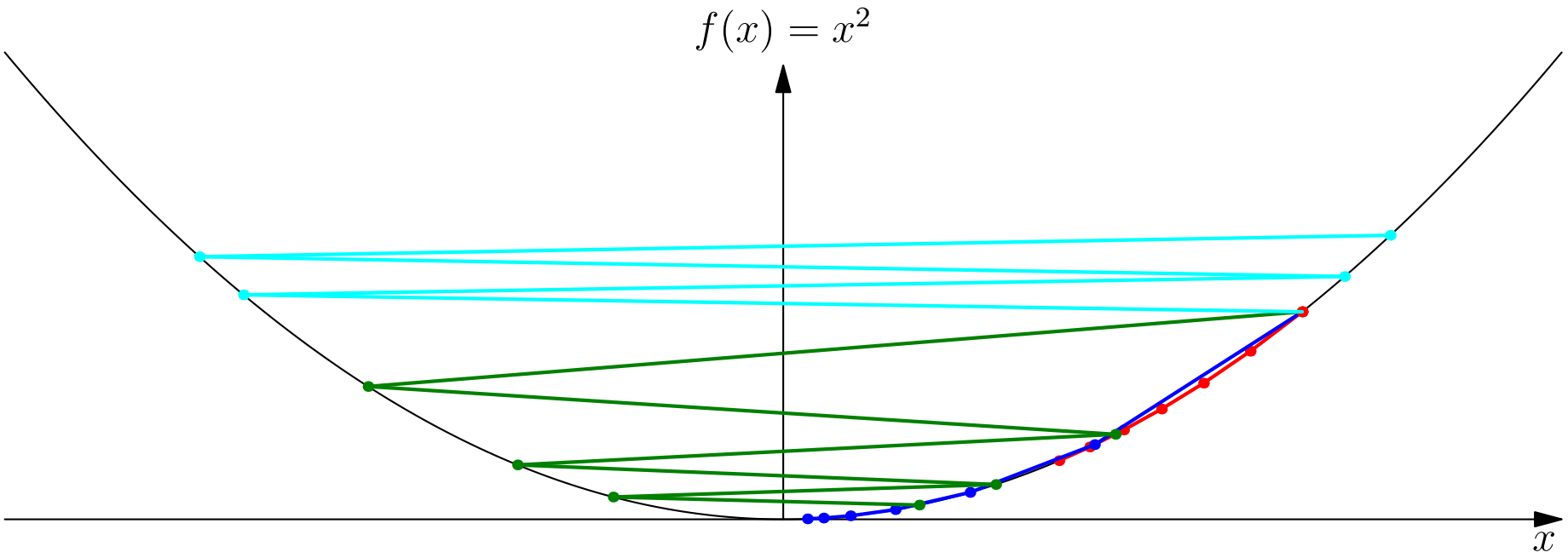
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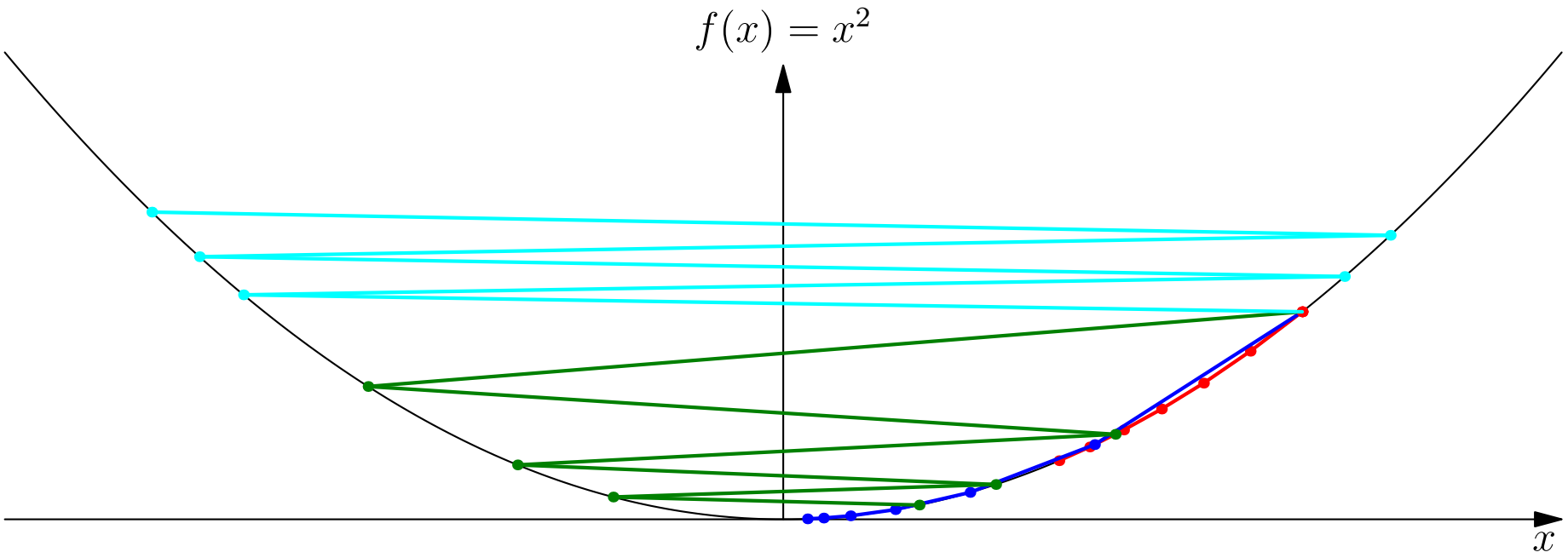
Step Size

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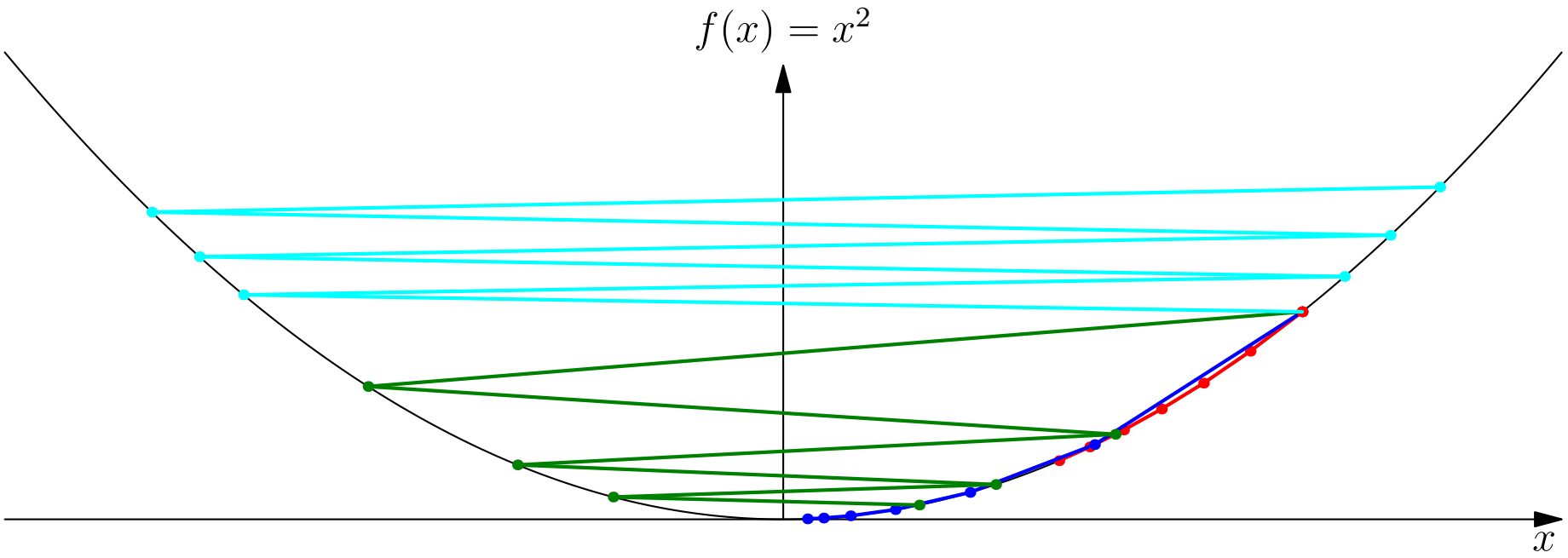
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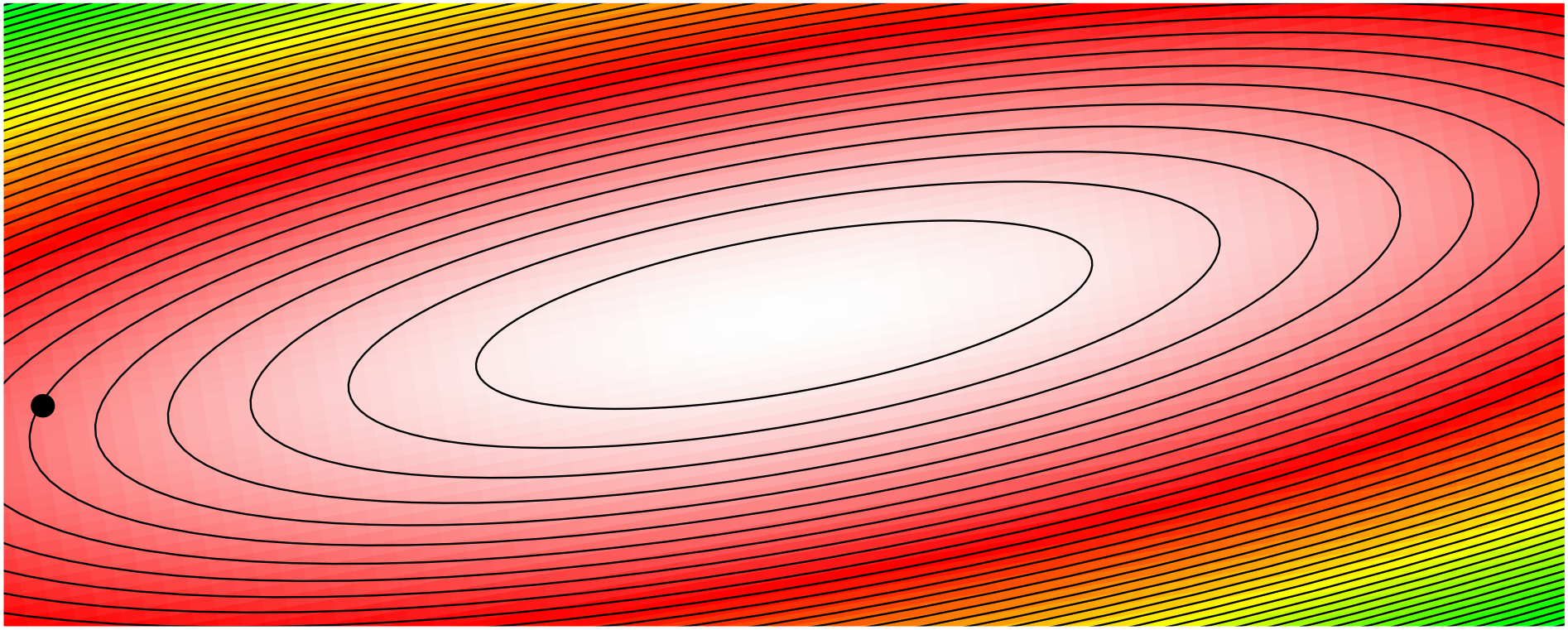
r $= 0.05$ r $= 0.2$ r $= 0.9$ r $= 1.02$

Higher Dimensions

- In higher dimensions the problem is that there are some directions you need to move a long way

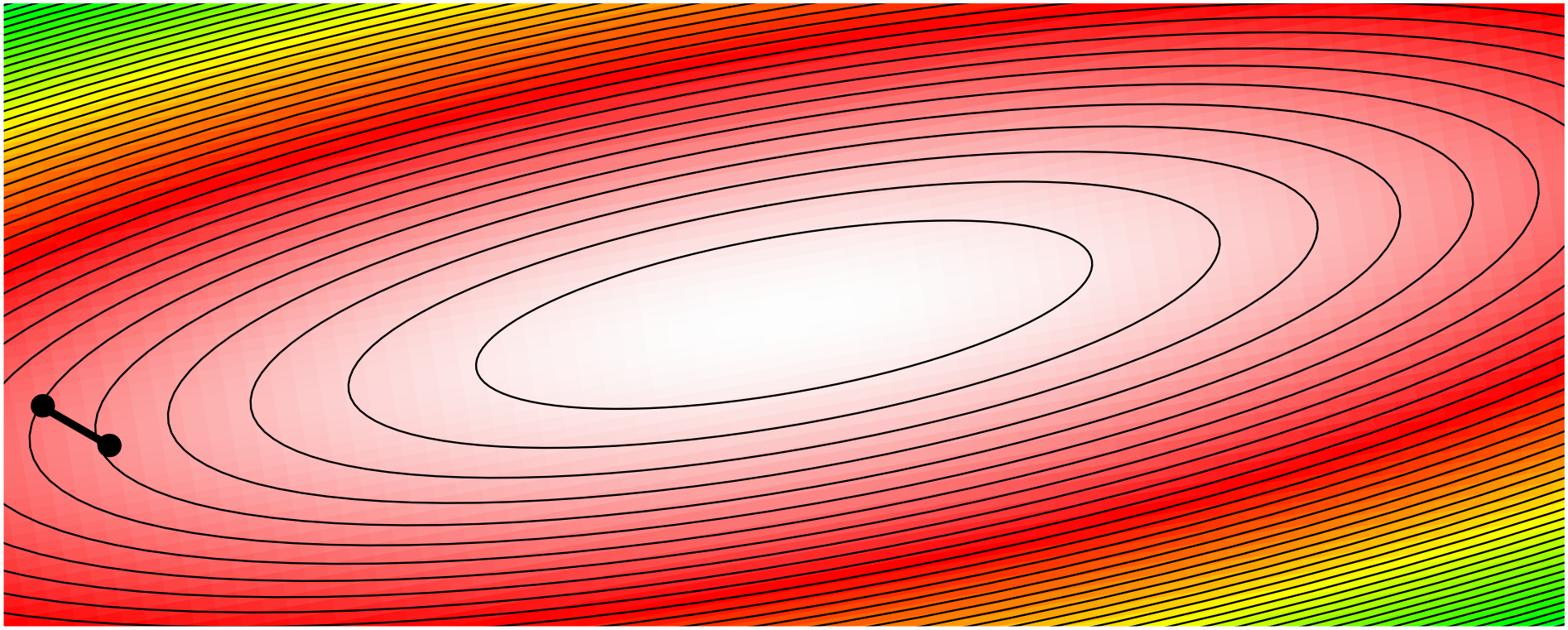
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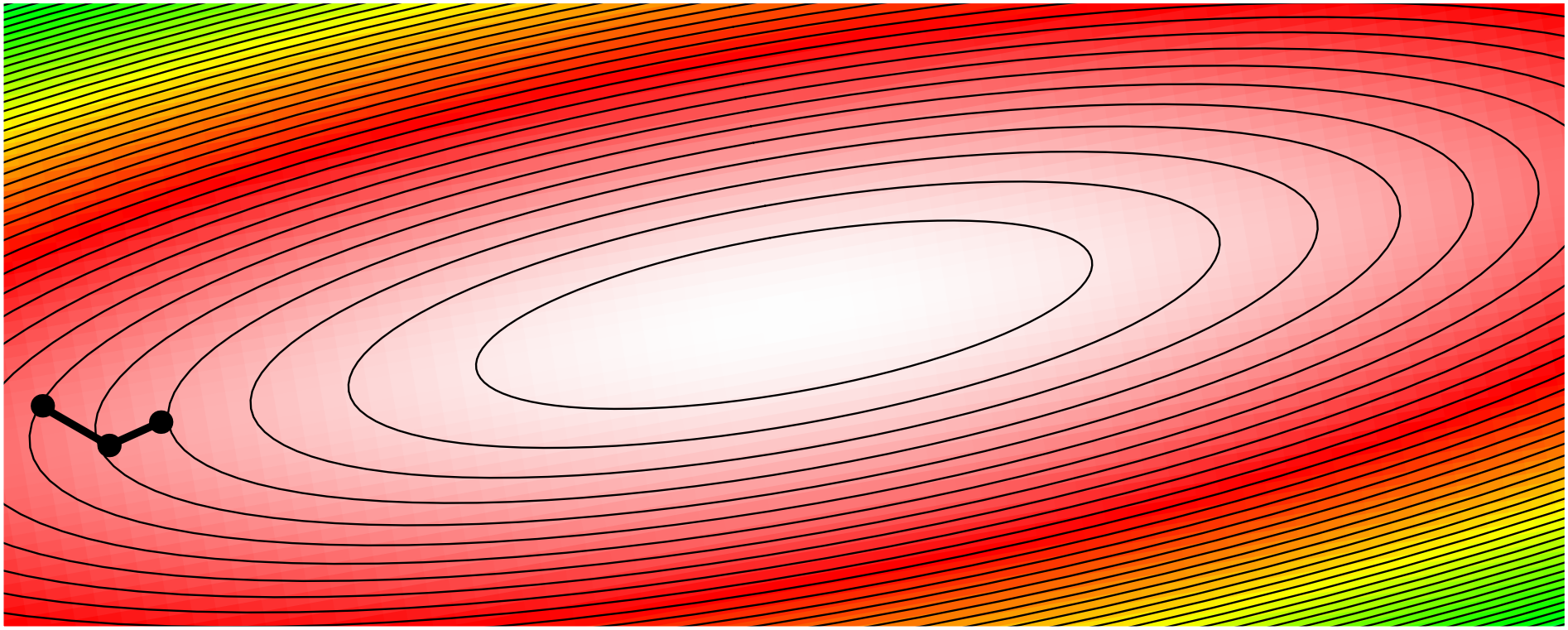
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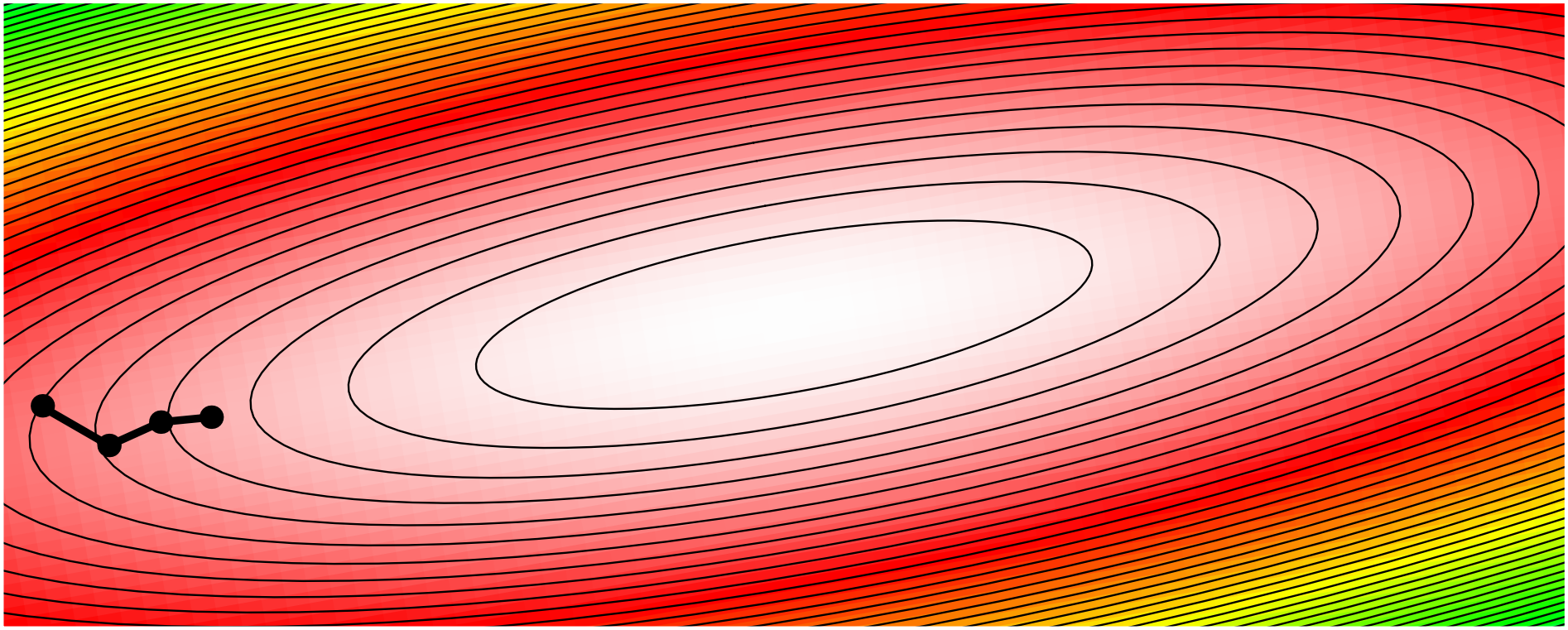
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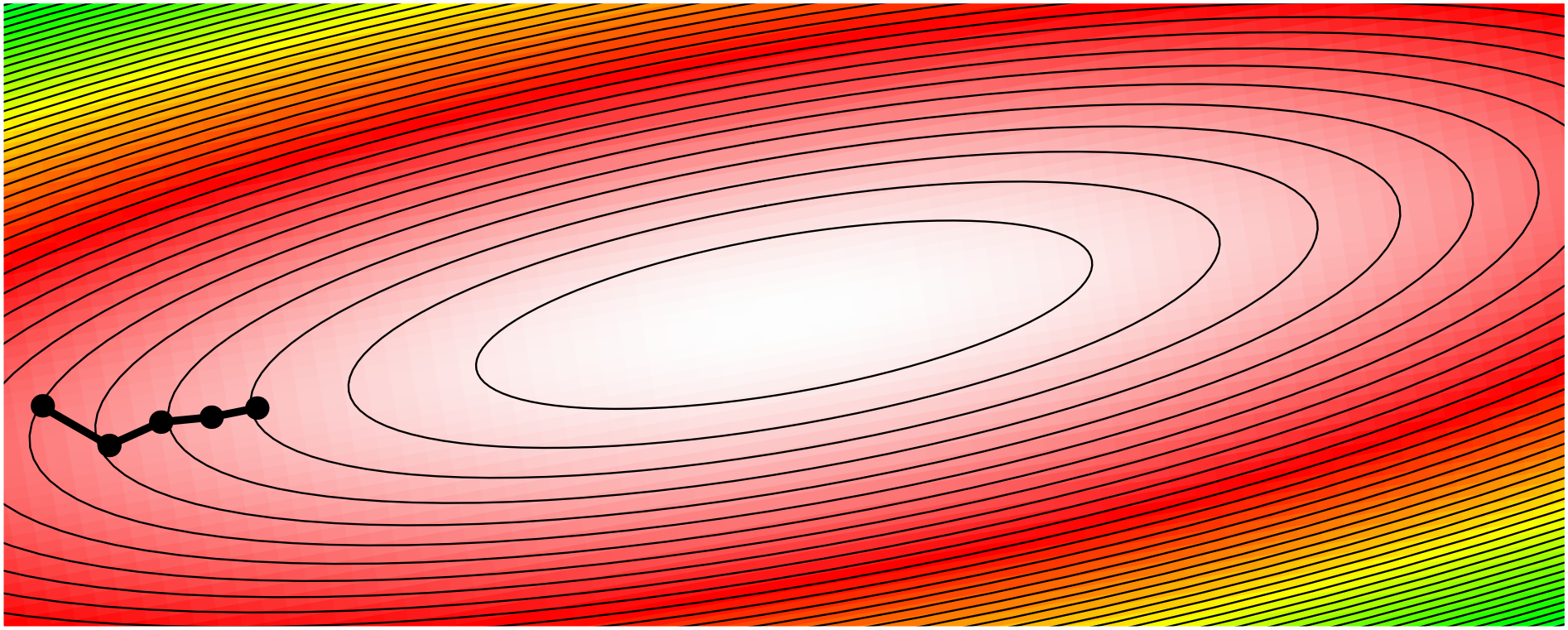
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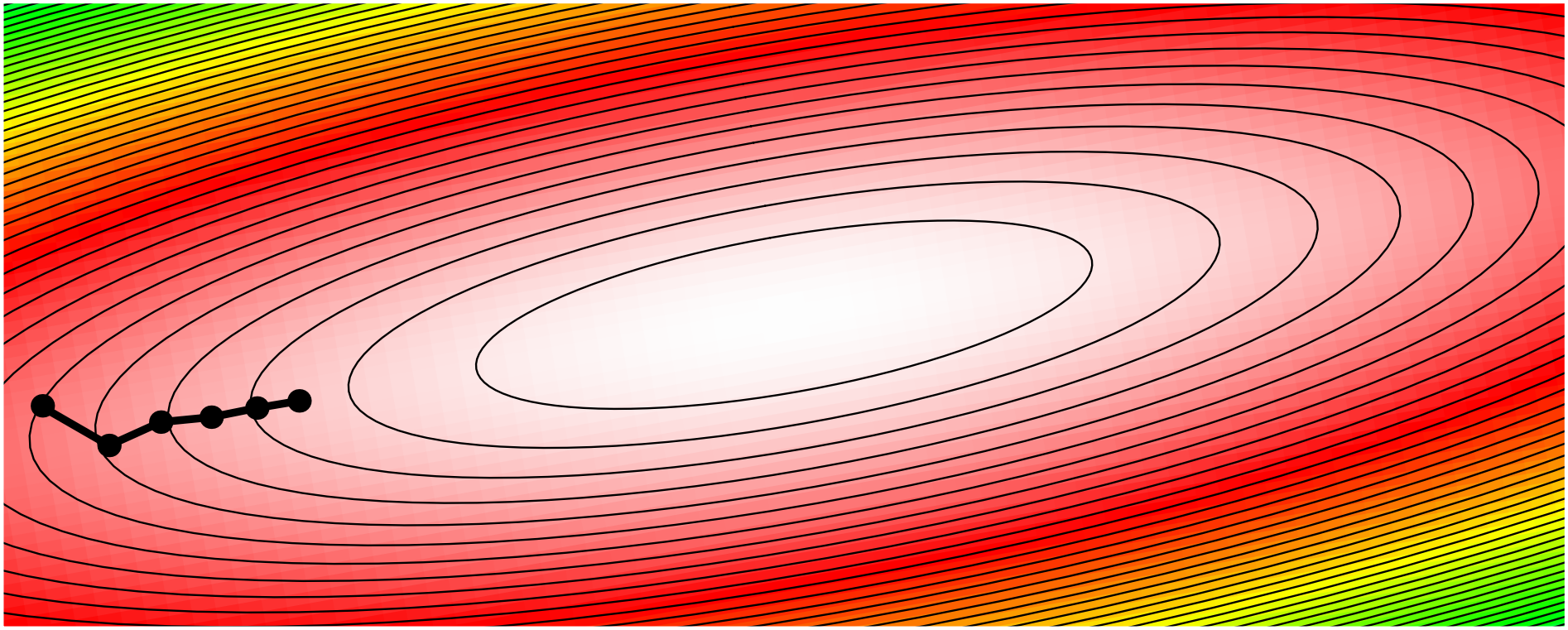
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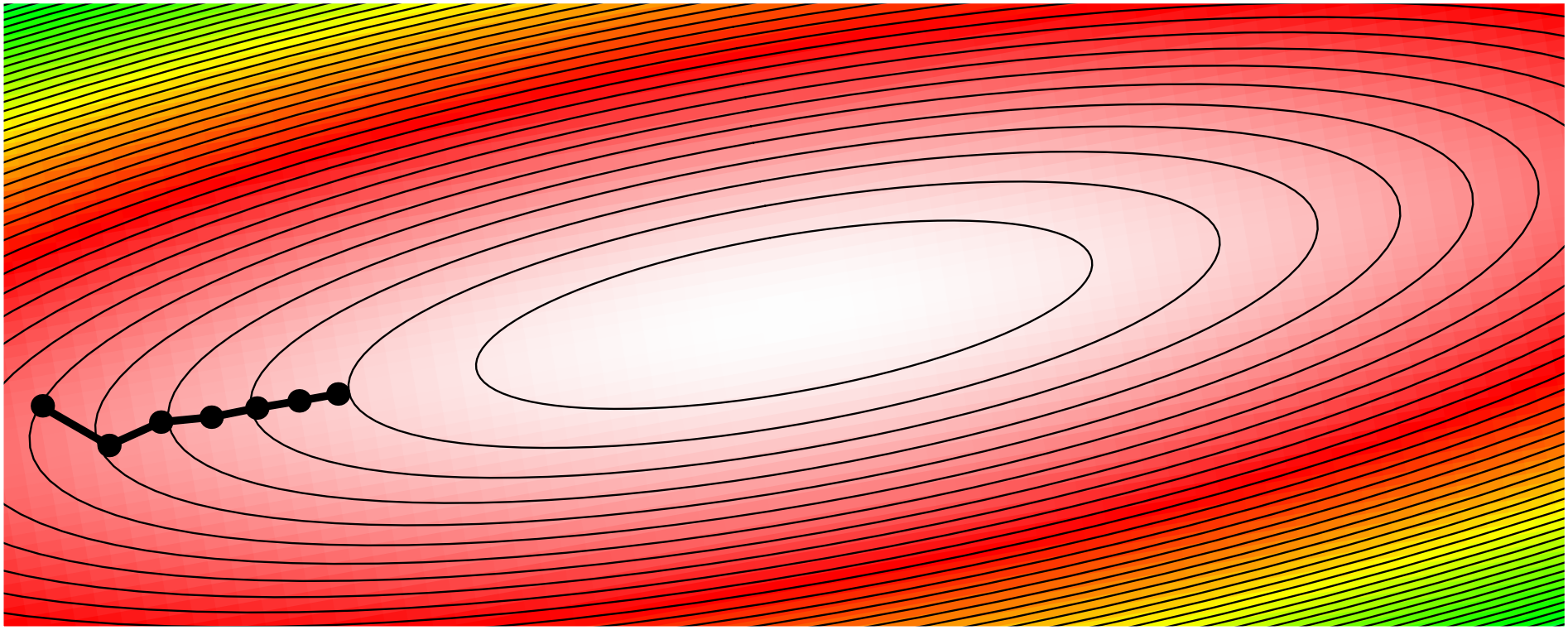
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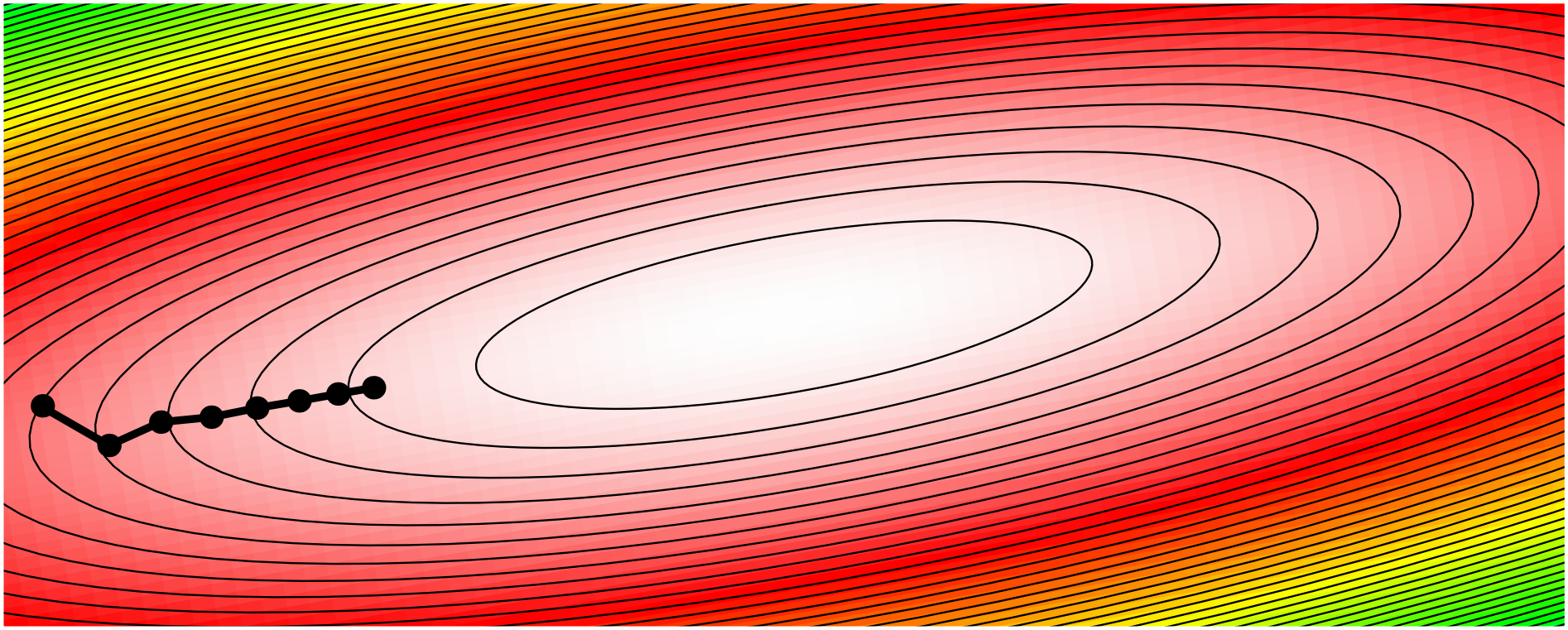
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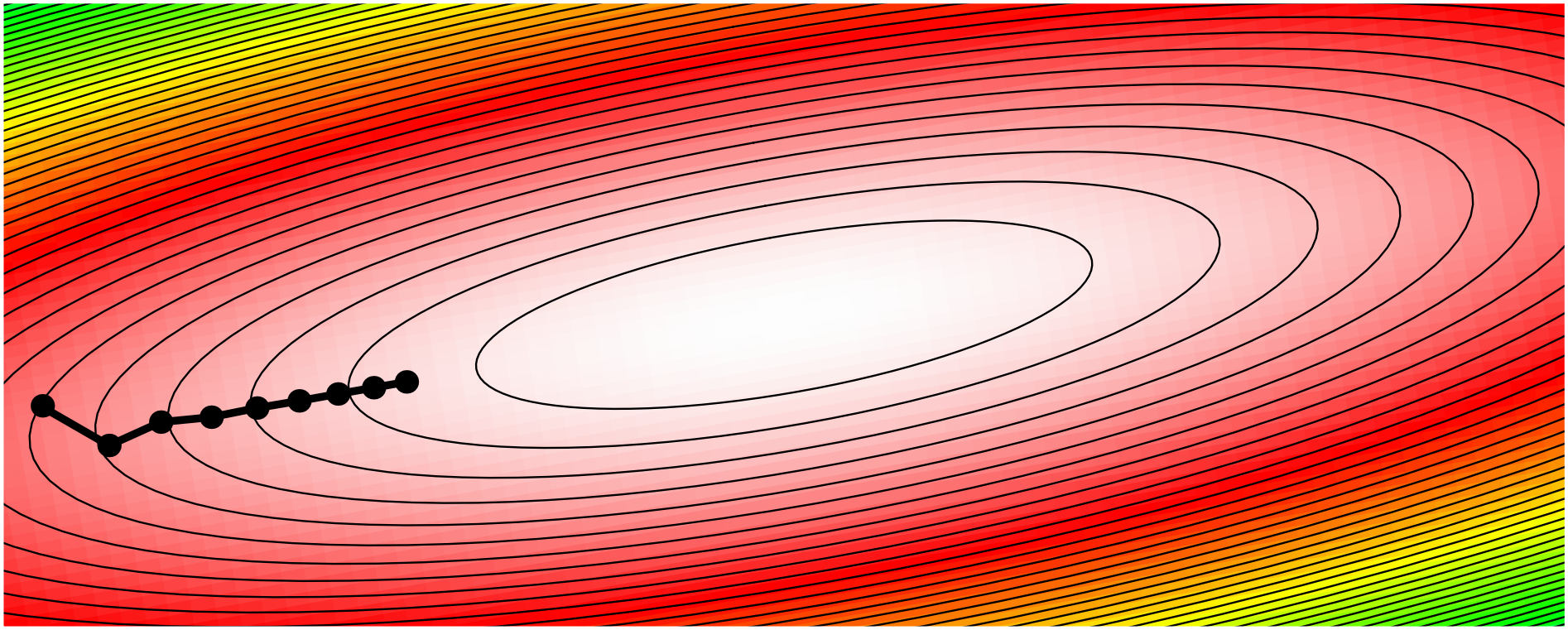
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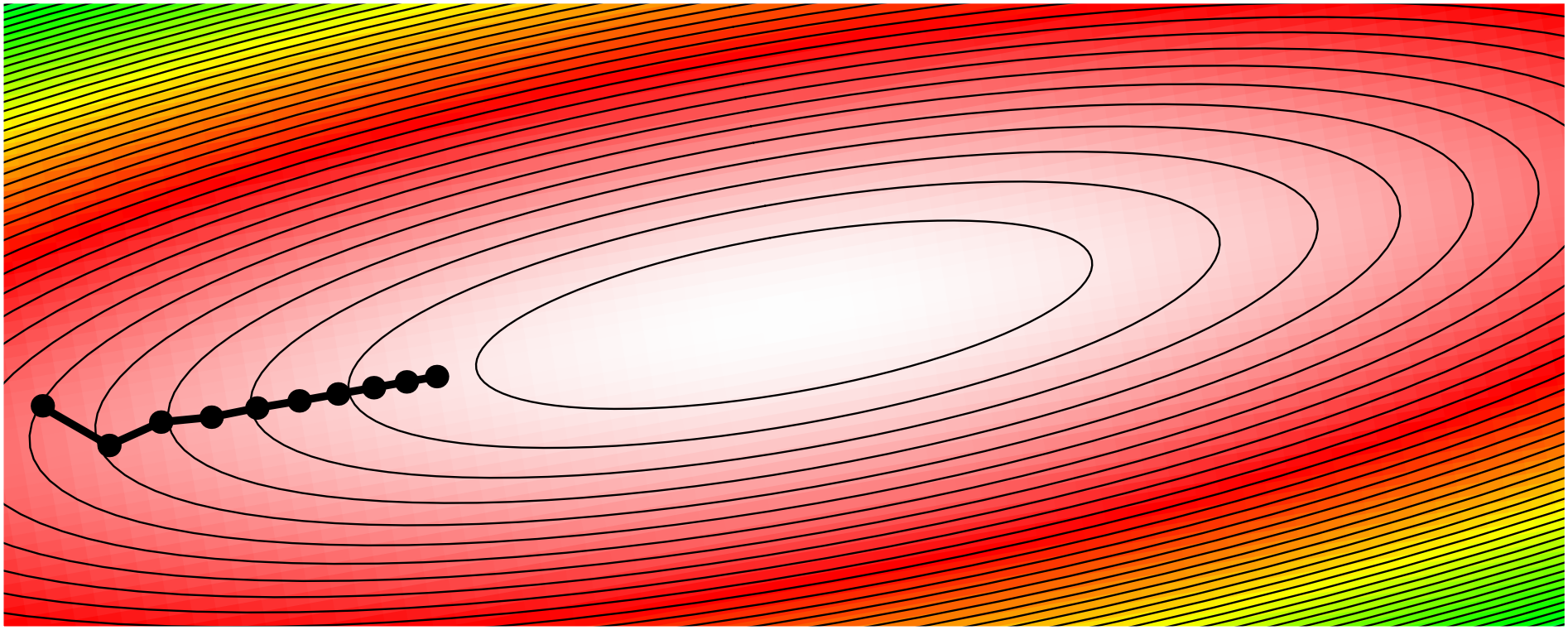
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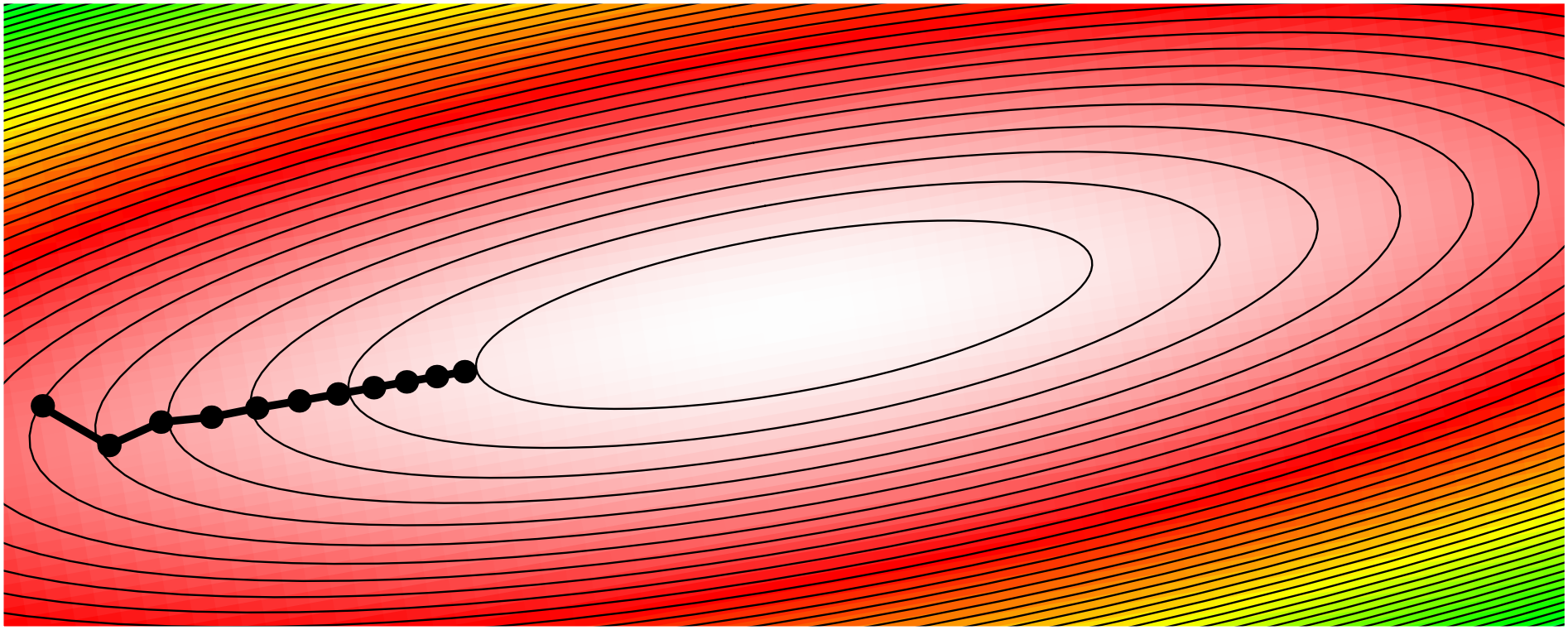
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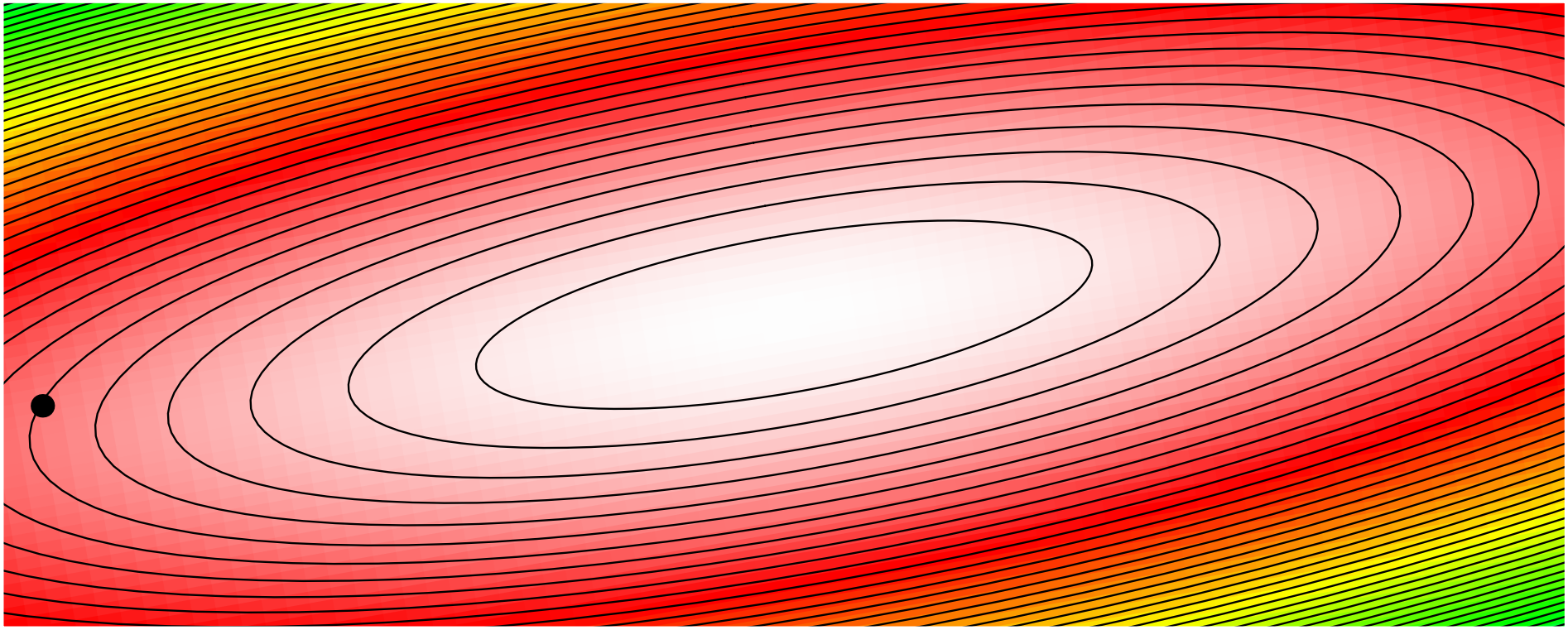


Getting There Quicker

- Increasing the step size speeds up convergence, but the direction of steepest descent doesn't point to the minimum

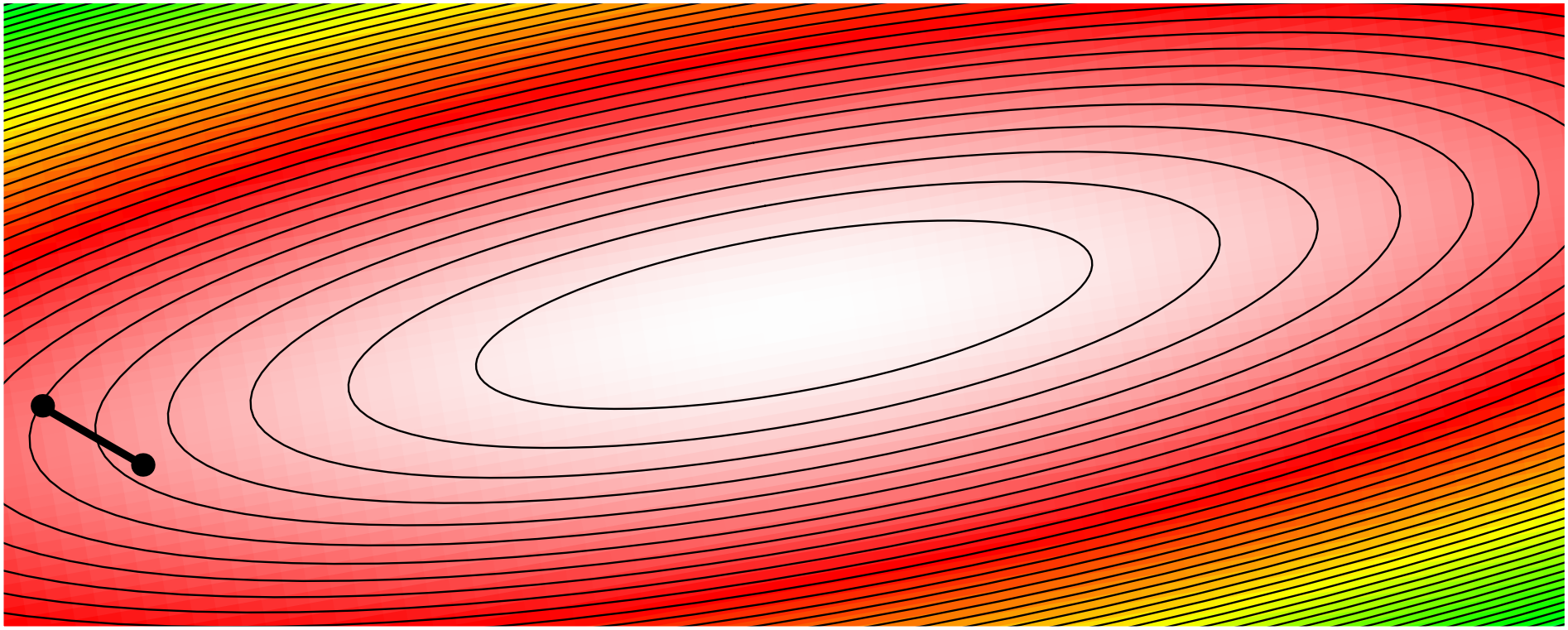
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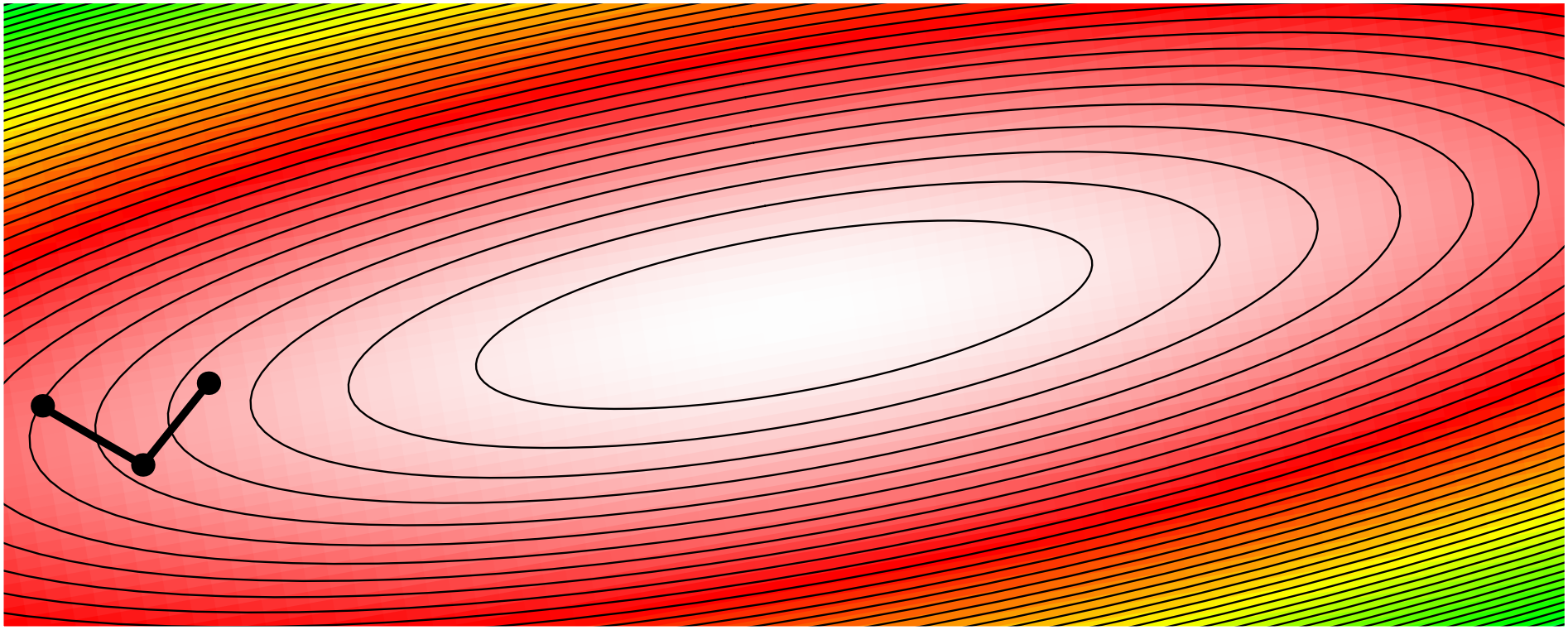
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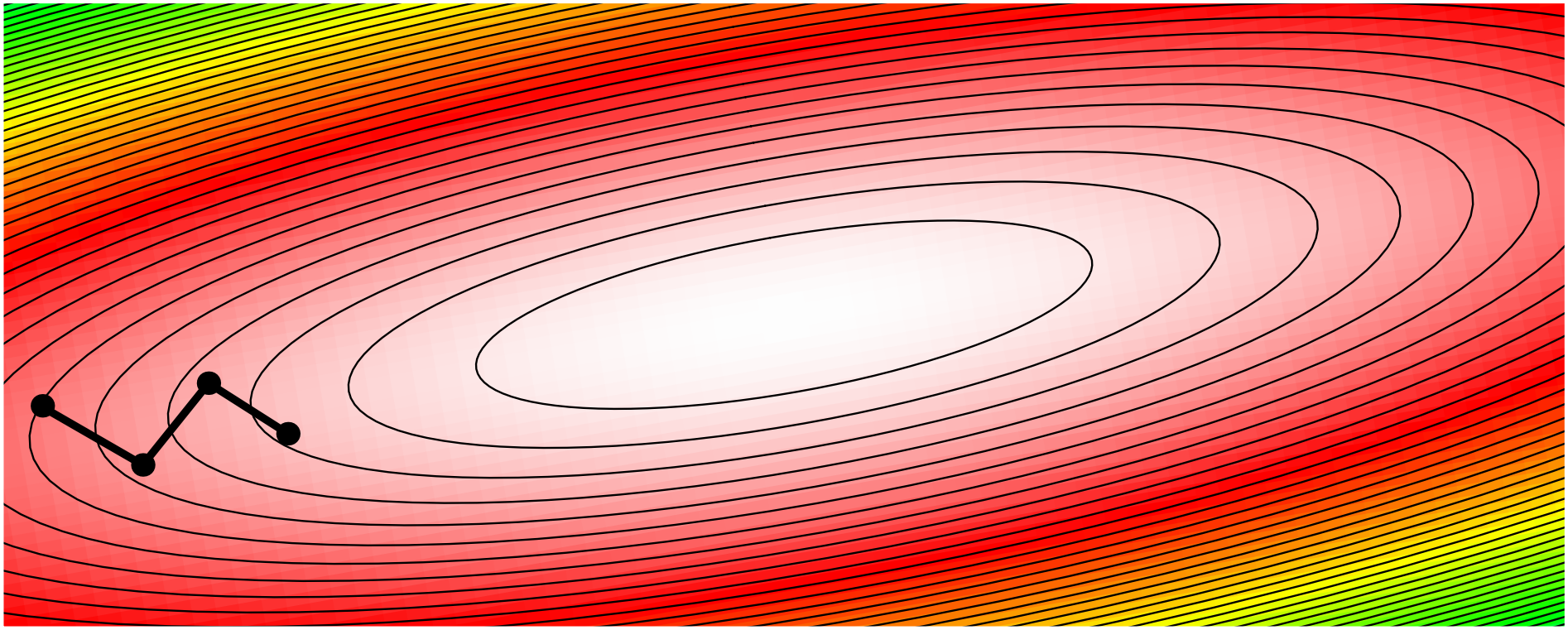
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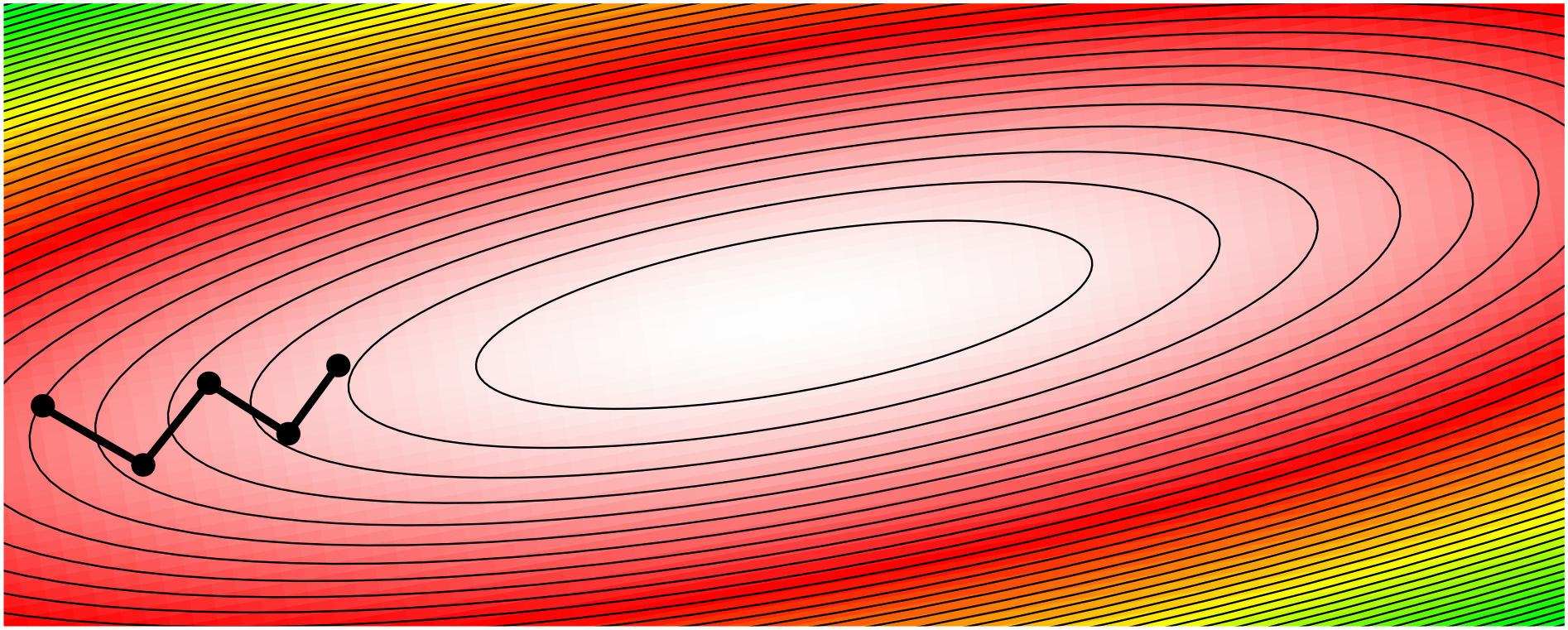
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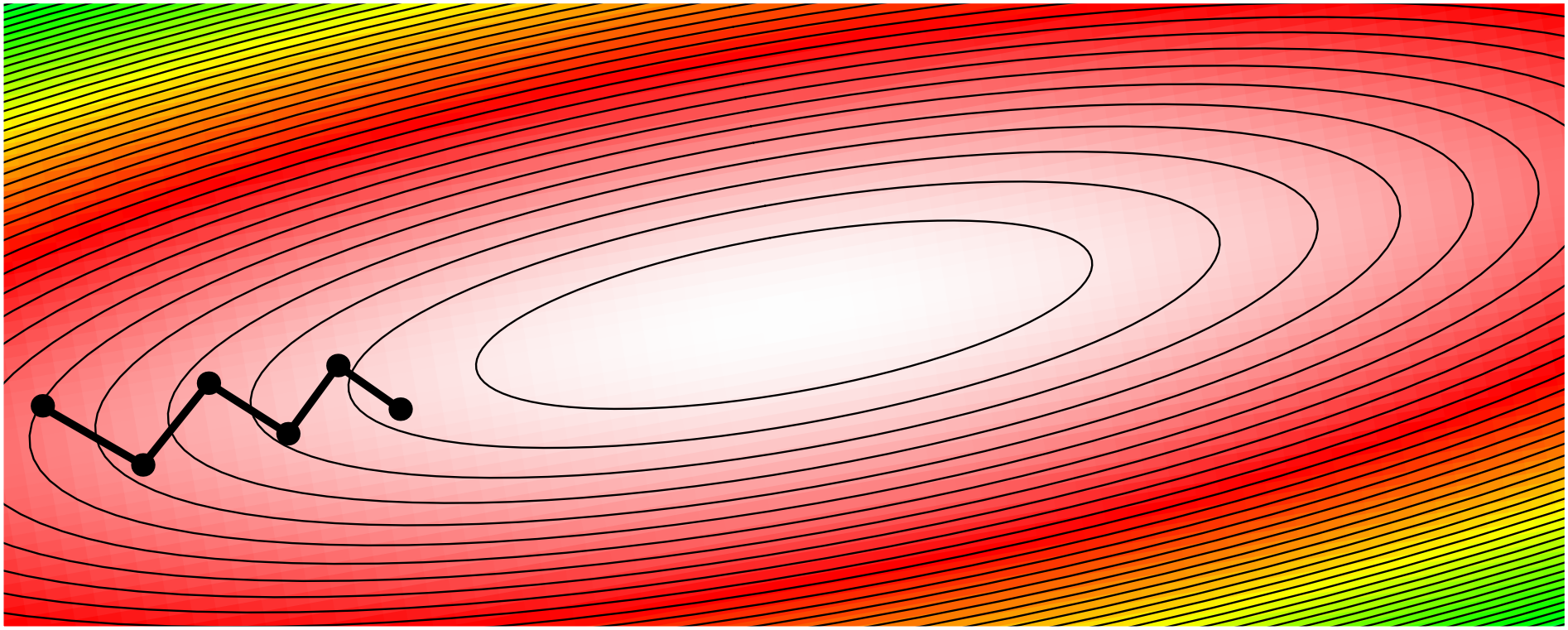
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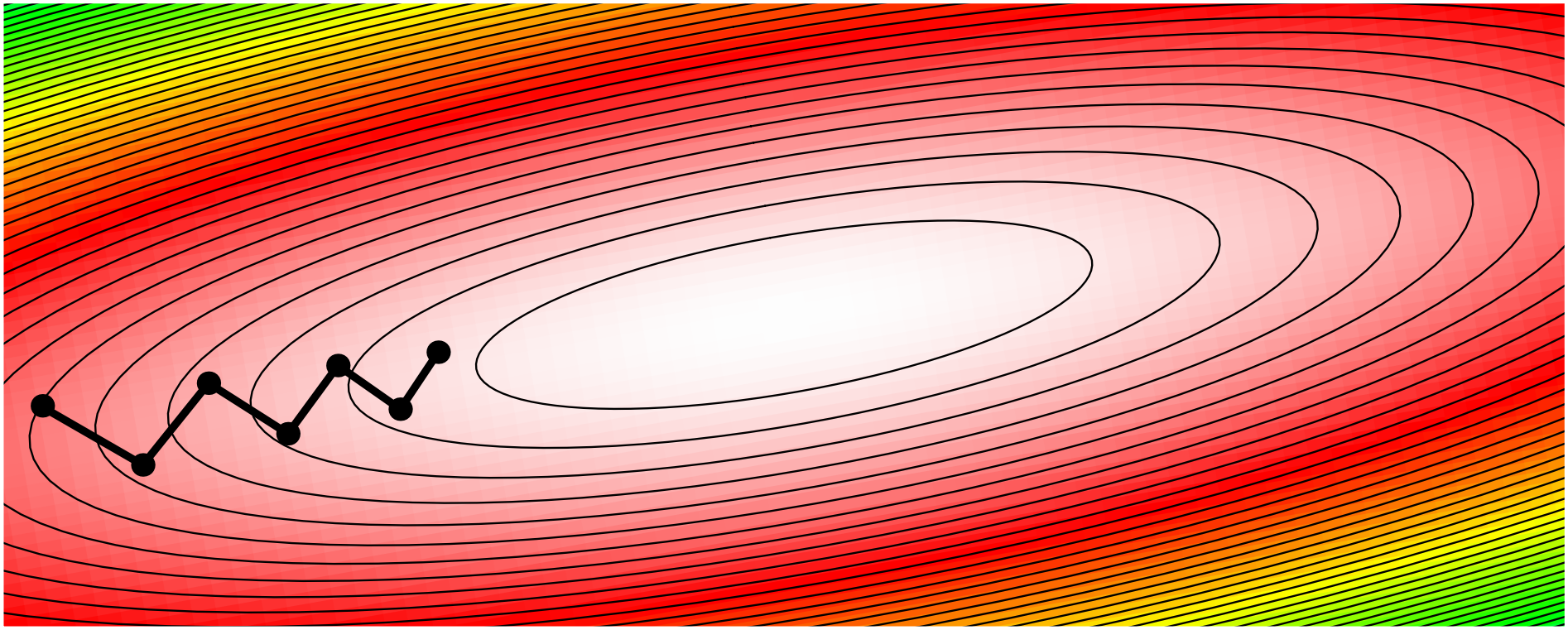
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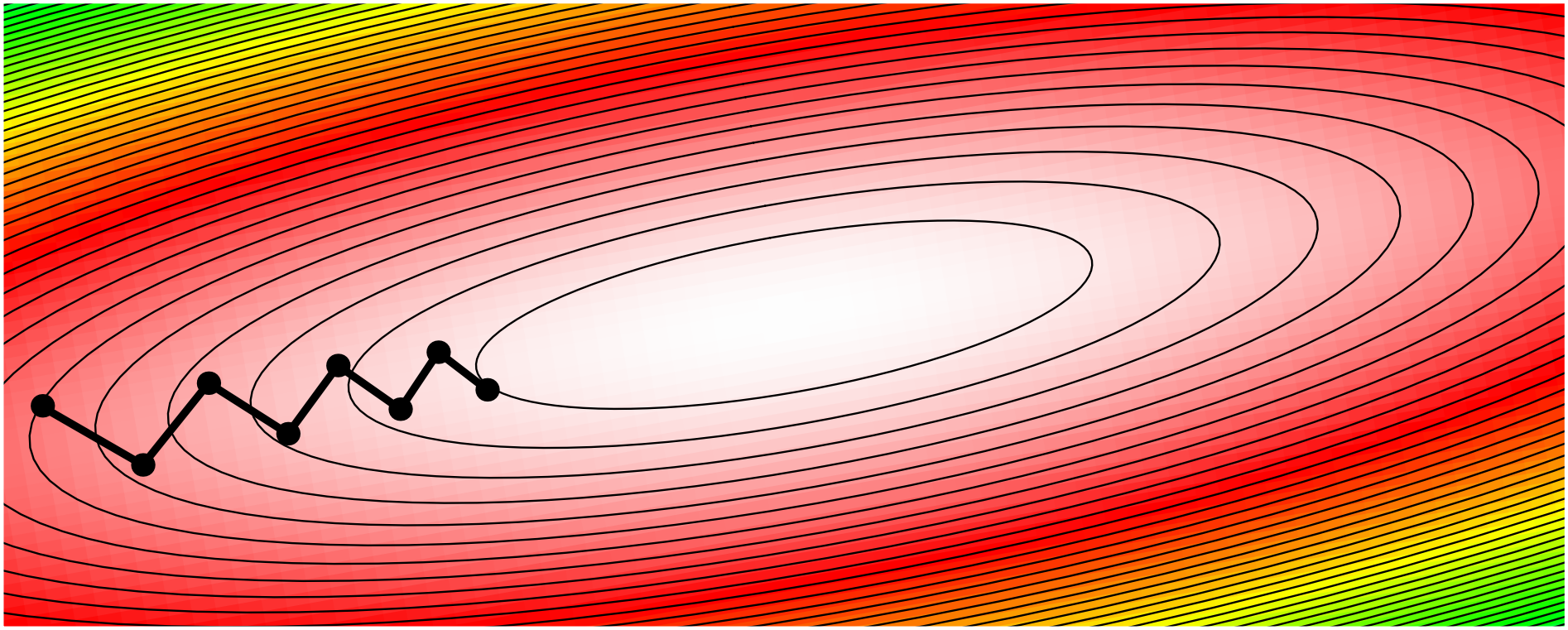
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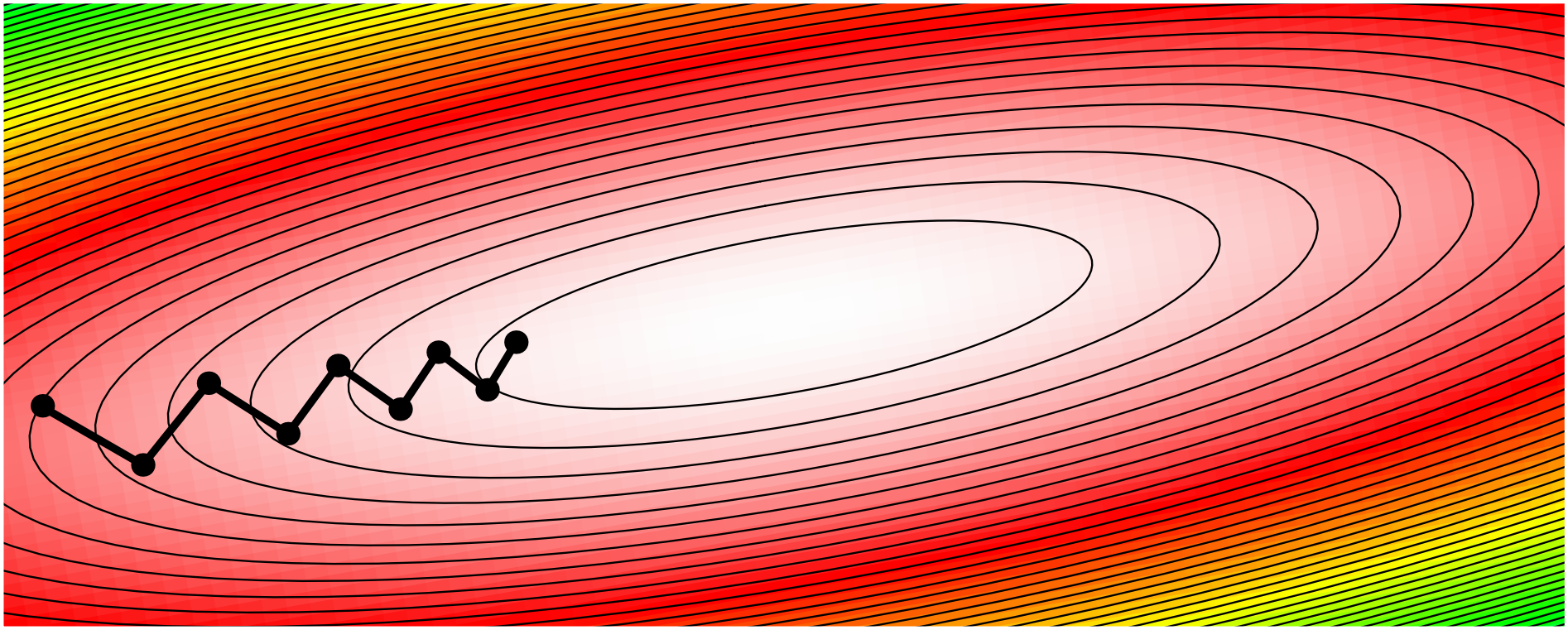
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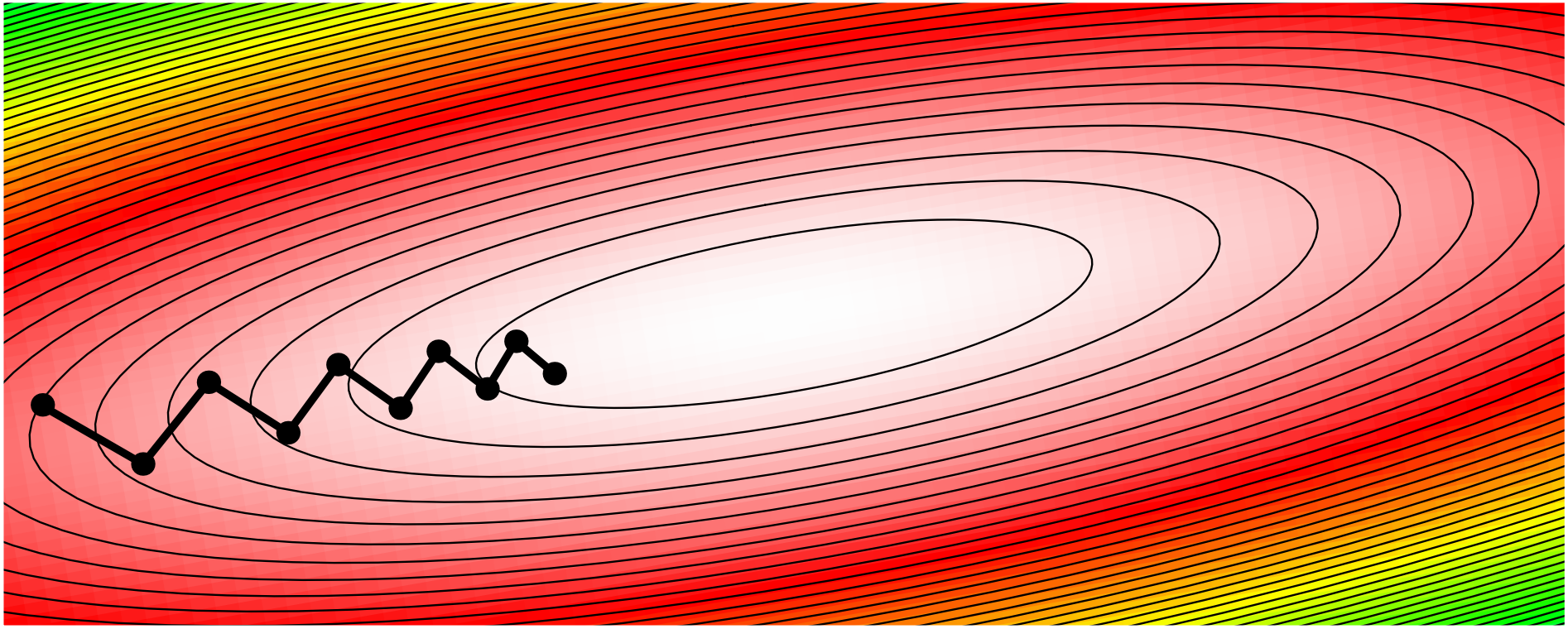
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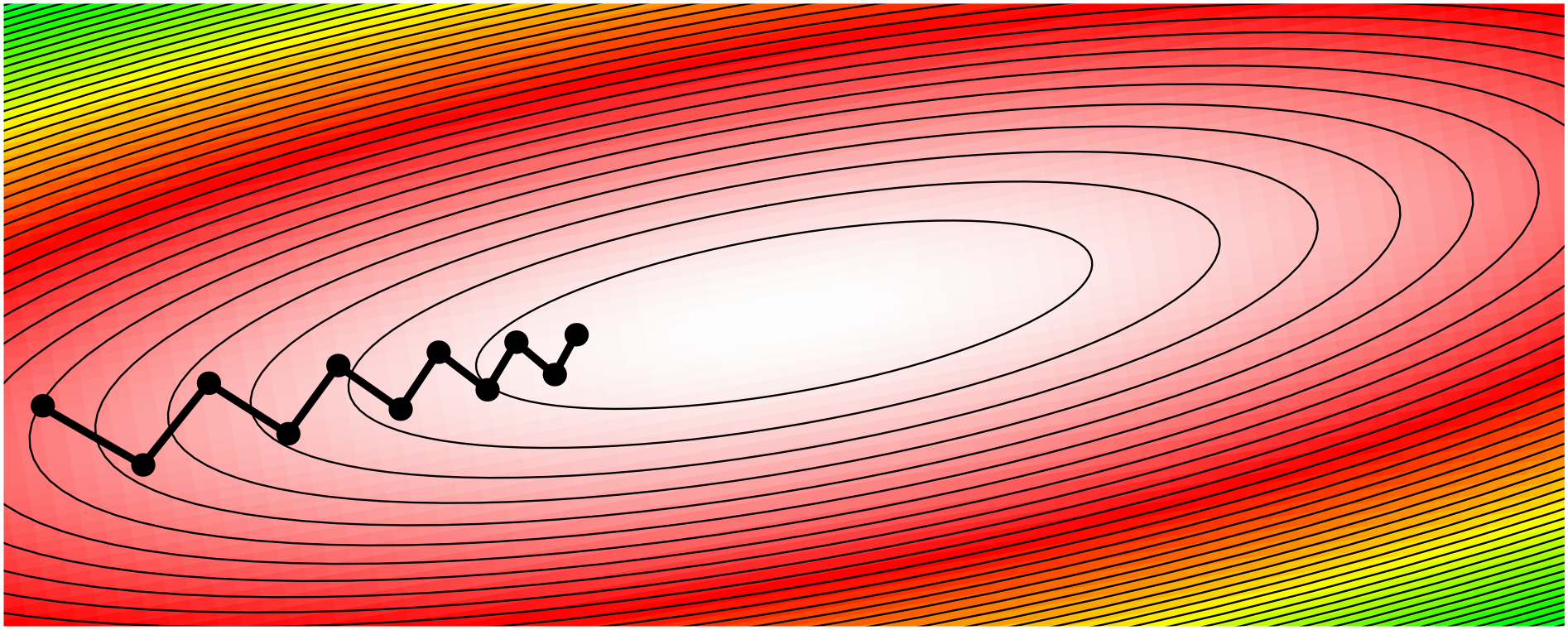
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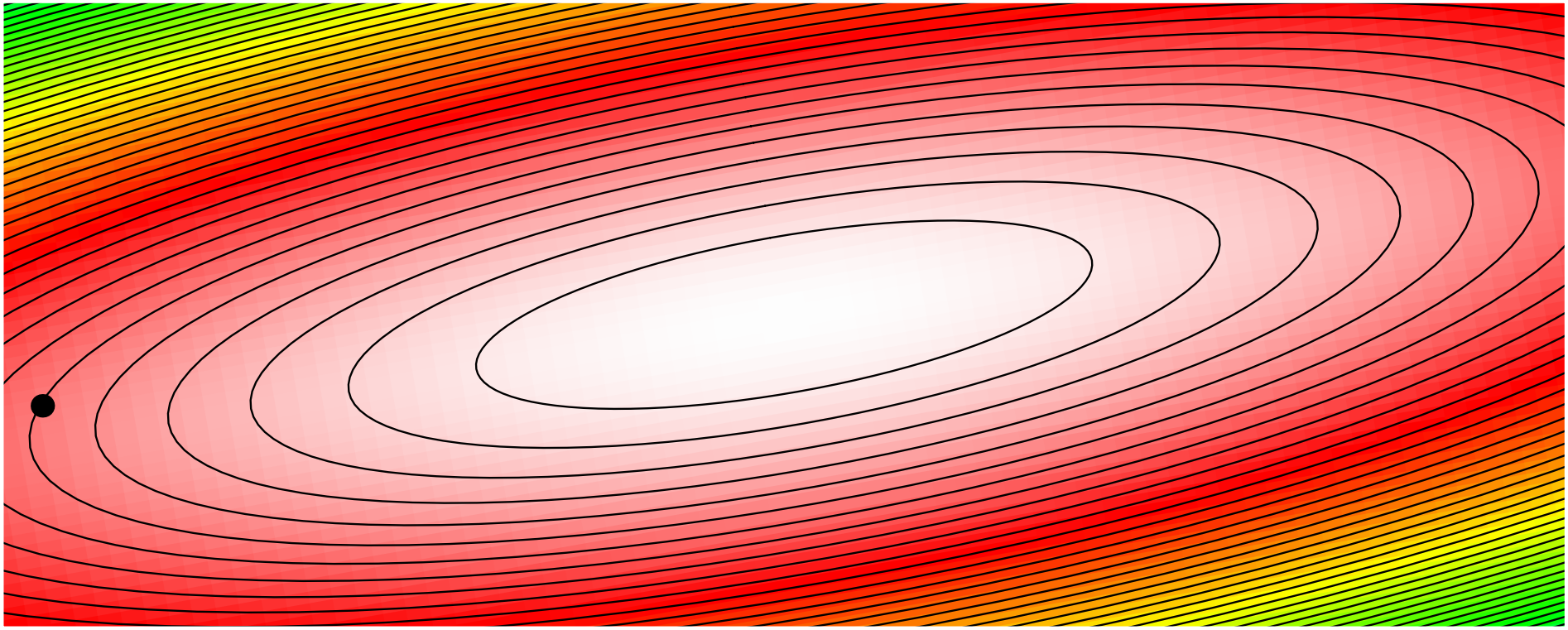


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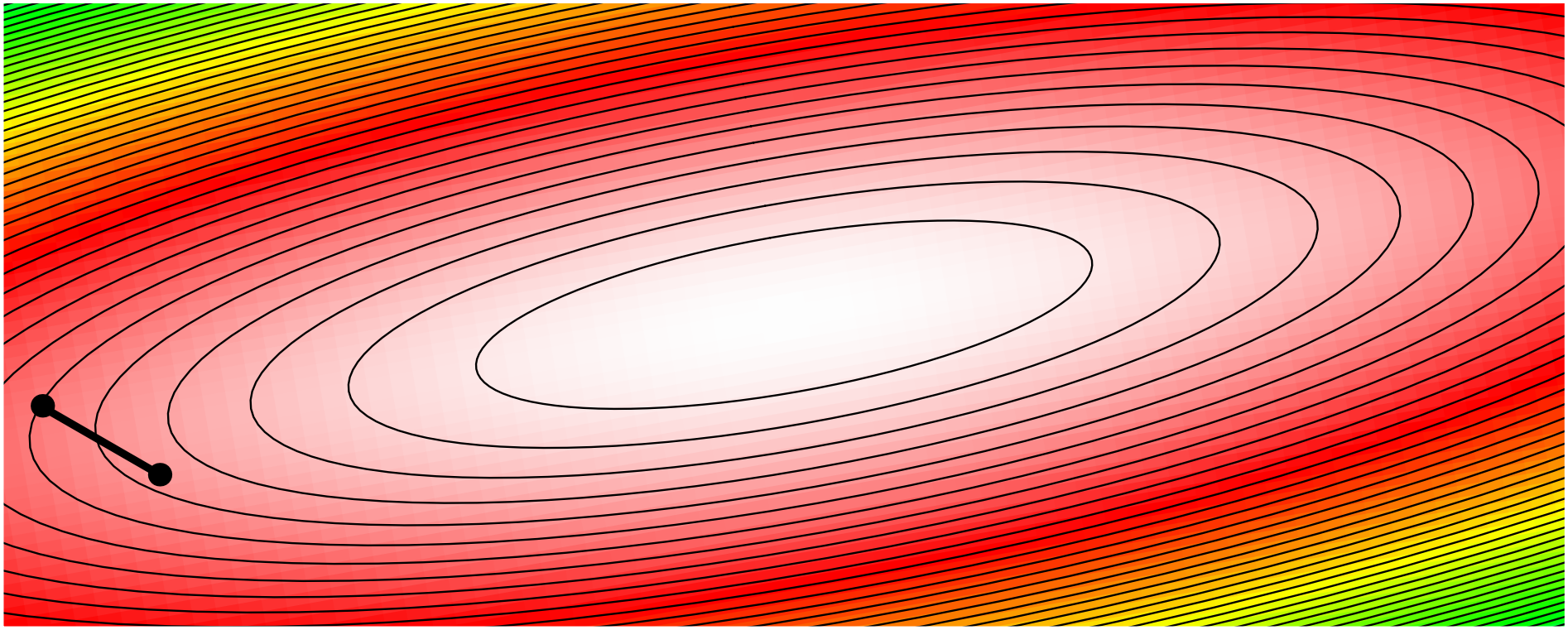
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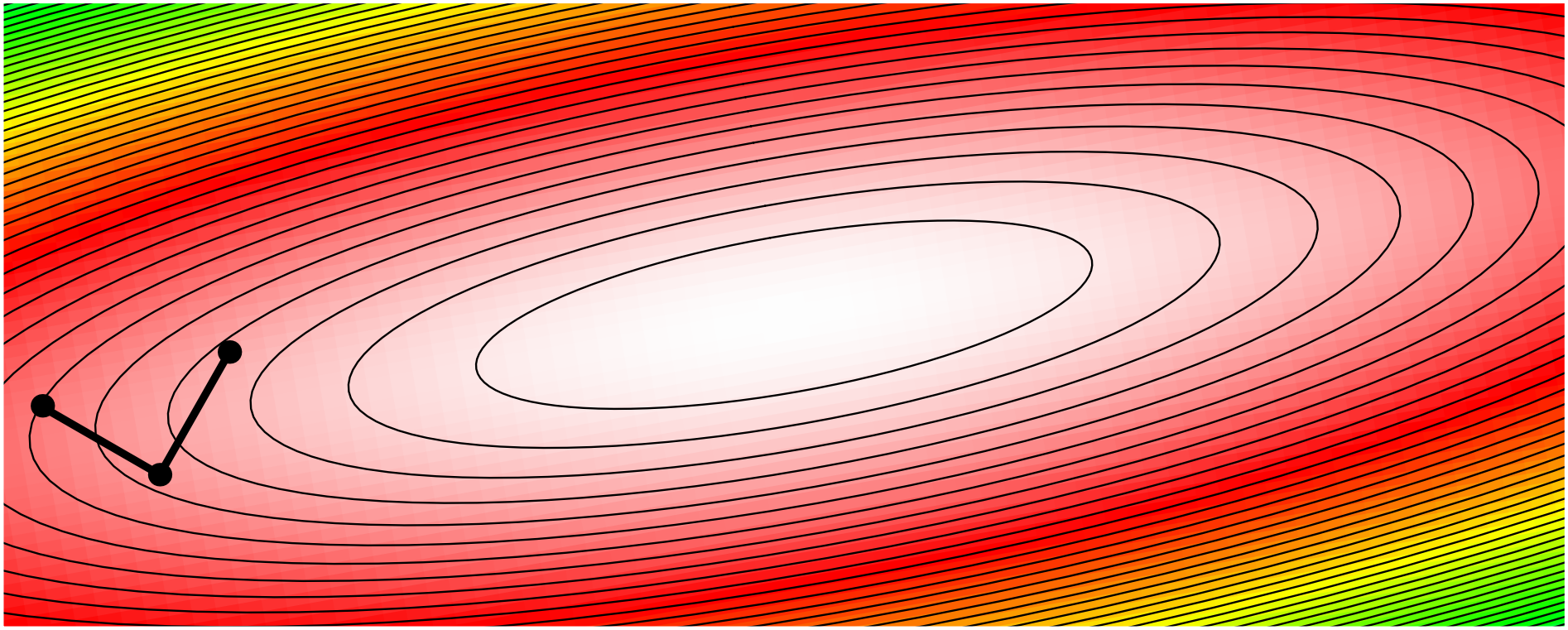
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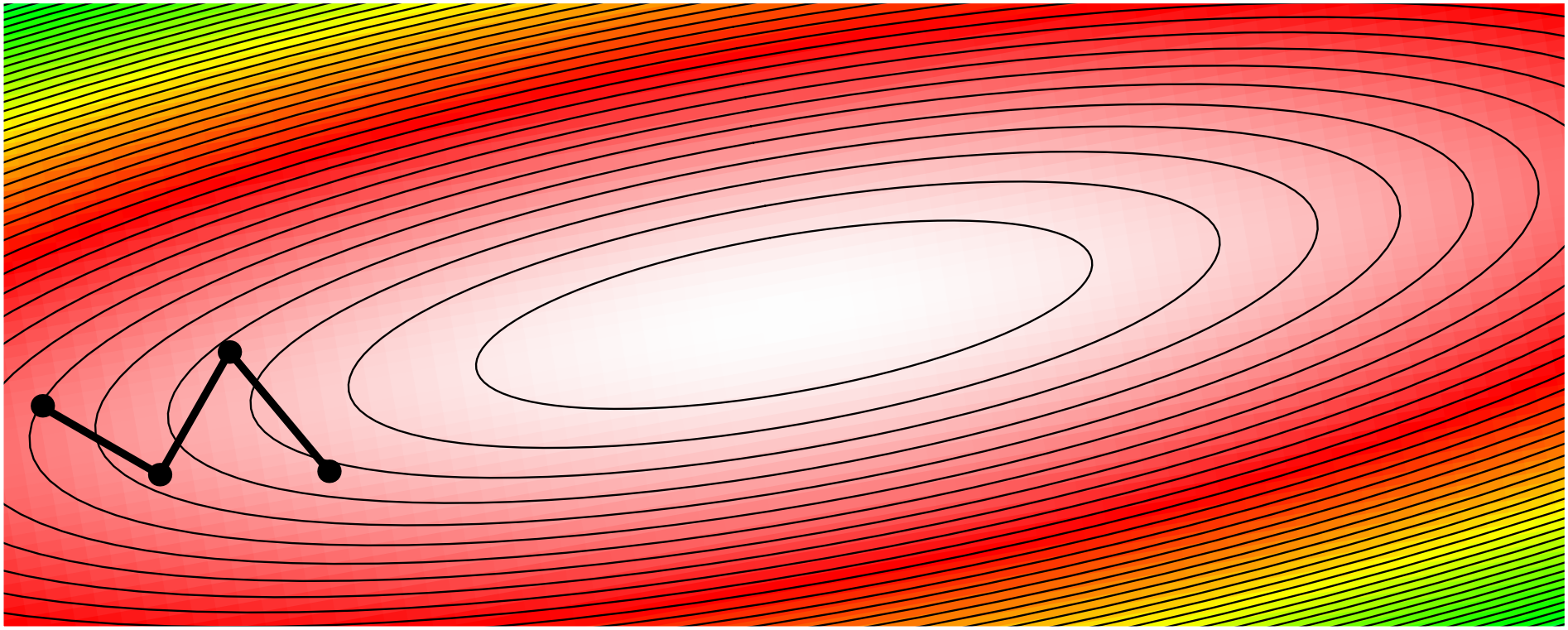
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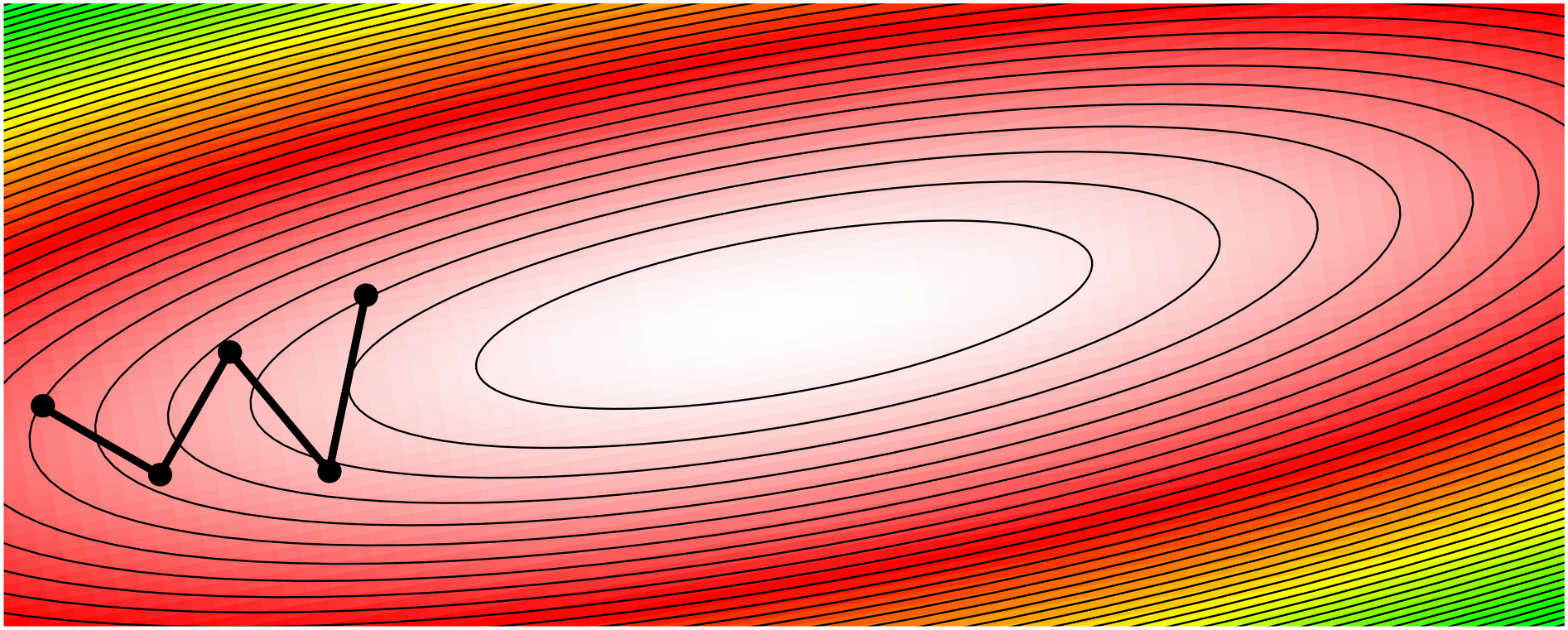
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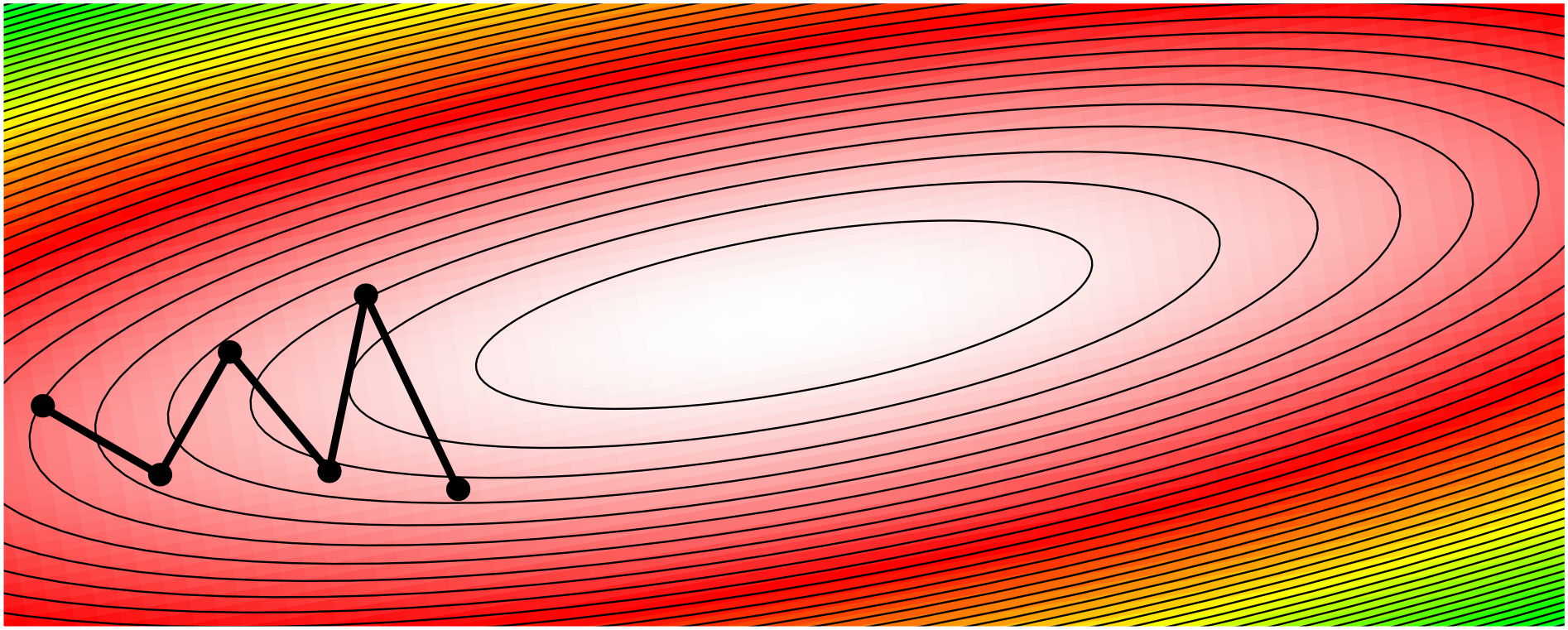
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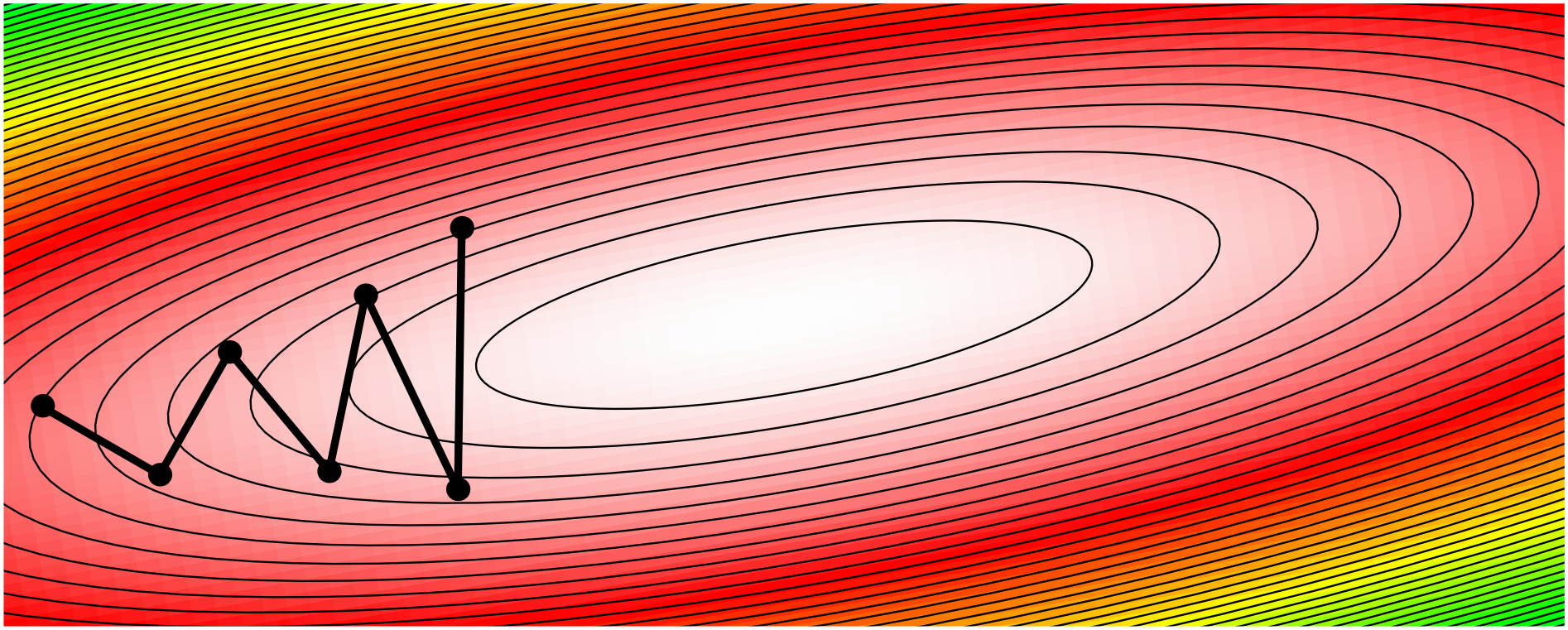
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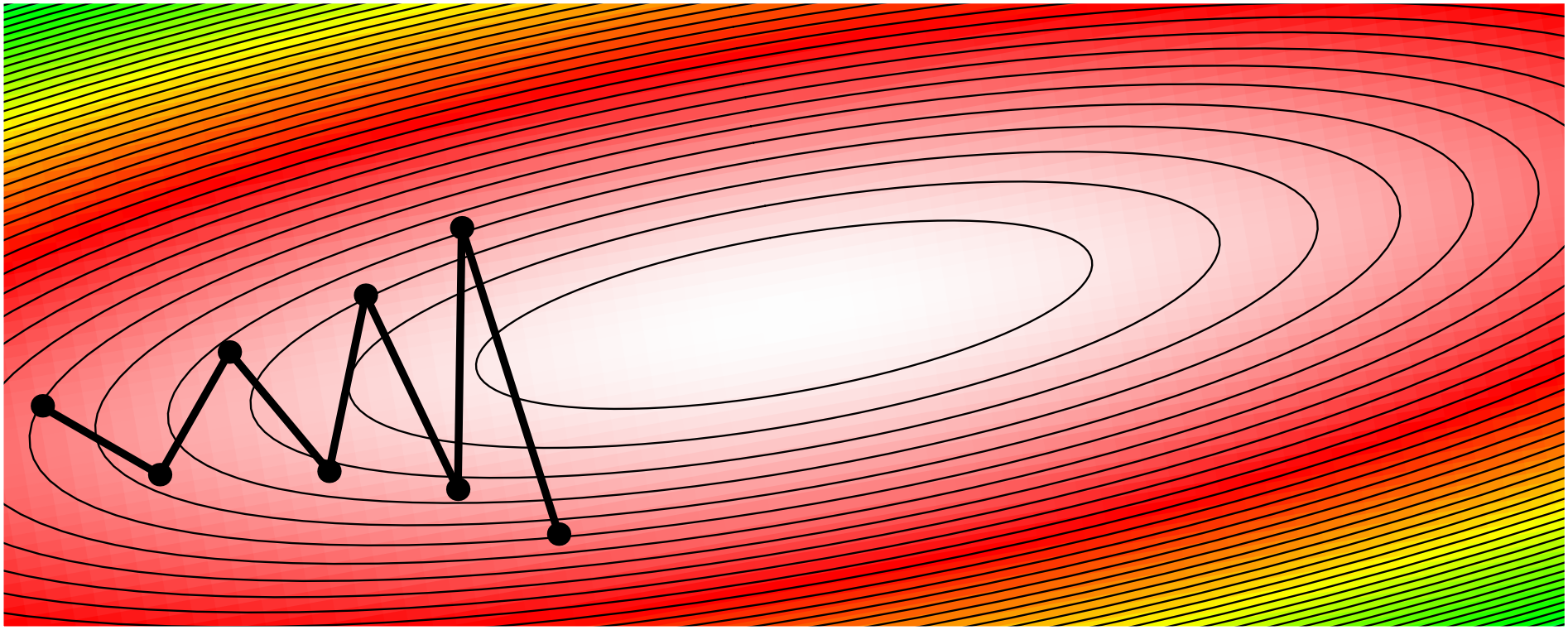
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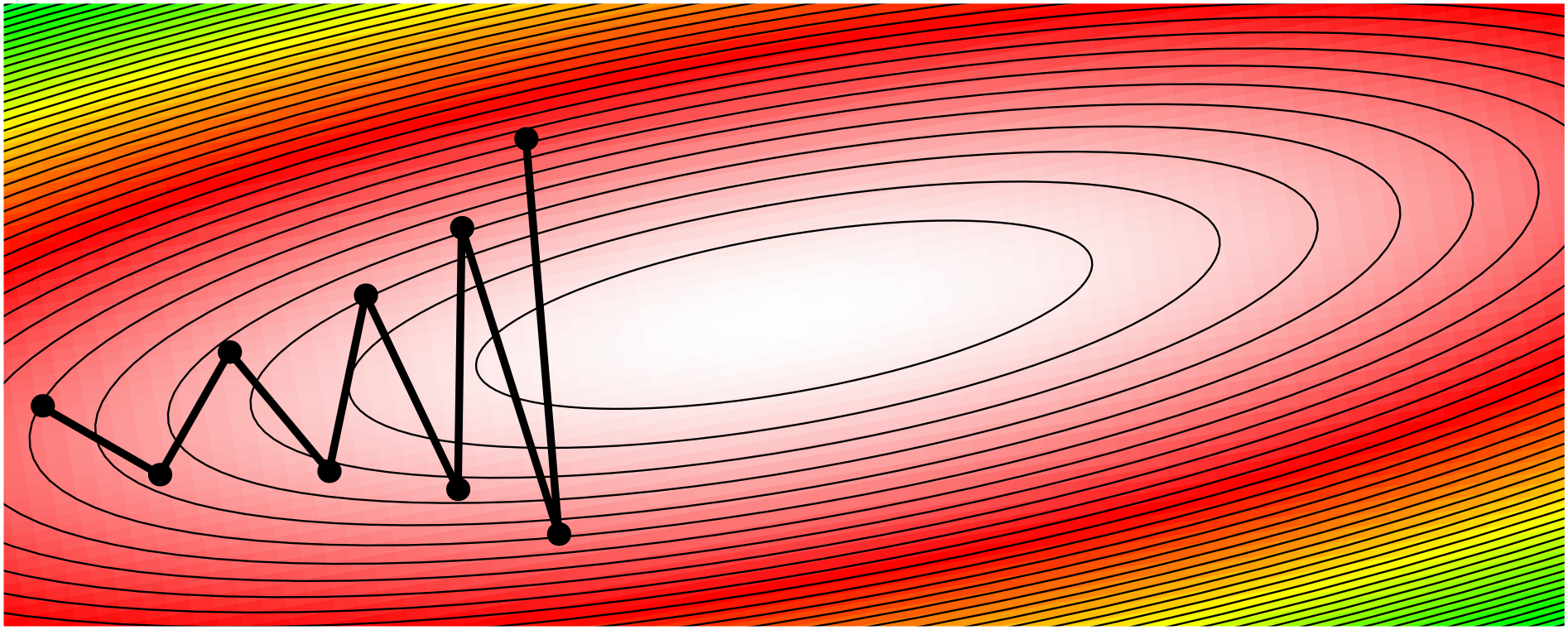
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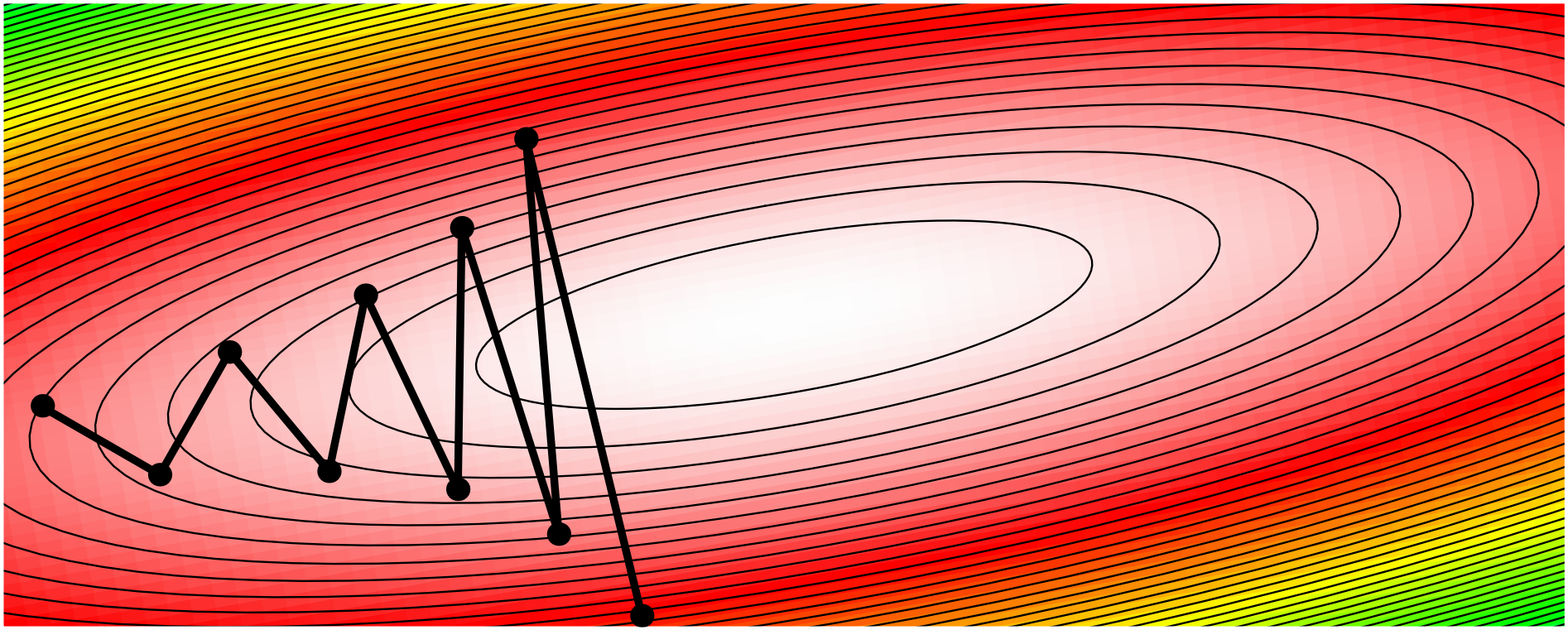
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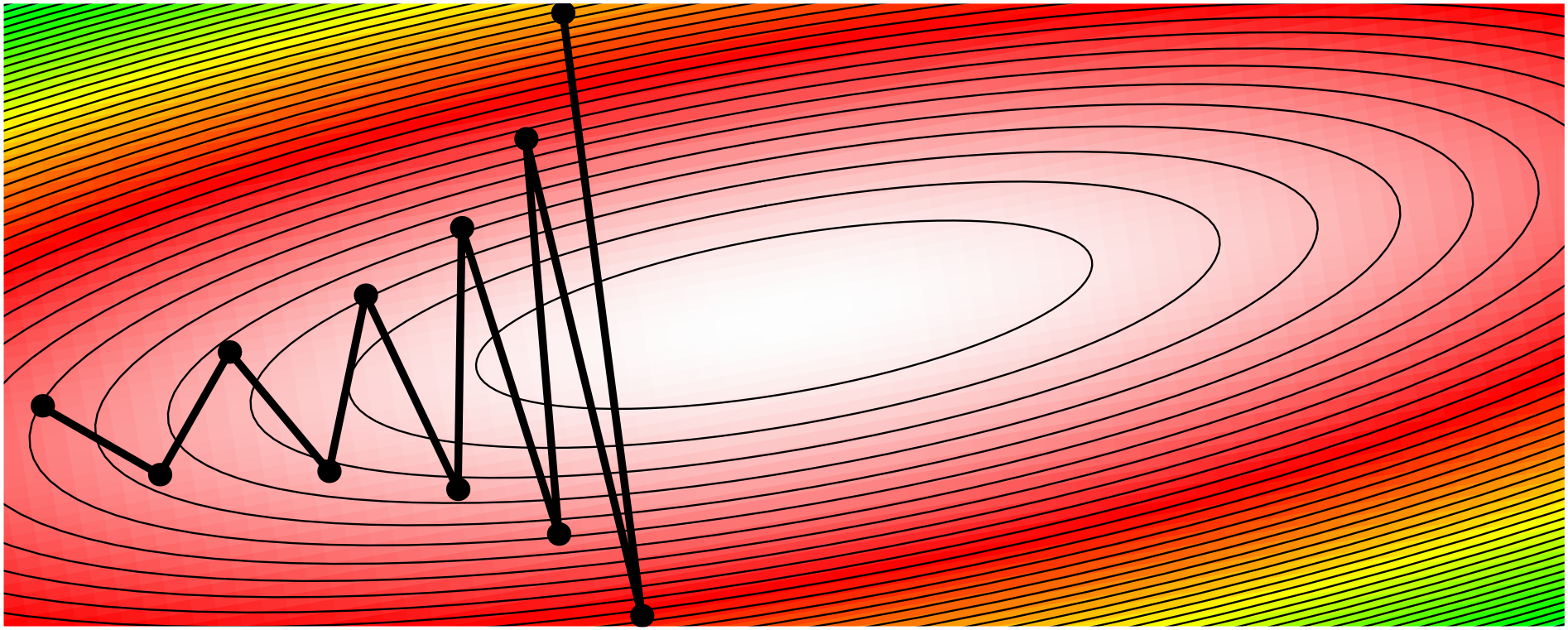
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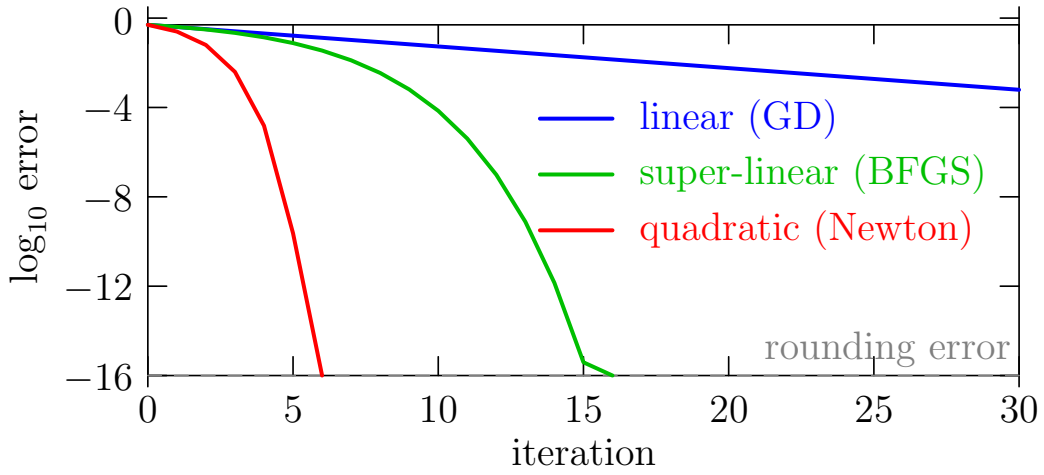
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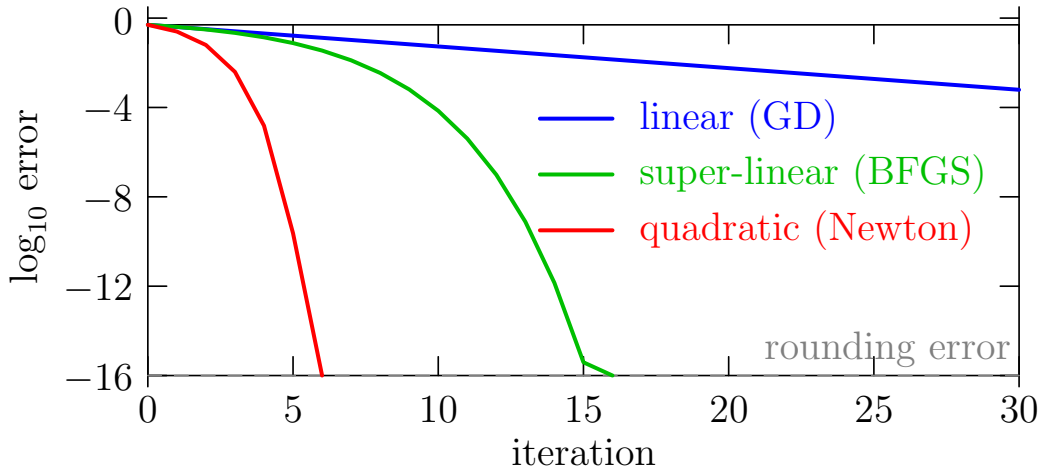
Linear and Quadratic Convergence

- Near a minimum, along each eigenvector of the Hessian, a gradient step multiplies the error by $(1 - r\lambda_i)$
- Stable only if $r < 2/\lambda_{\max}$, so the flattest direction shrinks by at best $1 - 2\lambda_{\min}/\lambda_{\max} = 1 - 2/\kappa$ per step (κ is the condition number)
- A constant factor per step is **linear** convergence; Newton squares the error: **quadratic** convergence



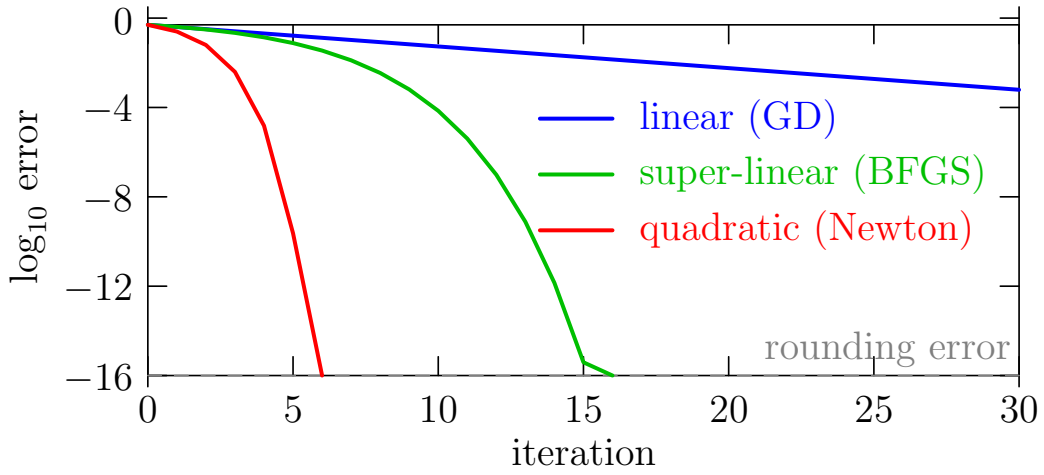
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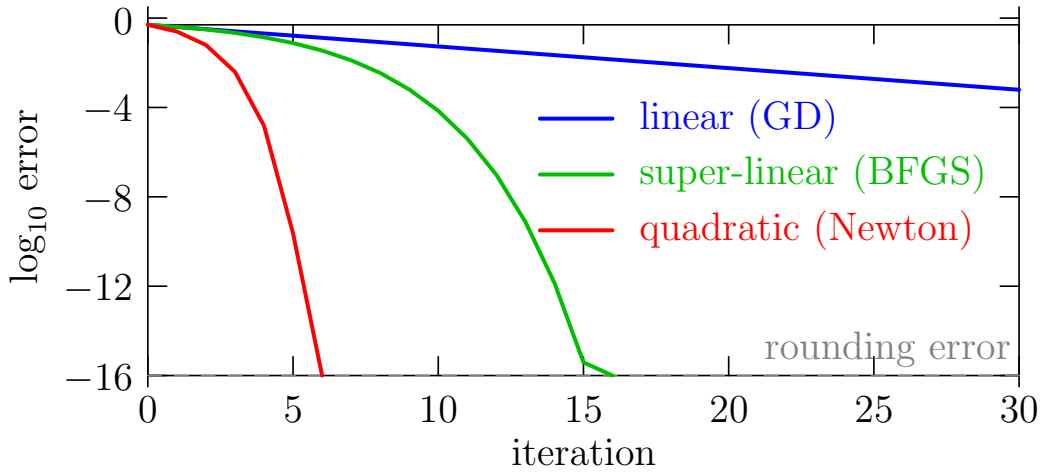
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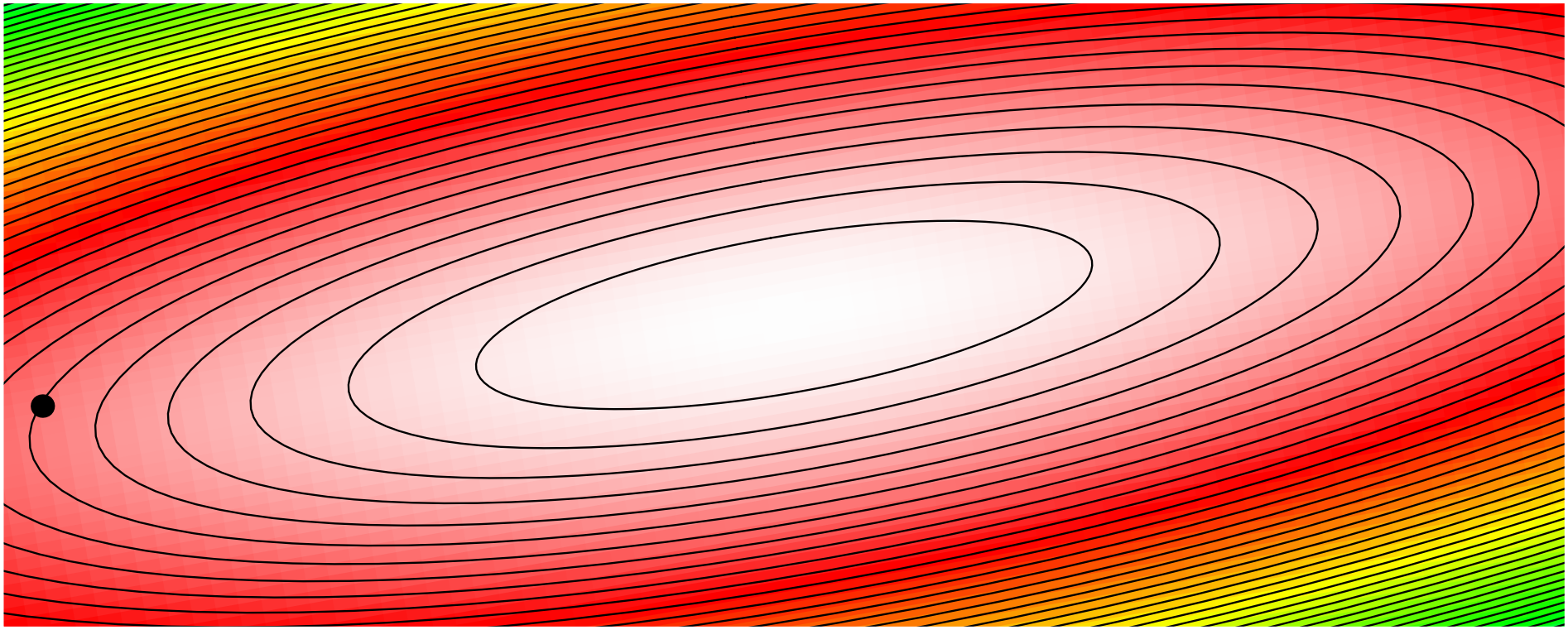
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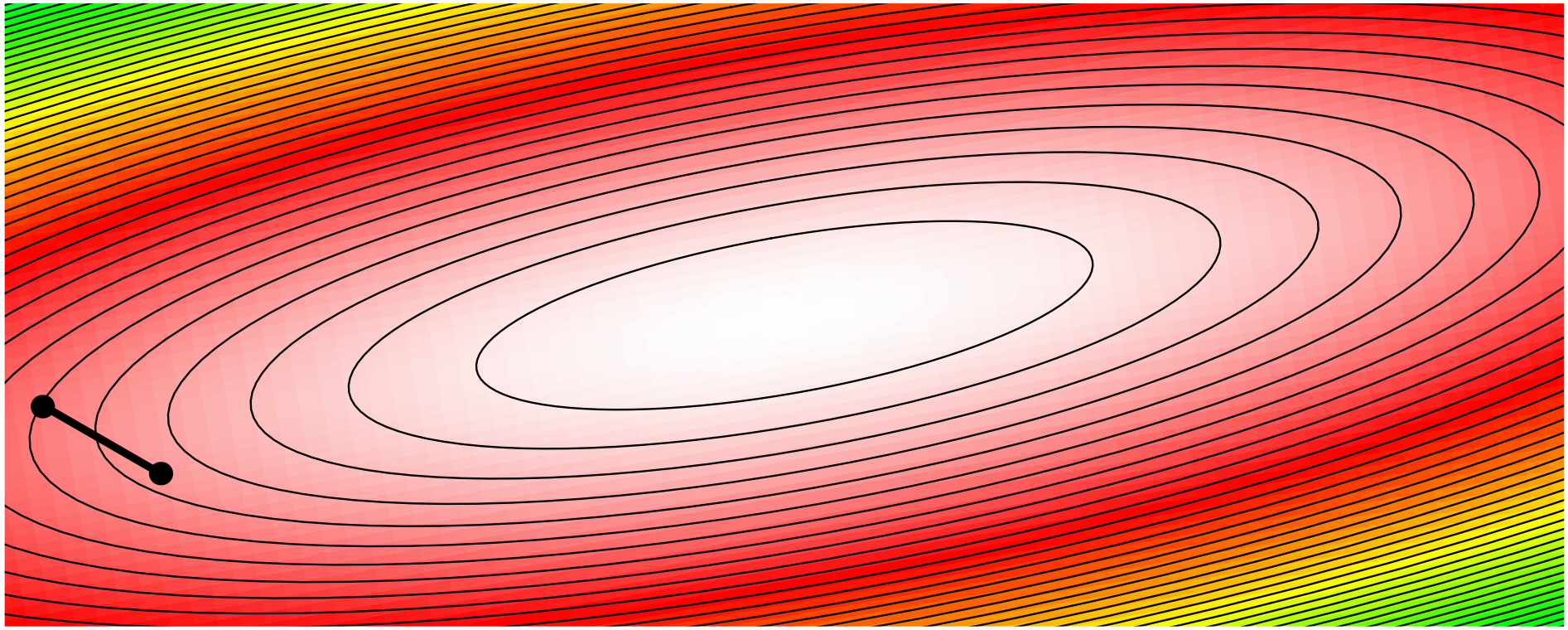
Line Minimisation

- We can systematically seek the minimum along a line of the gradient



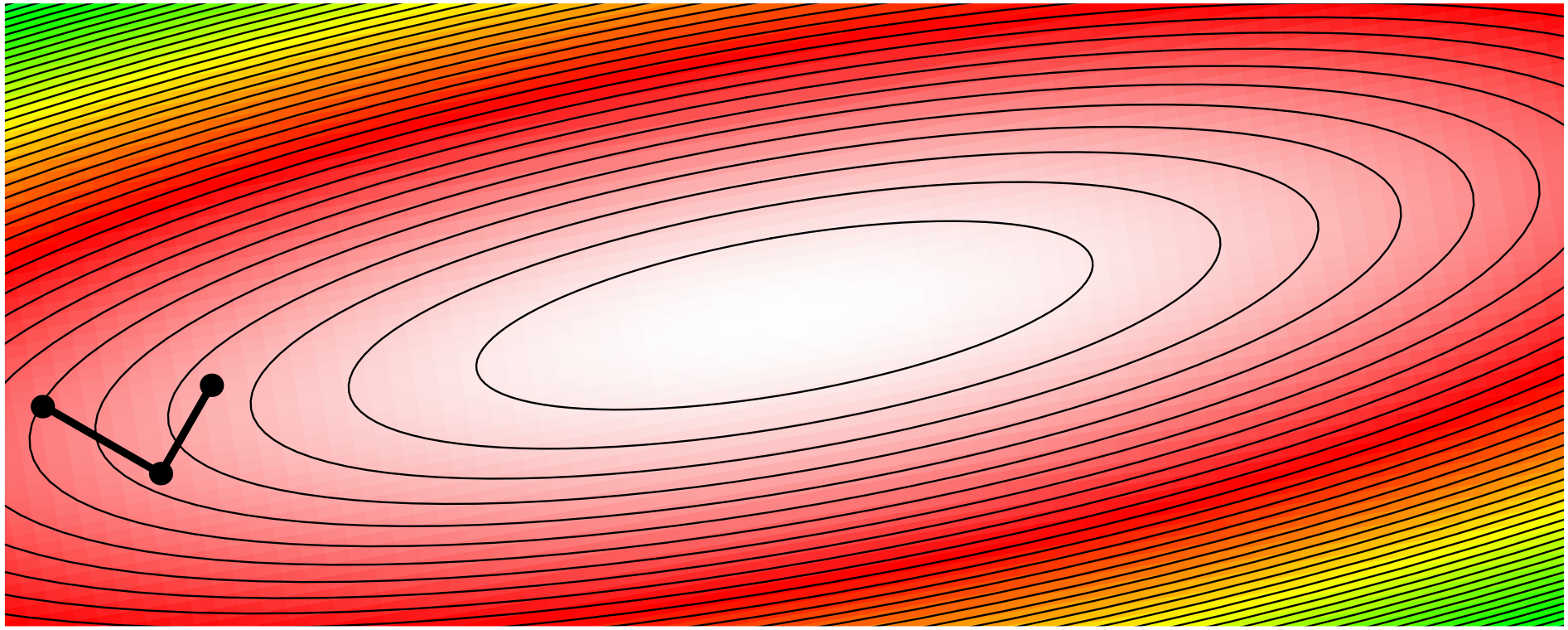
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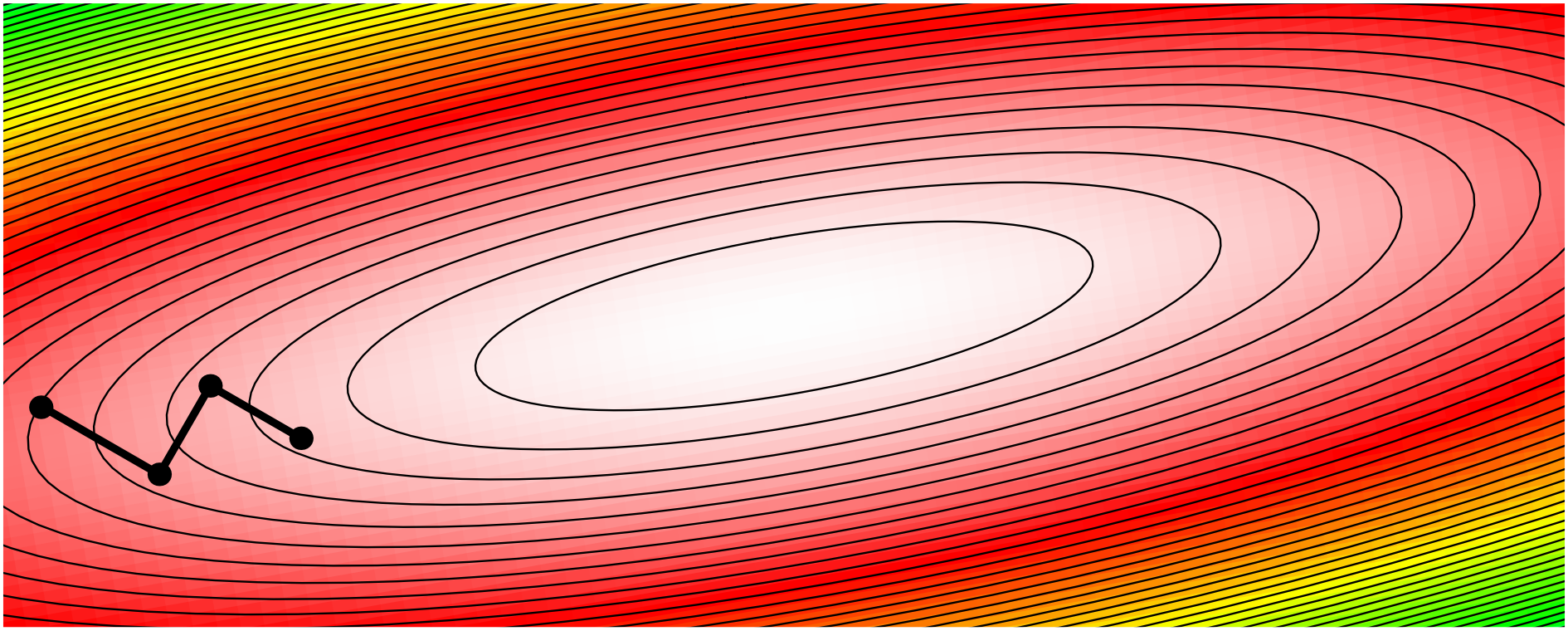
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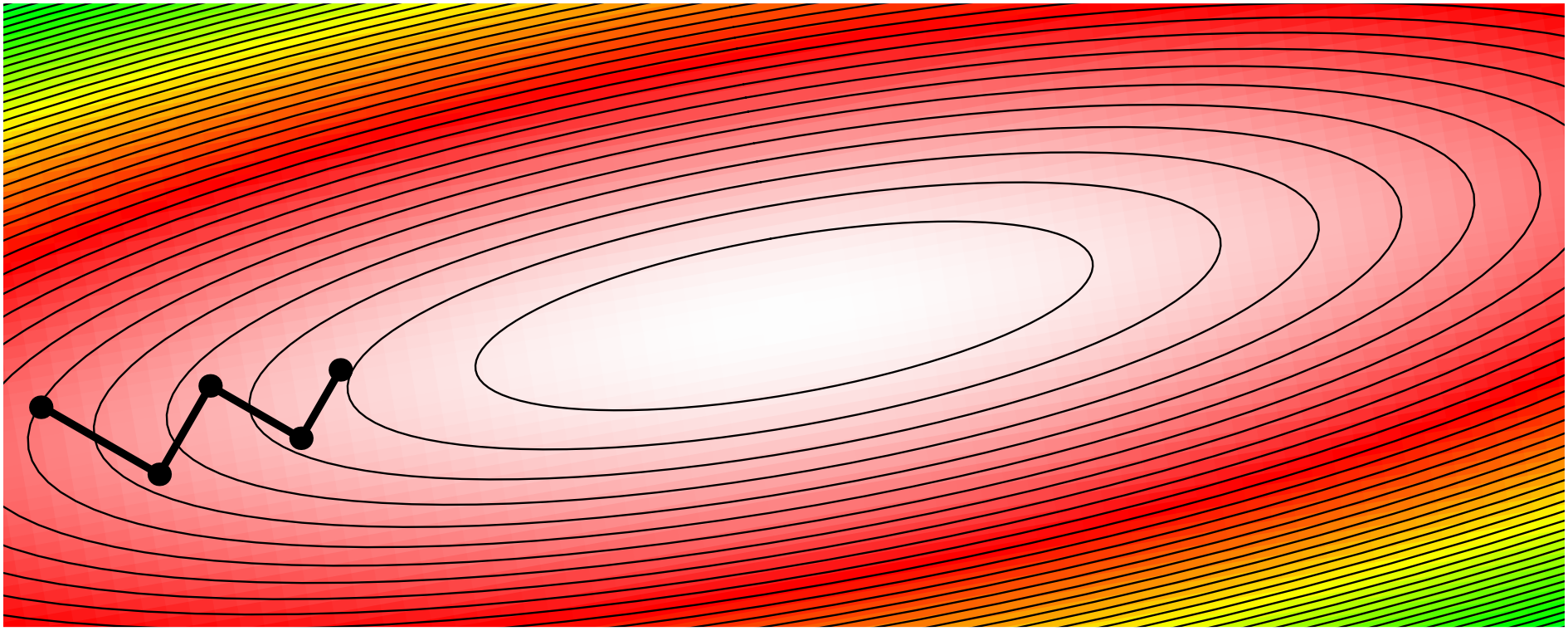
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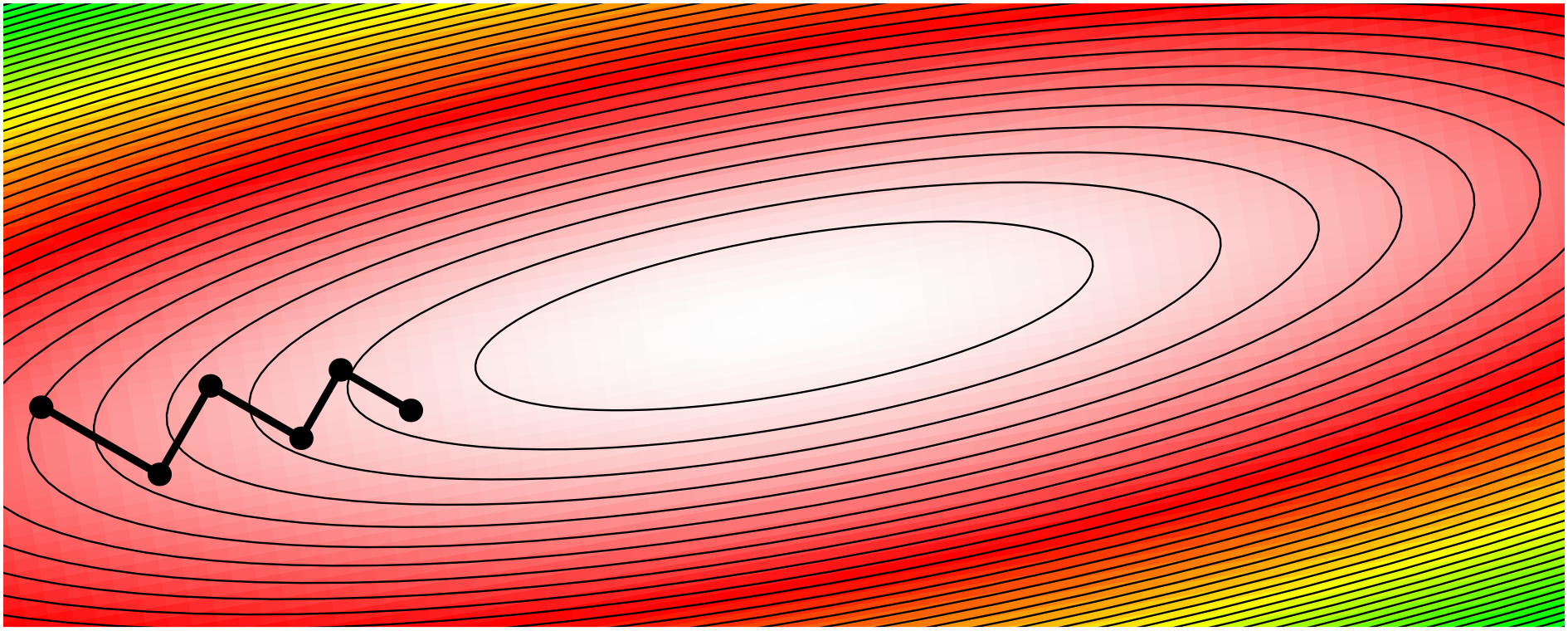
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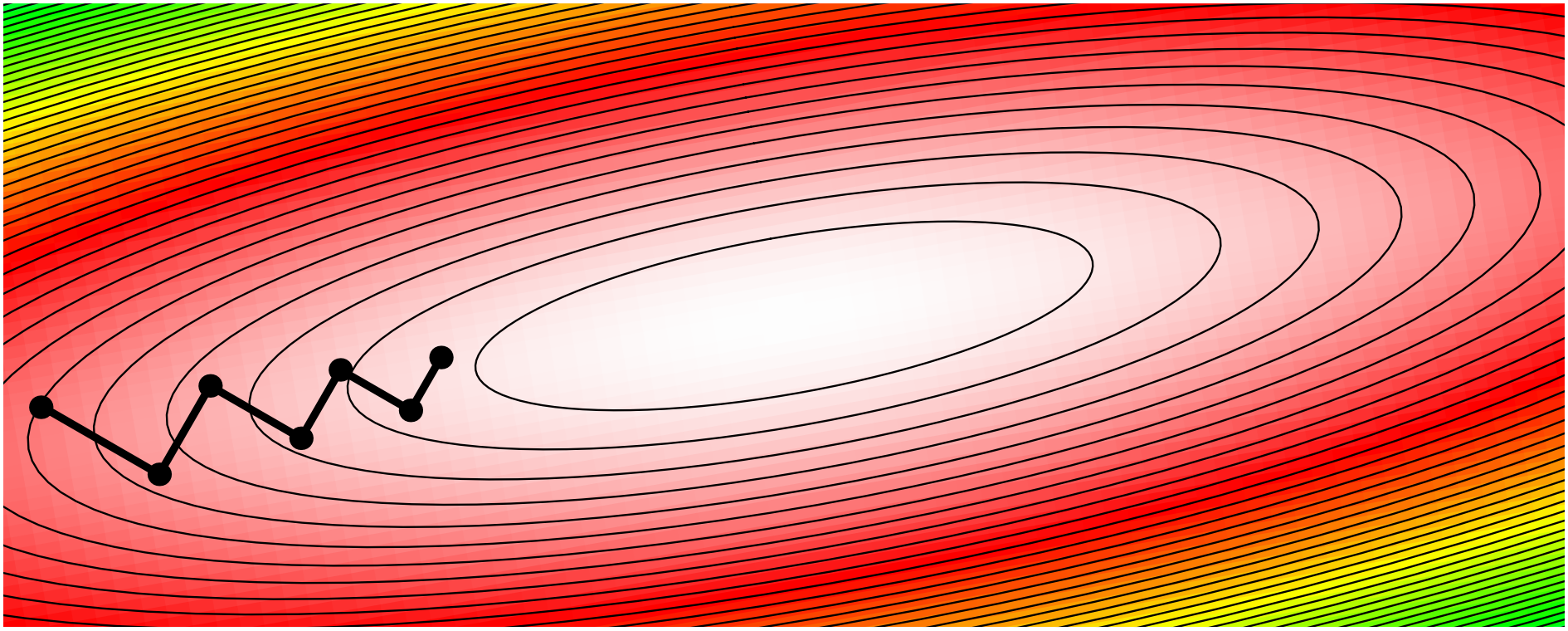
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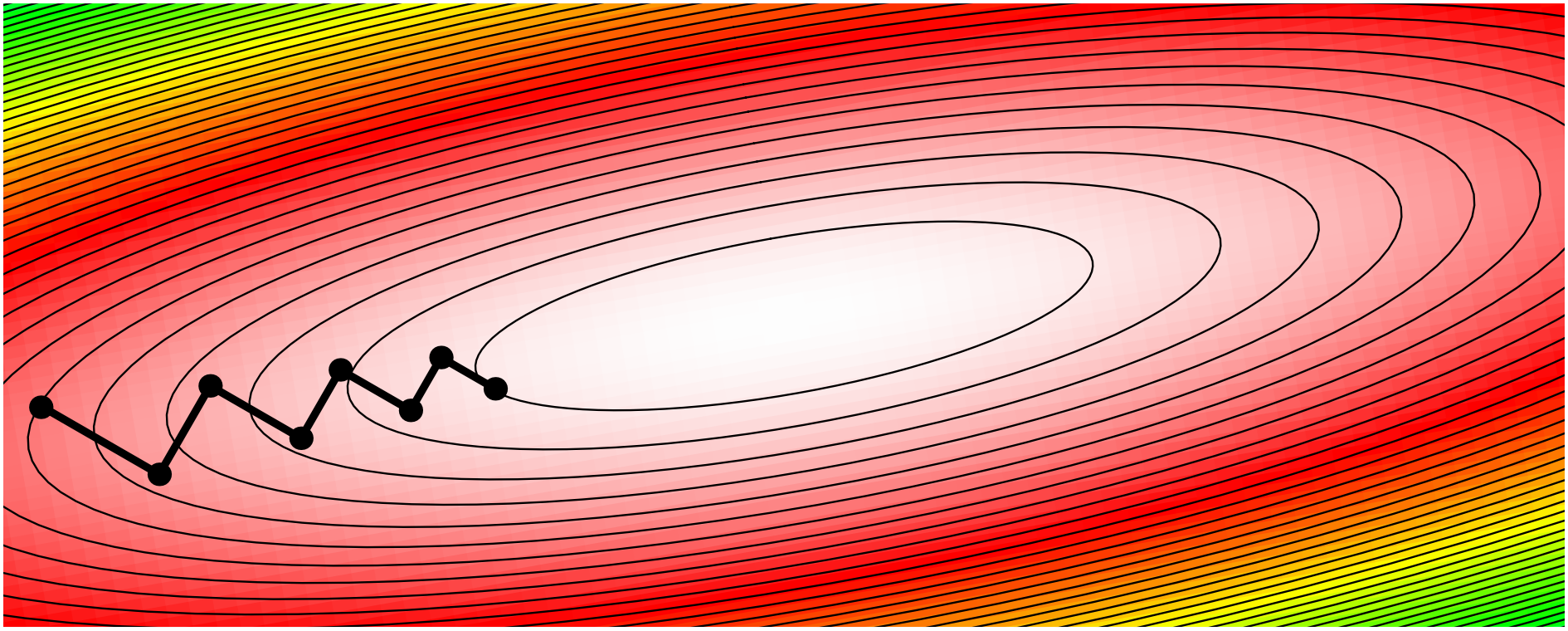
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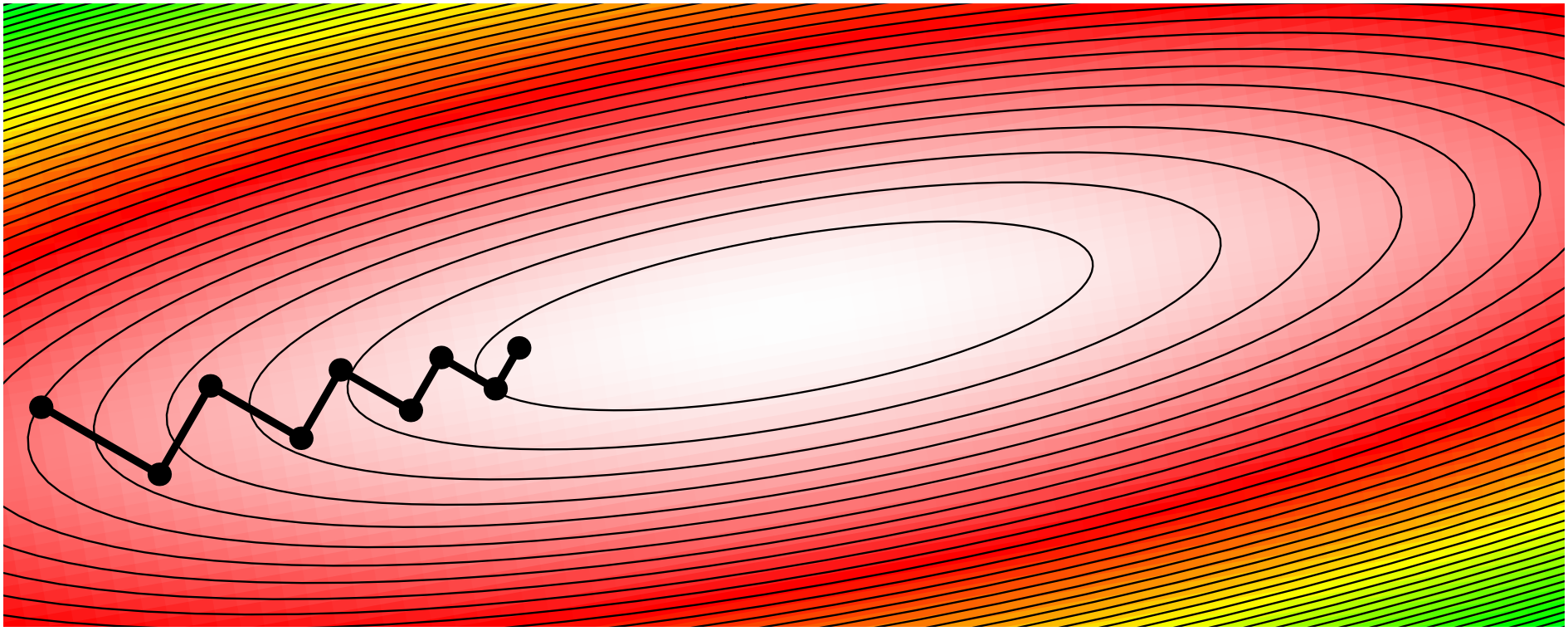
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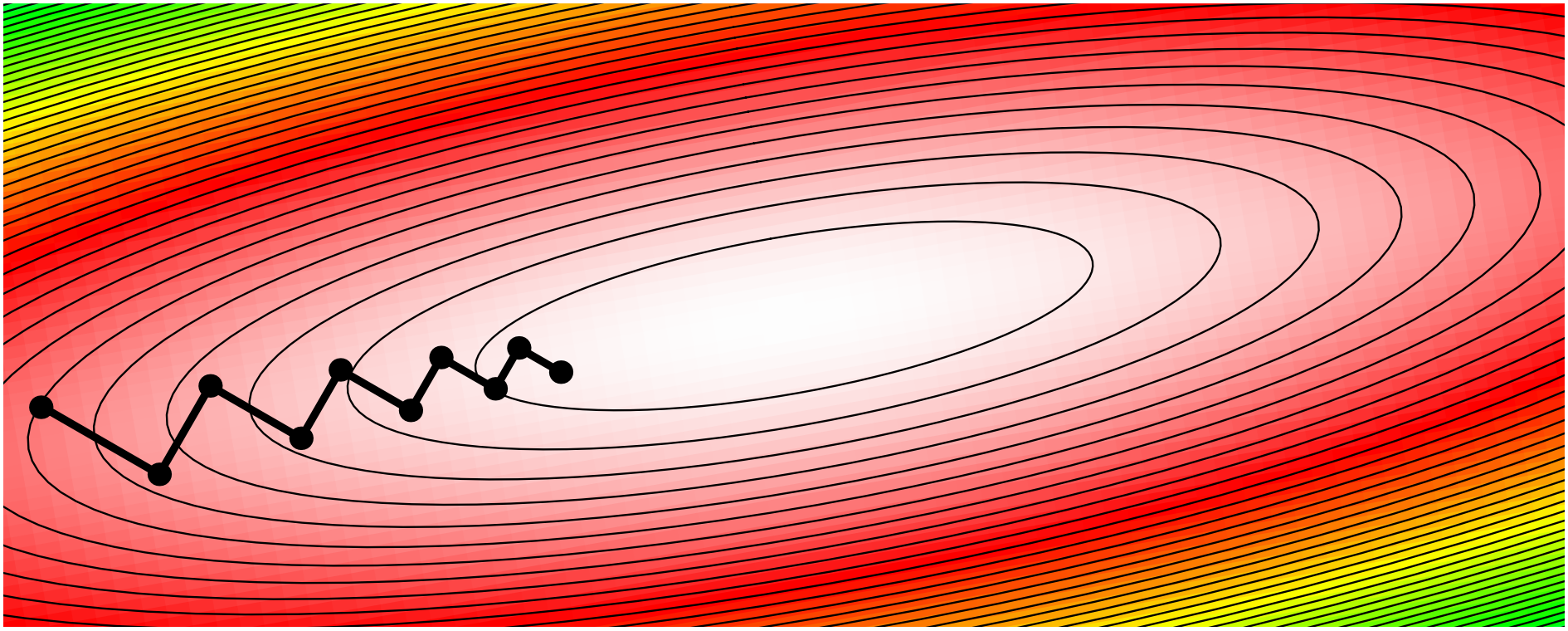
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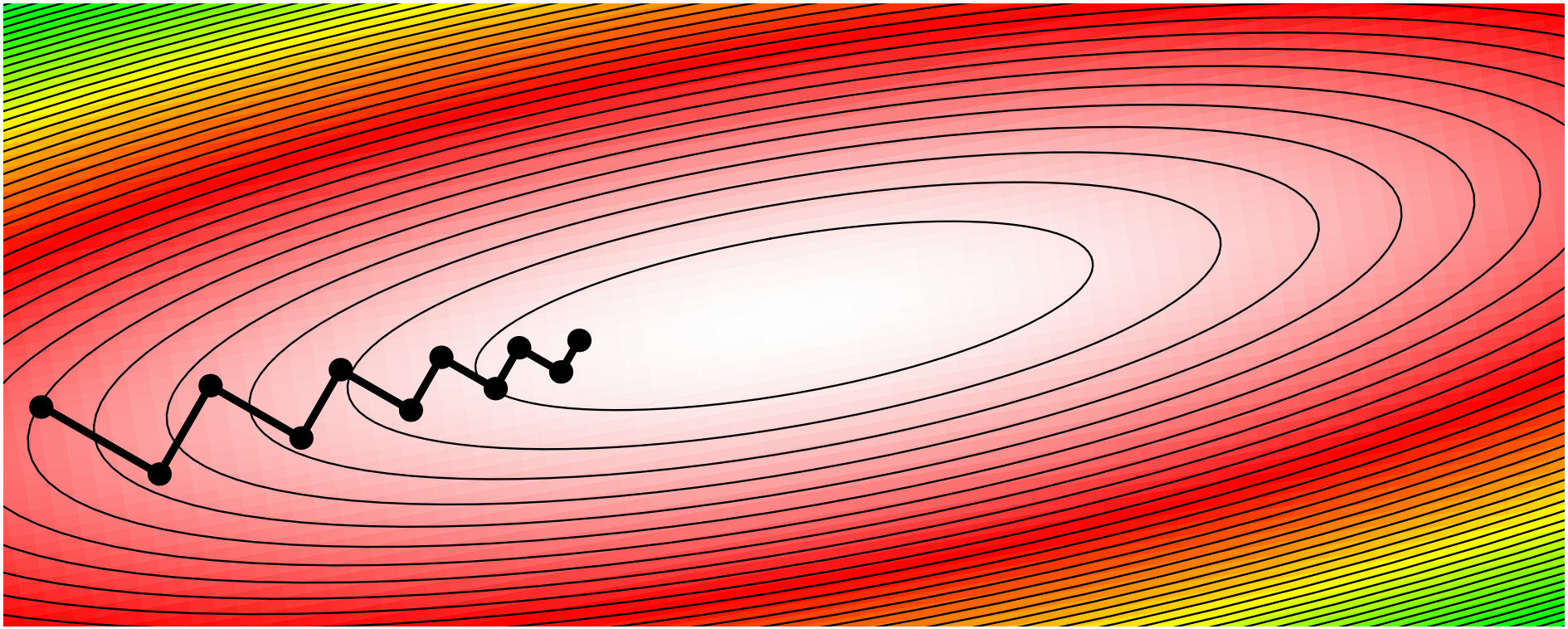
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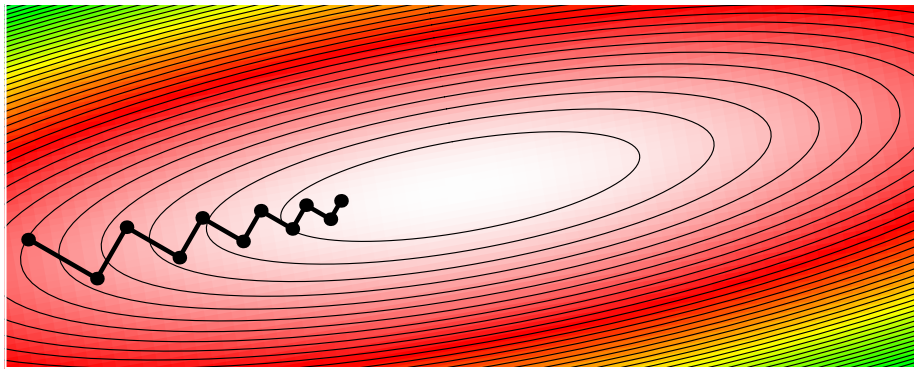
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Zig-Zag

- Note that in narrow valleys line minimisation zigzags (each step is at right angles to the last)

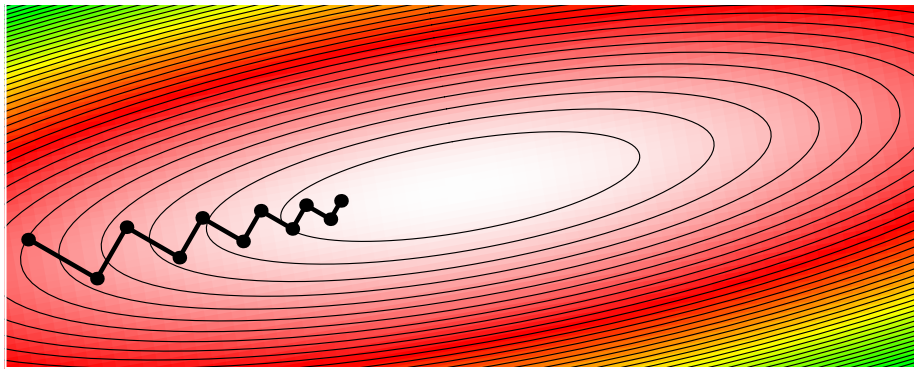


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- However computing the Hessian is time consuming and misleading if we are not in a quadratic potential (i.e. far from the optimum)

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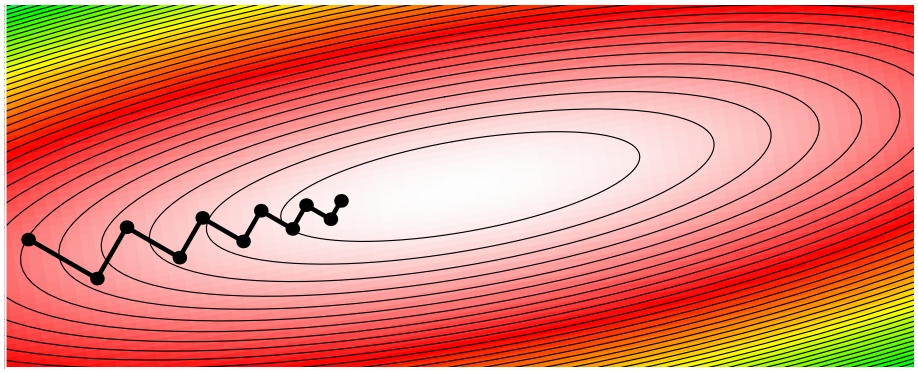


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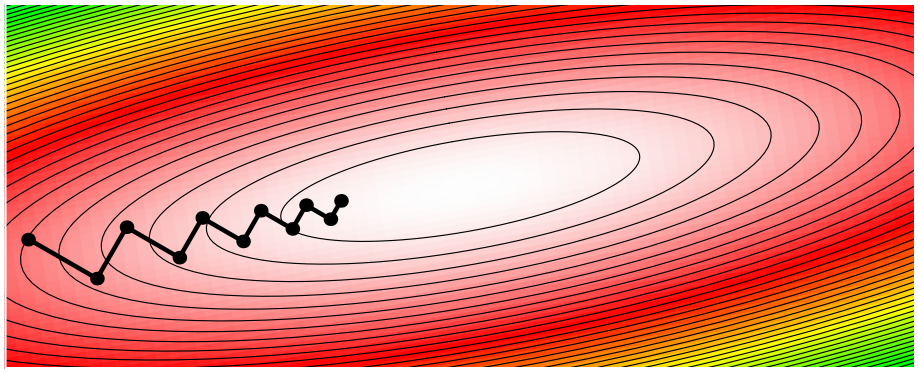


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Better Optimisation Algorithms

- Good optimisation algorithms use second-order (curvature) information without computing the full Hessian
- E.g. Conjugate gradient
  - ⋆ Performs Line Minimisation
  - ⋆ Uses gradient, but does not go along it
  - ⋆ For a quadratic minimum in d dimensions it reaches the minimum in d steps
- E.g. Levenberg-Marquardt
  - ⋆ Used on least squares problem only
  - ⋆ Uses linear approximation of function to approximate Hessian
  - ⋆ Adapts from hill-climbing to Newton method
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BFGS and L-BFGS

- BFGS (Broyden–Fletcher–Goldfarb–Shanno) is the most widely used quasi-Newton method
- It builds an approximation $\mathbf{B} \approx \mathbf{H}^{-1}$ from how the gradient changes from one step to the next

$$\mathbf{B}\left(\boldsymbol{\nabla} f(\boldsymbol{x}_{t+1}) - \boldsymbol{\nabla} f(\boldsymbol{x}_t)\right) = \boldsymbol{x}_{t+1} - \boldsymbol{x}_t$$

- Steps $\boldsymbol{x}_{t+1} = \boldsymbol{x}_t - \alpha \mathbf{B} \boldsymbol{\nabla} f(\boldsymbol{x}_t)$, with a line search for α
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- Want to minimise $\|\boldsymbol{\epsilon}(\boldsymbol{w})\|^2$ where $\epsilon_i(\boldsymbol{w}) = f(\boldsymbol{x}_i|\boldsymbol{w}) - y_i$
- Use linear approximation

$$\epsilon_i(\boldsymbol{w}) \approx \epsilon_i(\boldsymbol{w}^{(k)}) + (\boldsymbol{w} - \boldsymbol{w}^{(k)})^{\mathsf{T}} \boldsymbol{\nabla} \epsilon_i(\boldsymbol{w}^{(k)})$$

  with $\boldsymbol{\nabla} \epsilon_i(\boldsymbol{w}^{(k)}) = \boldsymbol{\nabla} f(\boldsymbol{x}_i|\boldsymbol{w}^{(k)})$

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Trust Region

- Solution given by $\boldsymbol{\nabla}_{\boldsymbol{w}} L_{approx}(\boldsymbol{w}) = 0$ gives

$$\boldsymbol{w}^{(k+1)} = \boldsymbol{w}^{(k)} - \left(\mathbf{J}^{\mathsf{T}}\mathbf{J}\right)^{-1}\mathbf{J}^{\mathsf{T}}\boldsymbol{\epsilon}(\boldsymbol{w}^{(k)})$$

- Can lead us in the wrong direction
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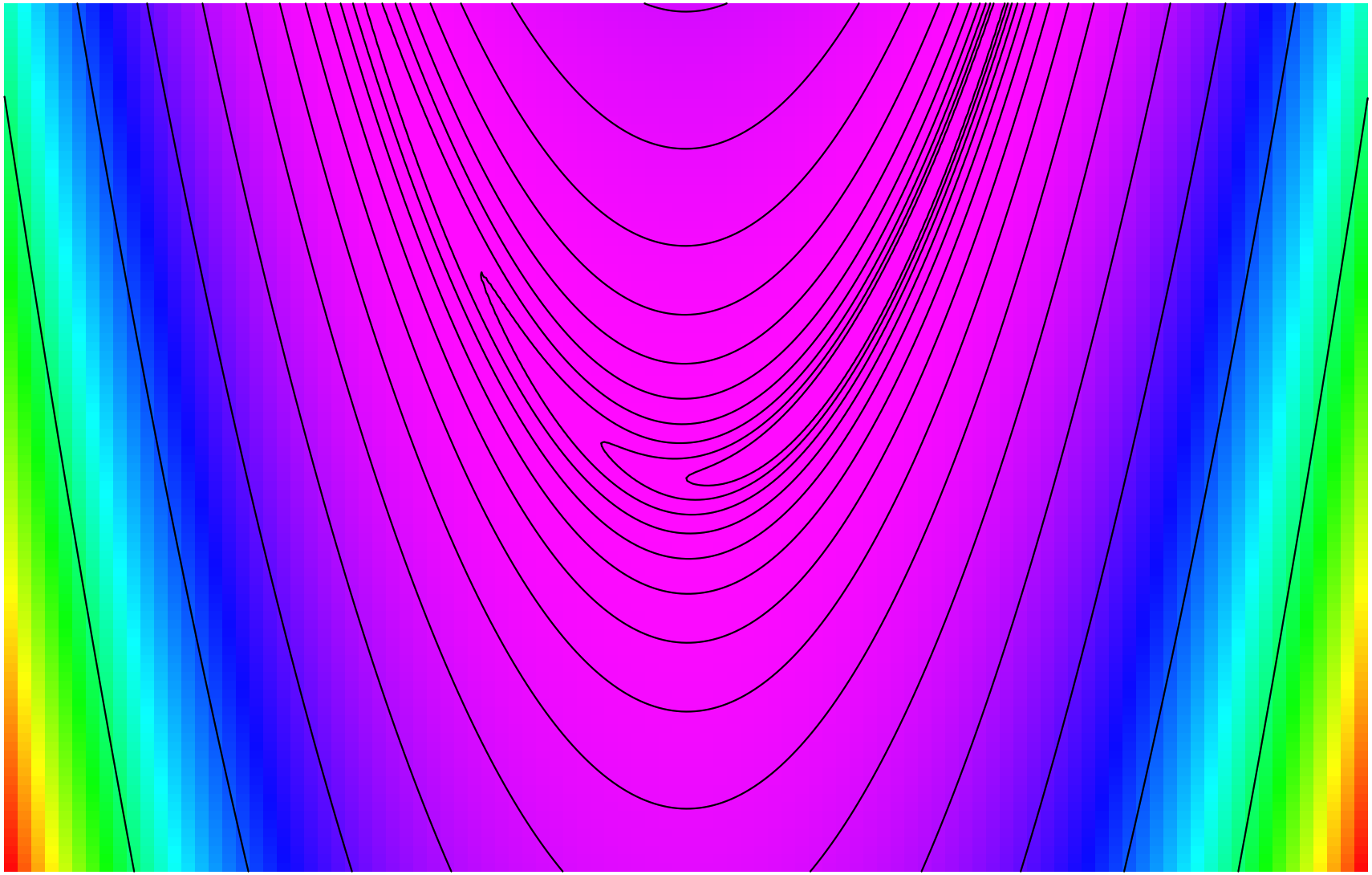
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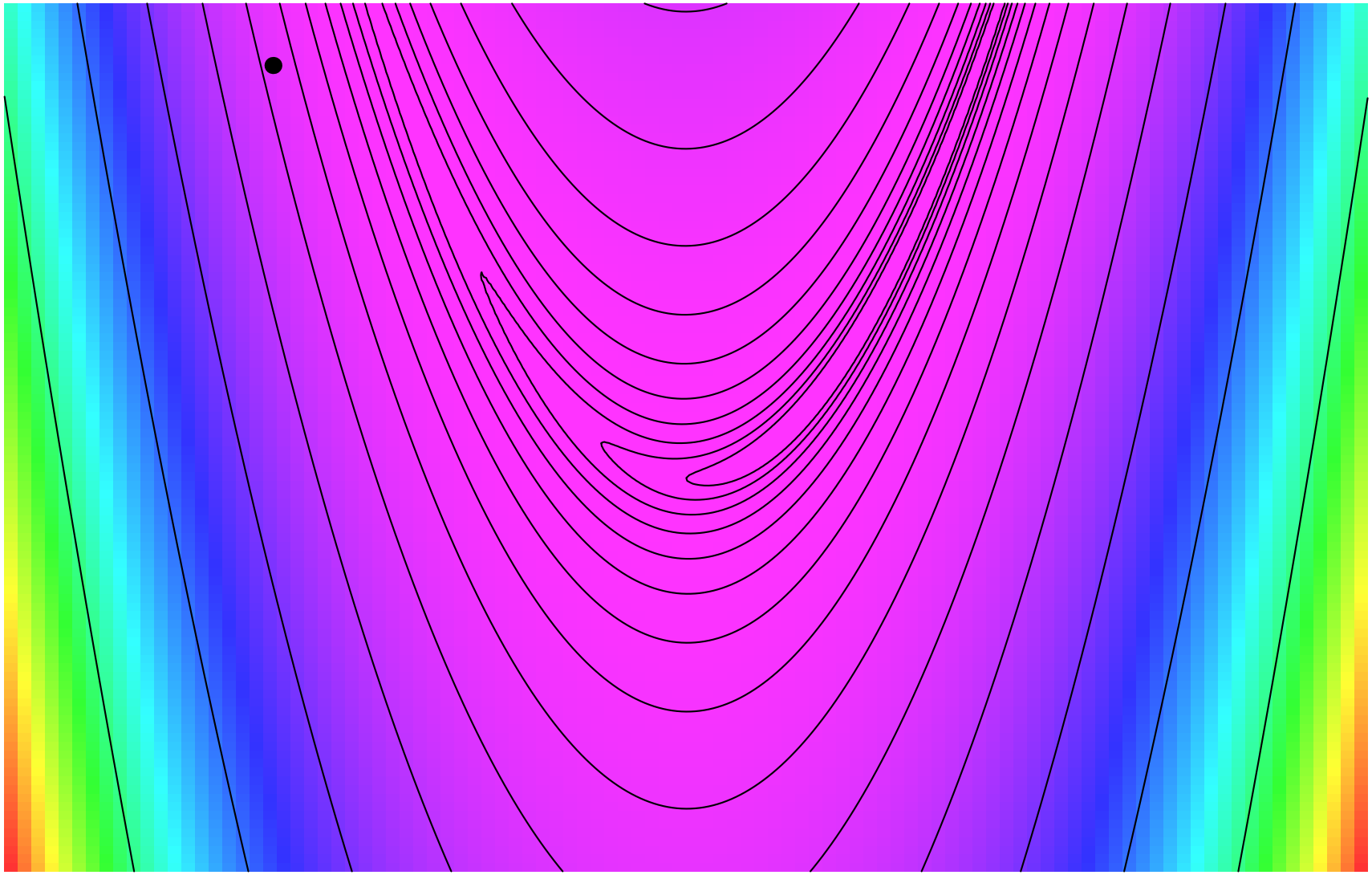
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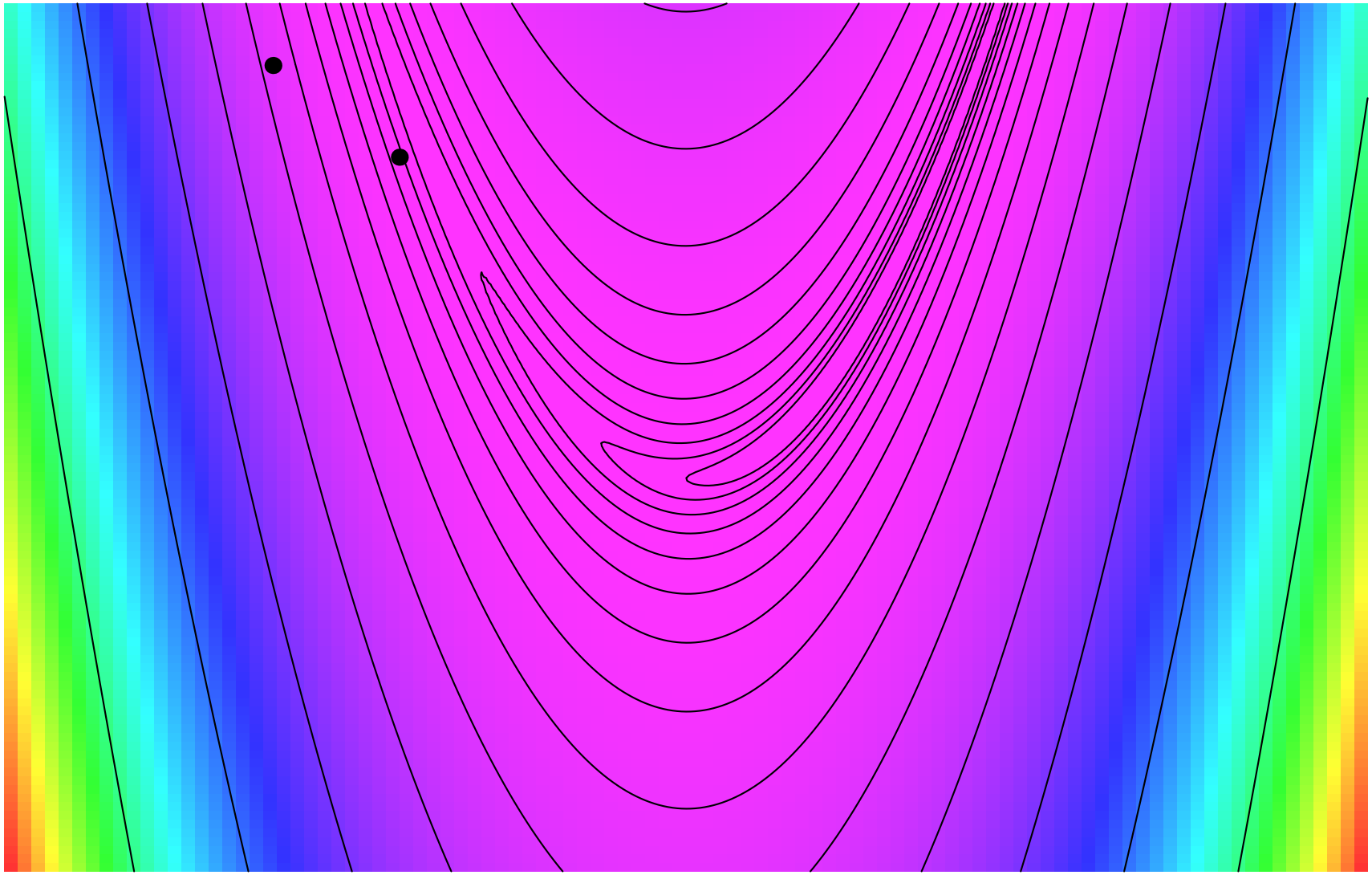
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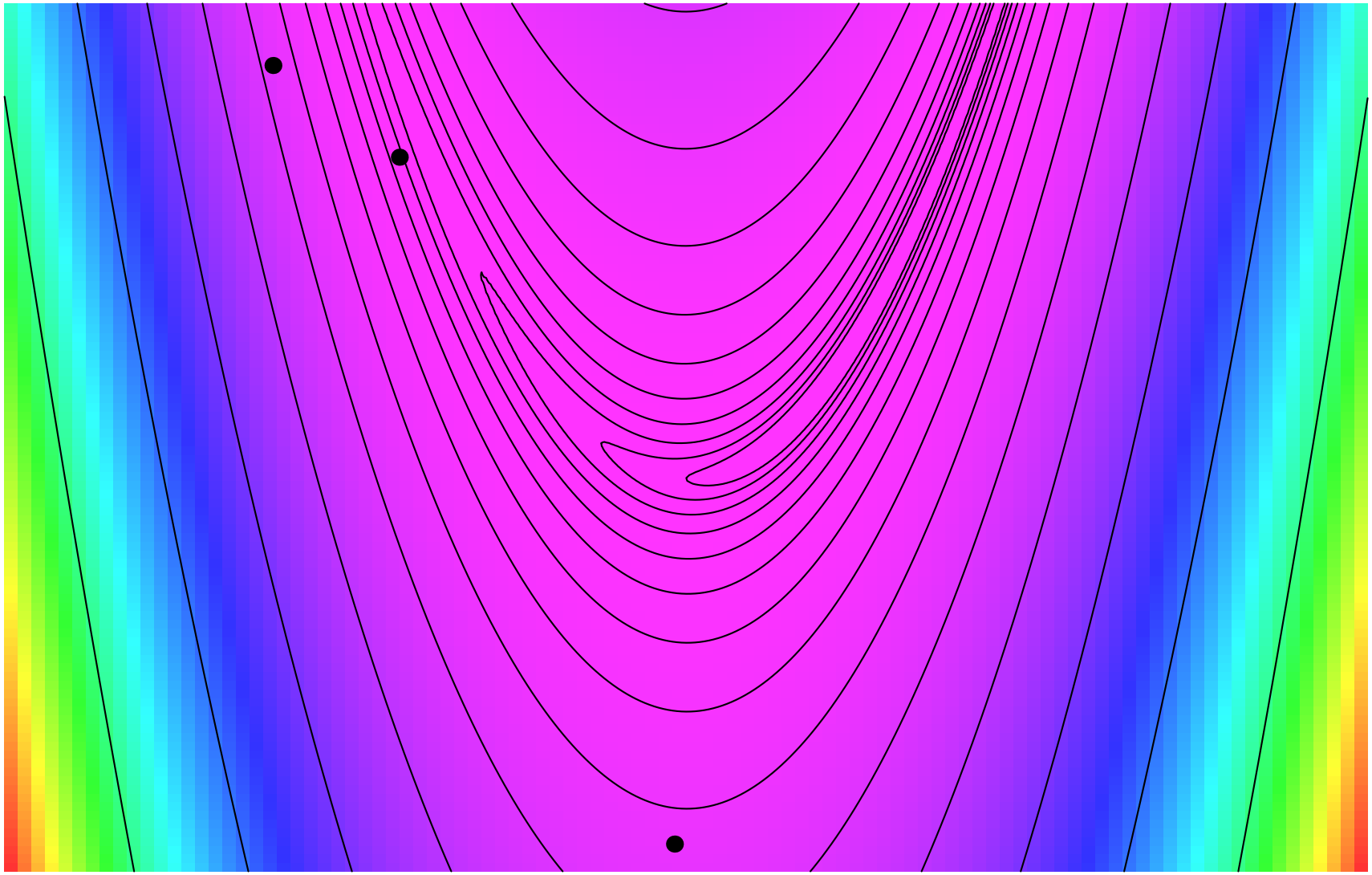
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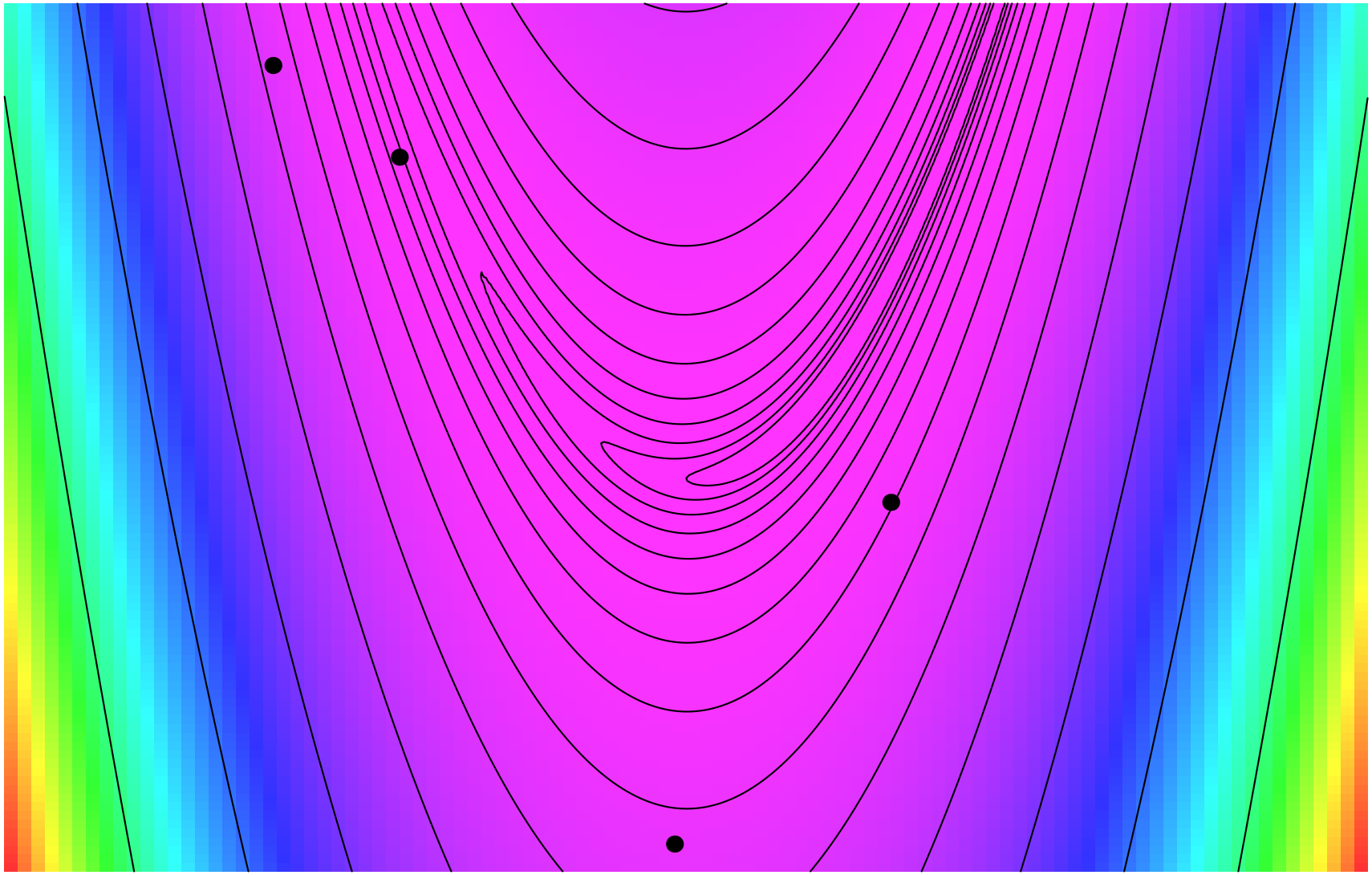
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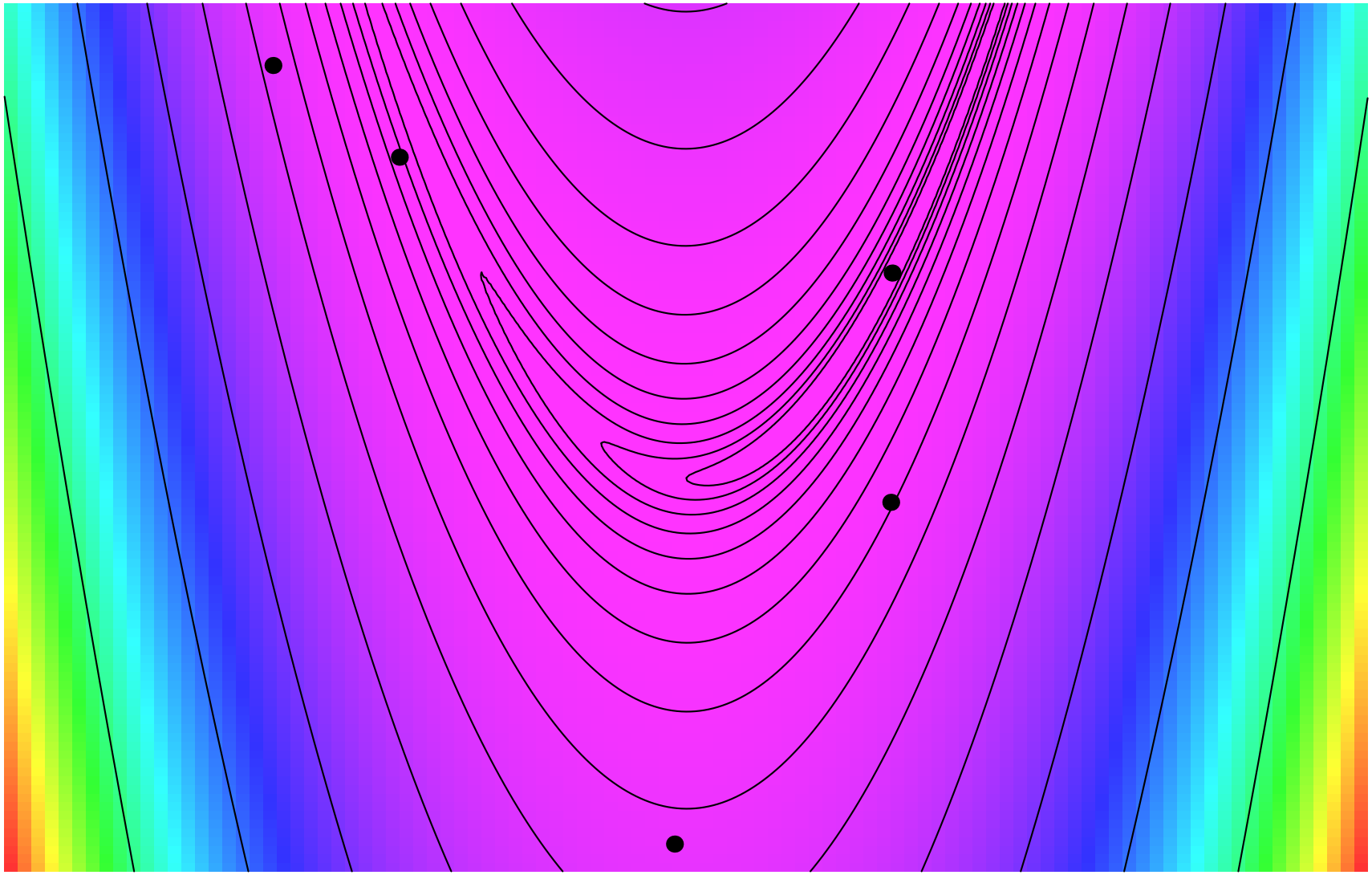
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Where Are These Used?

- For up to thousands of parameters (L-BFGS: millions), with gradients computed from all the data
- **Logistic regression**: scikit-learn's default solver is L-BFGS; Newton's method for generalised linear models
- **Gaussian processes**: fitting the kernel hyper-parameters with L-BFGS
- **Curve fitting** (non-linear least squares): Levenberg–Marquardt, e.g. SciPy's `curve_fit`
- **Support vector machines**: a quadratic programme (later lectures)
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